

UNDETERMINISTIC SEARCH EXAMPLE

INTRODUCTION TO ARTIFICIAL INTELLIGENT (AI)



REV: 22.04.2024

Example: The goal of the puzzle is to reach the red square from the most upper-left square by moving horizontally or vertically up to the number in the square that you are present. However, if you present on a square labelled by 2 and have a legal move, then you may sometimes move only one square. The agent does not remember its past.

Under this conditions, search on tree by using

1. Breadth-First search
2. Greedy search (use Manhattan Distance)

2	1	2
3	1	1
3	3	

Ex 1: Breadth First Search



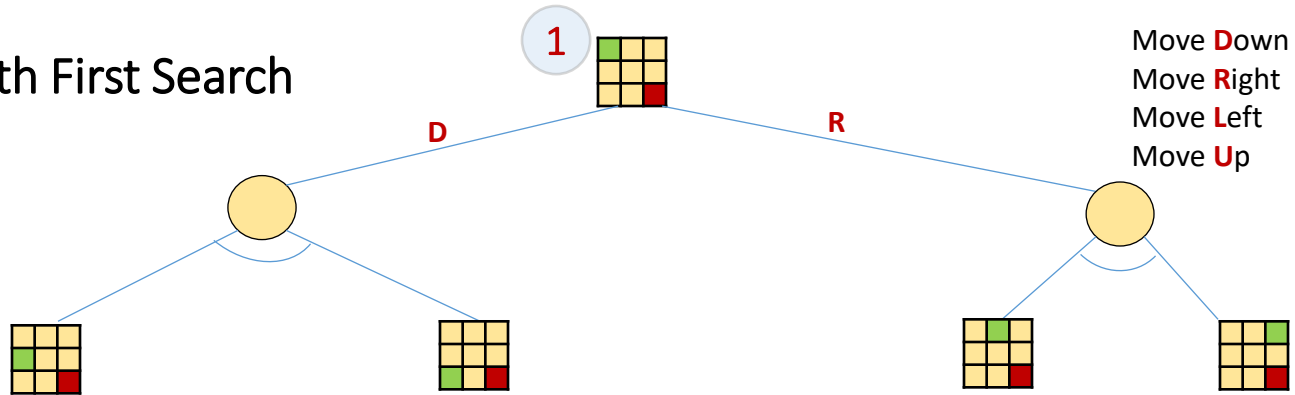
Move **D**own
Move **R**ight
Move **L**eft
Move **U**p

2	1	2
3	1	1
3	3	

Sonuca ulaştıysak ve bu sonuç tek çizgili bir yapraktaysa yani kesin hamlelerden biriyle ulaşılıyorsa burada 1 kademe yukarı çıkabiliriz.

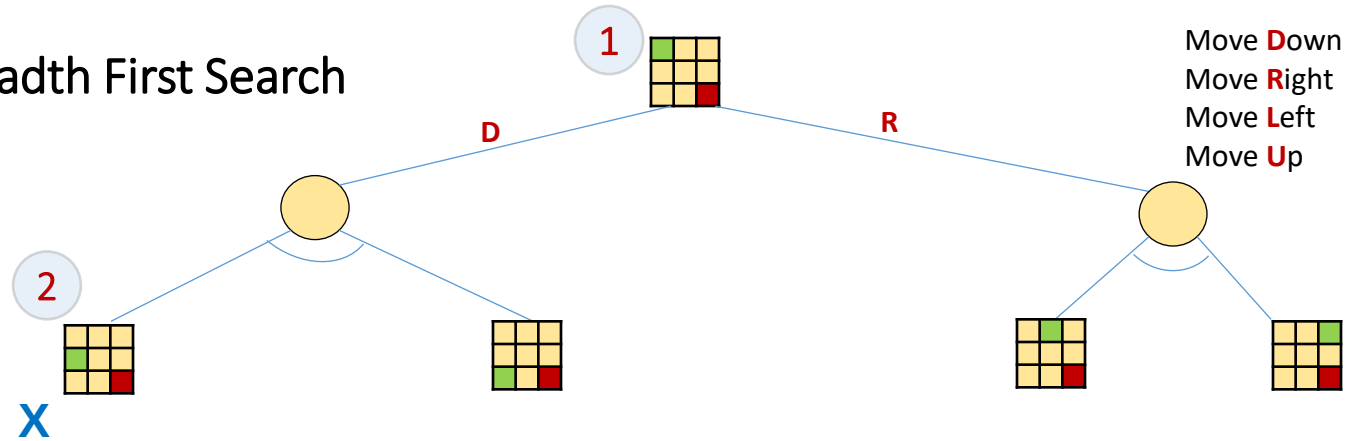
Eğer kesin hamlelerden değil buradaki 2 gibi 1 adım ya da 2 adım attıracak şekildeyse yani çemberli yapraktaysa, yapraktaki kardeşi de onunla beraber yukarı çıkabilmeli tek başına çıkamaz.

Ex 1: Breadth First Search



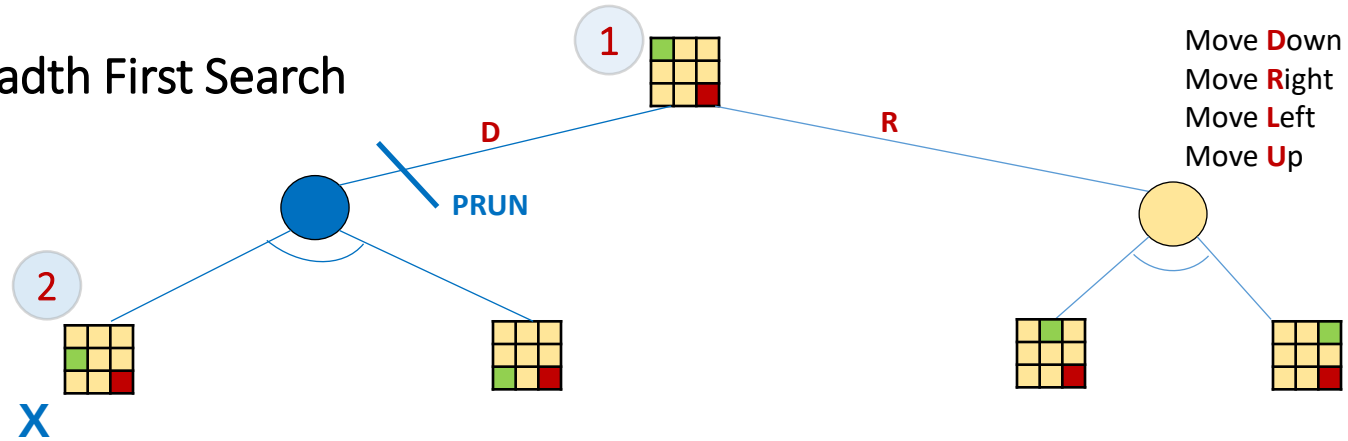
2	1	2
3	1	1
3	3	

Ex 1: Breadth First Search



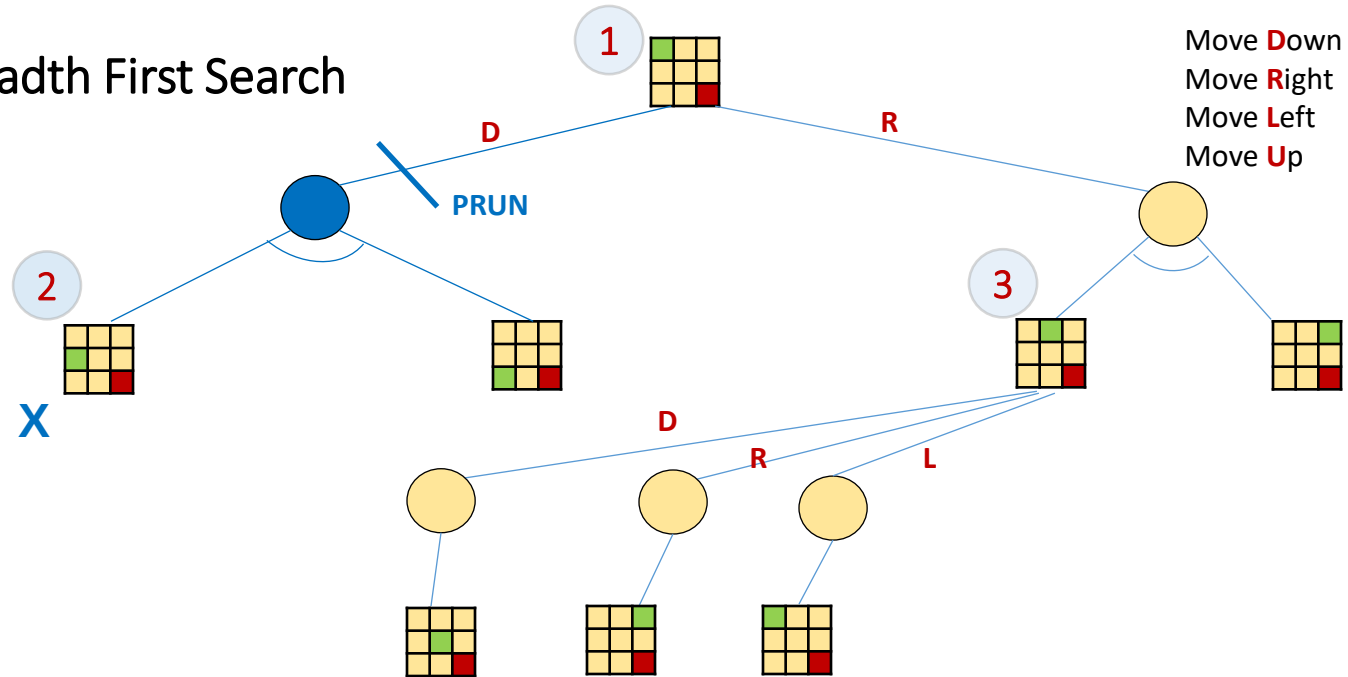
2	1	2
3	1	1
3	3	

Ex 1: Breadth First Search



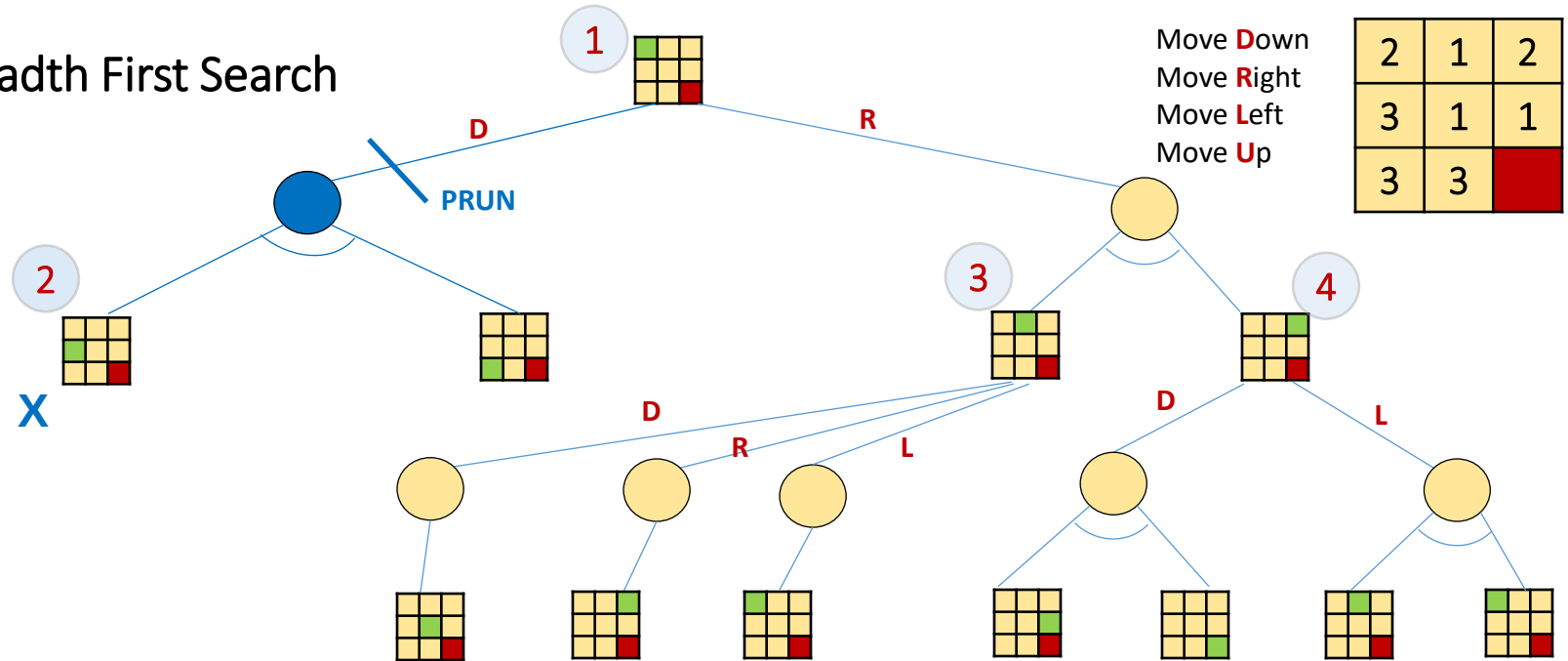
2	1	2
3	1	1
3	3	

Ex 1: Breadth First Search

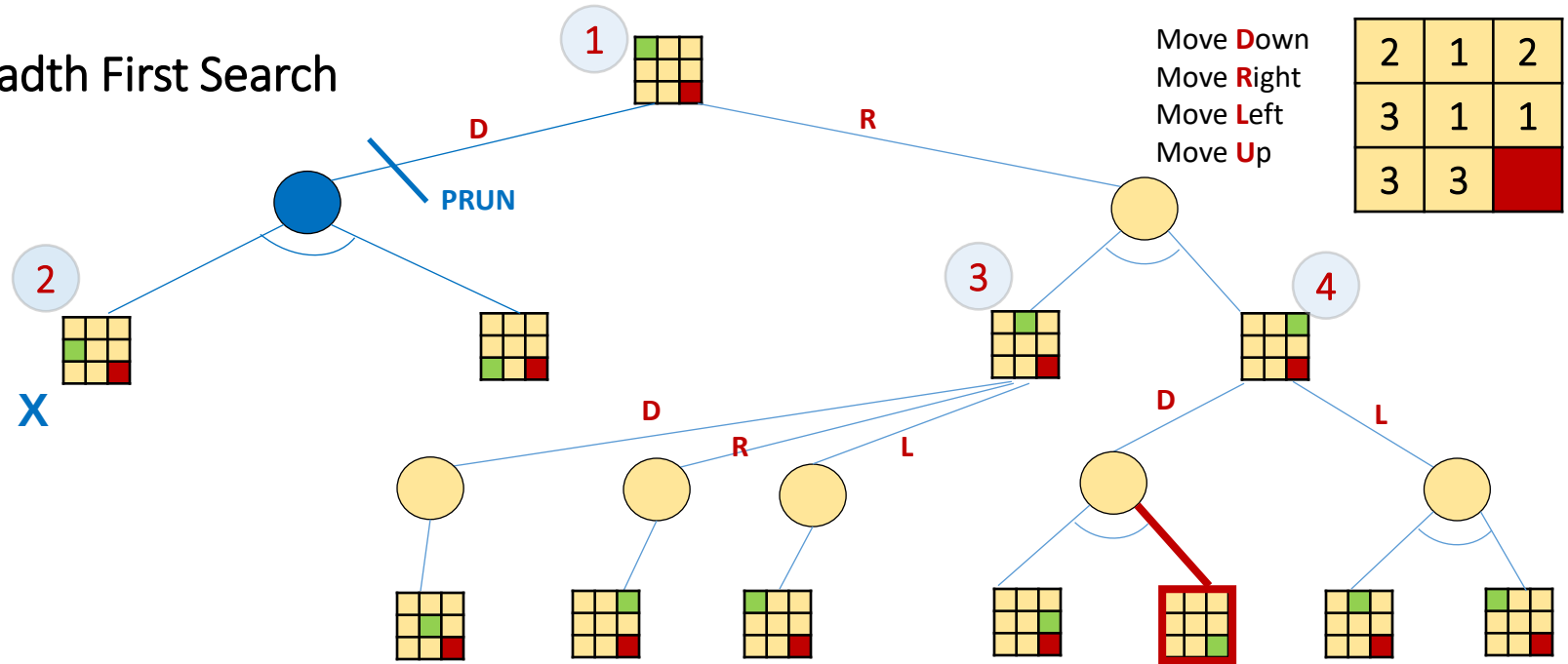


2	1	2
3	1	1
3	3	

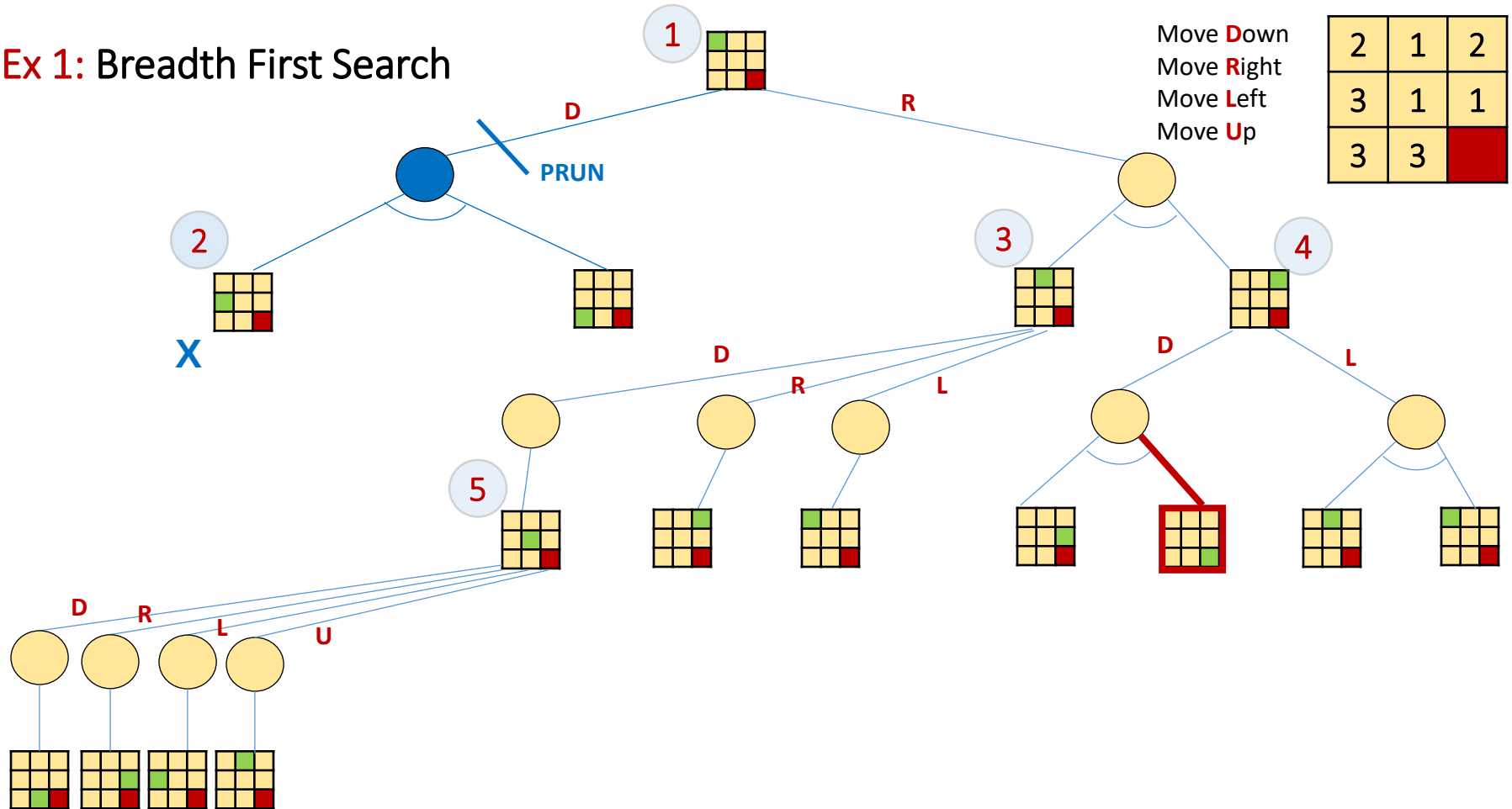
Ex 1: Breadth First Search



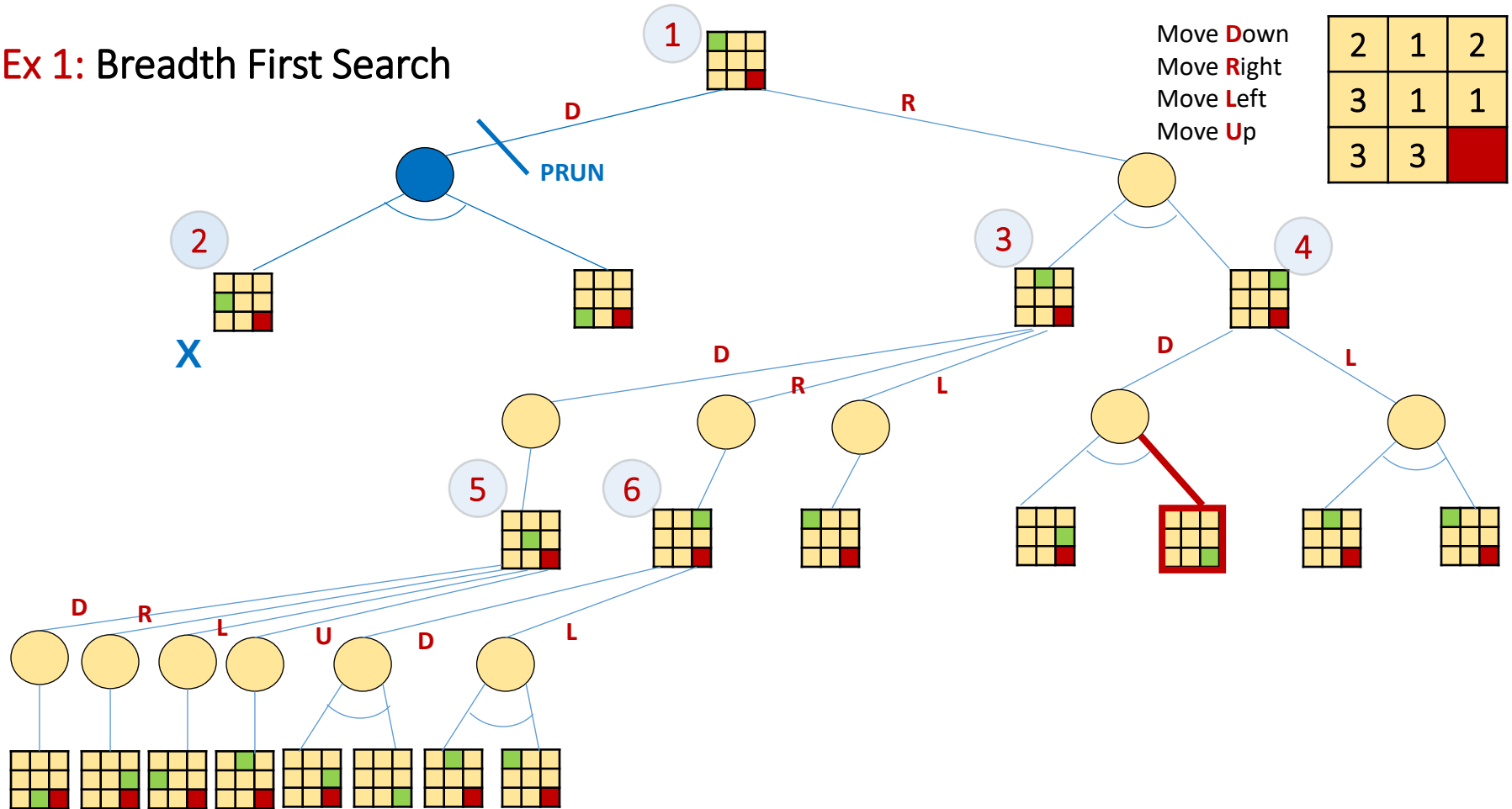
Ex 1: Breadth First Search



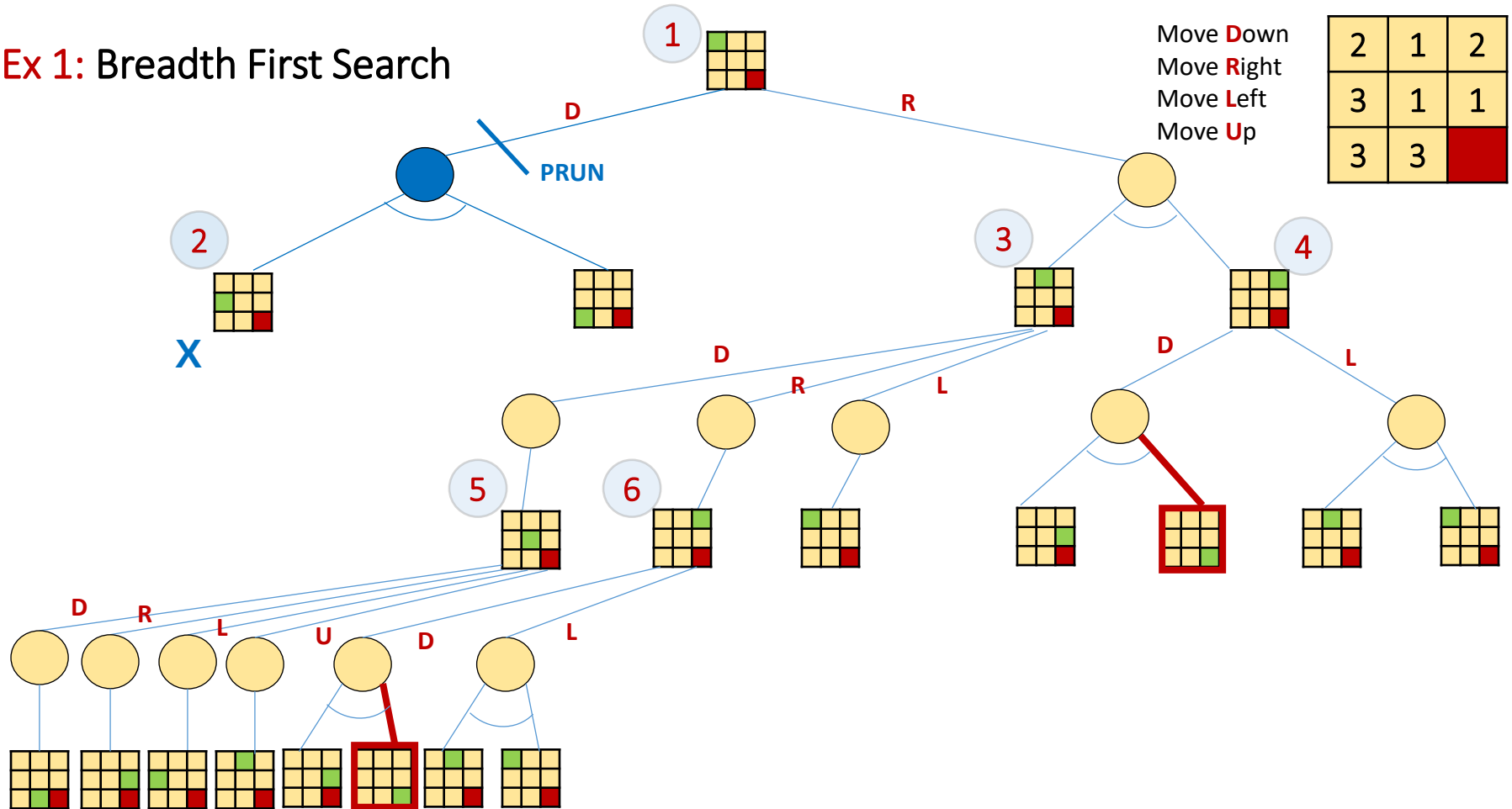
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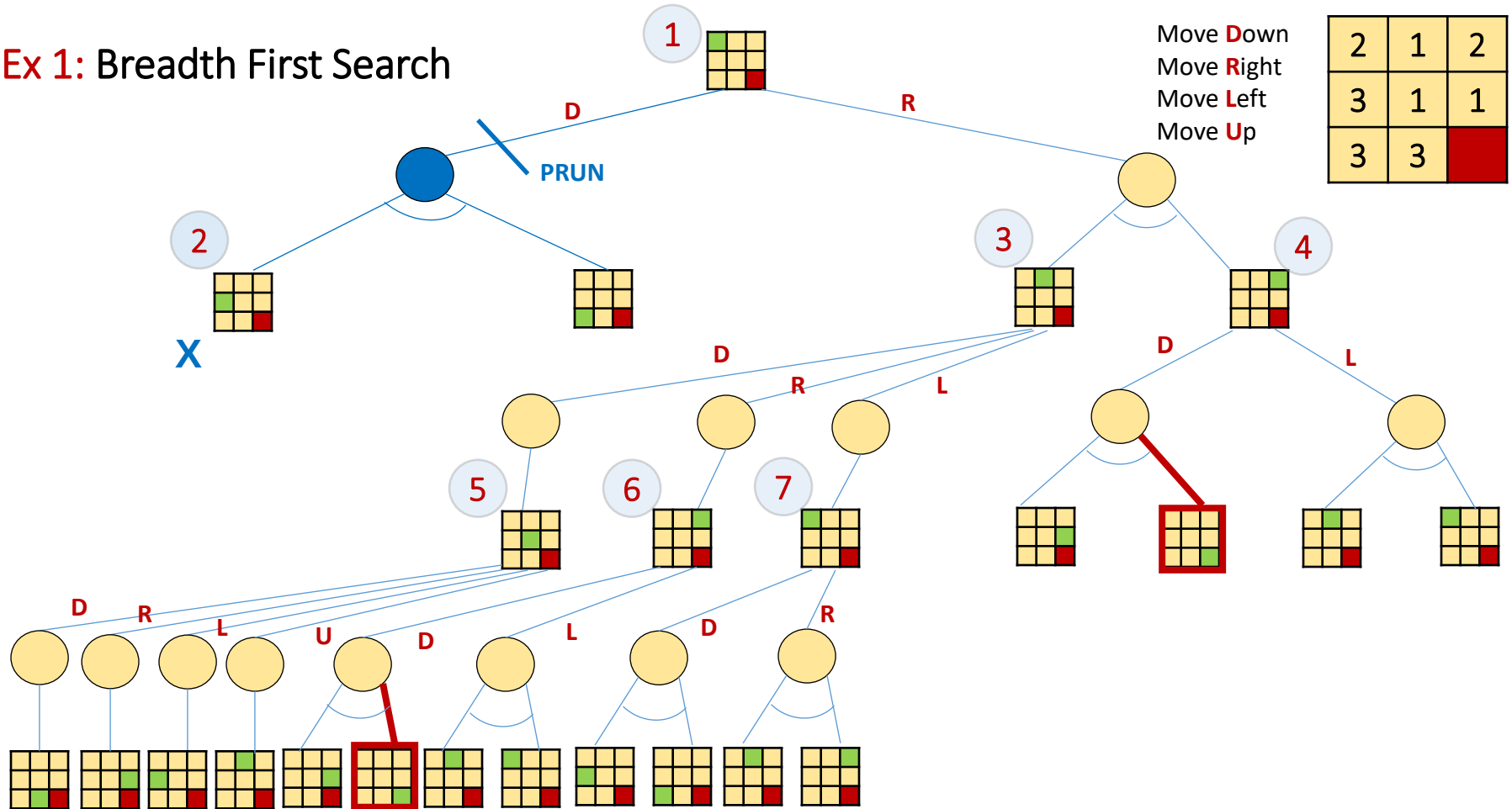
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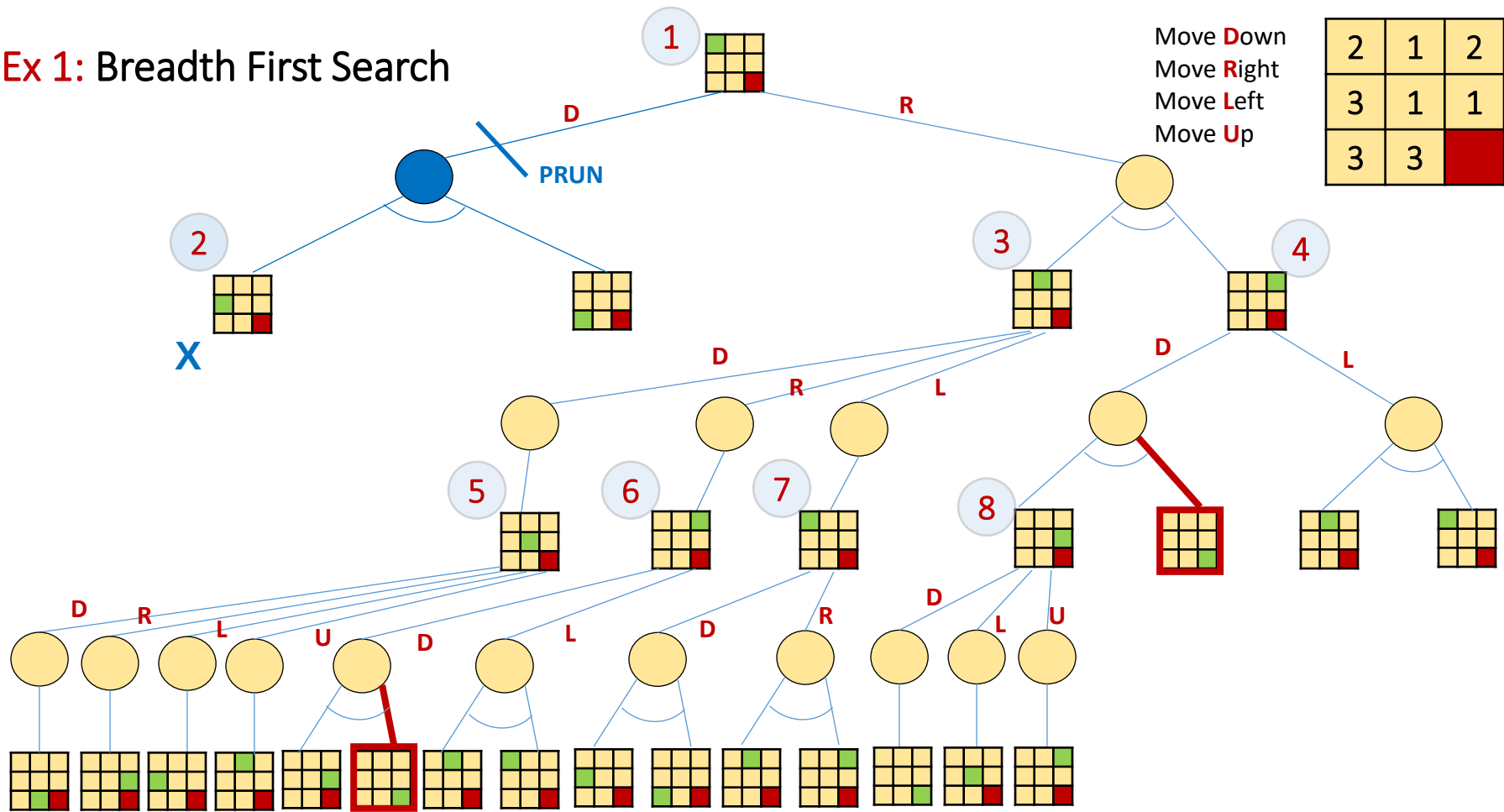
Ex 1: Breadth First Search



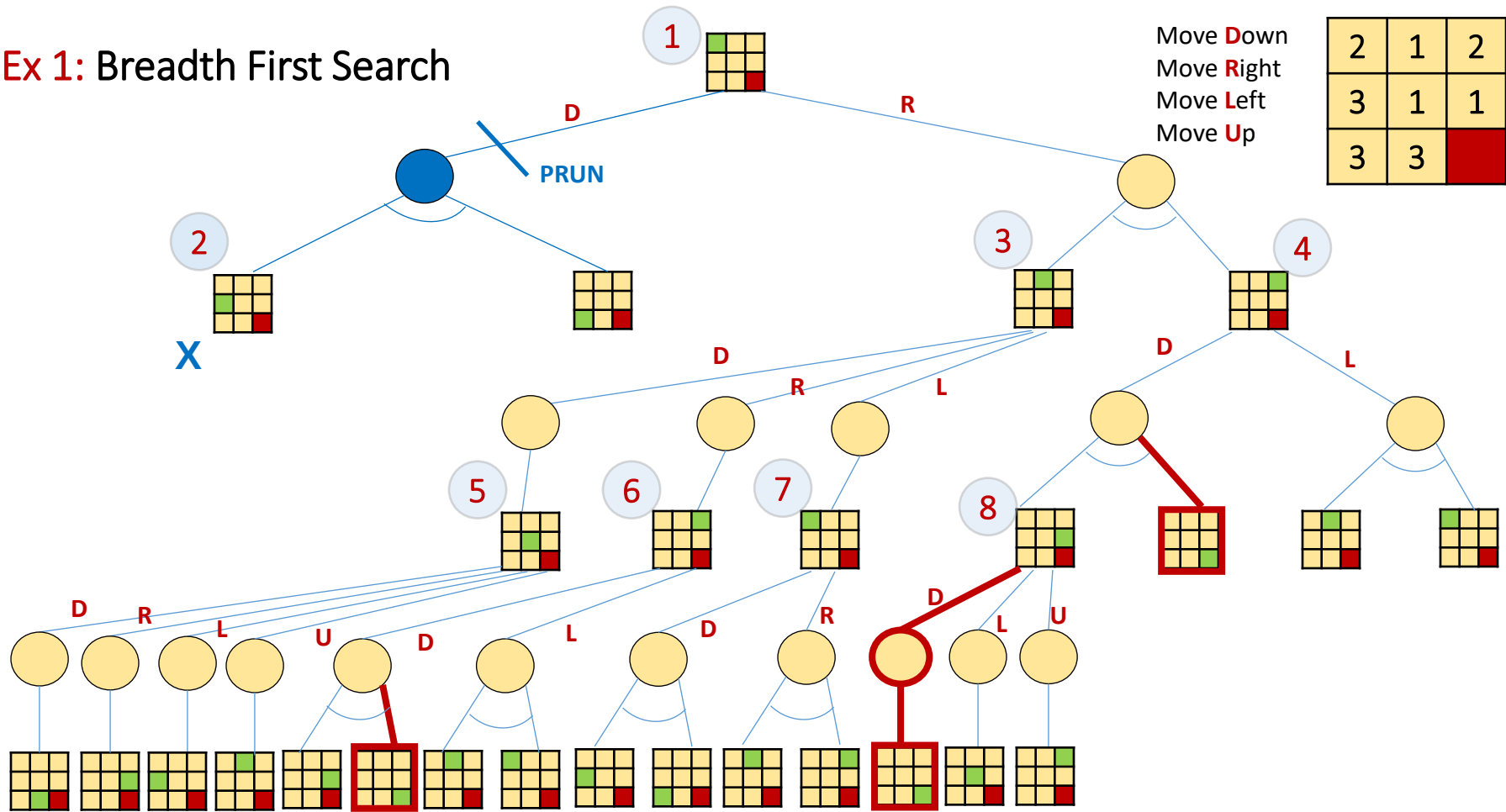
Ex 1: Breadth First Search



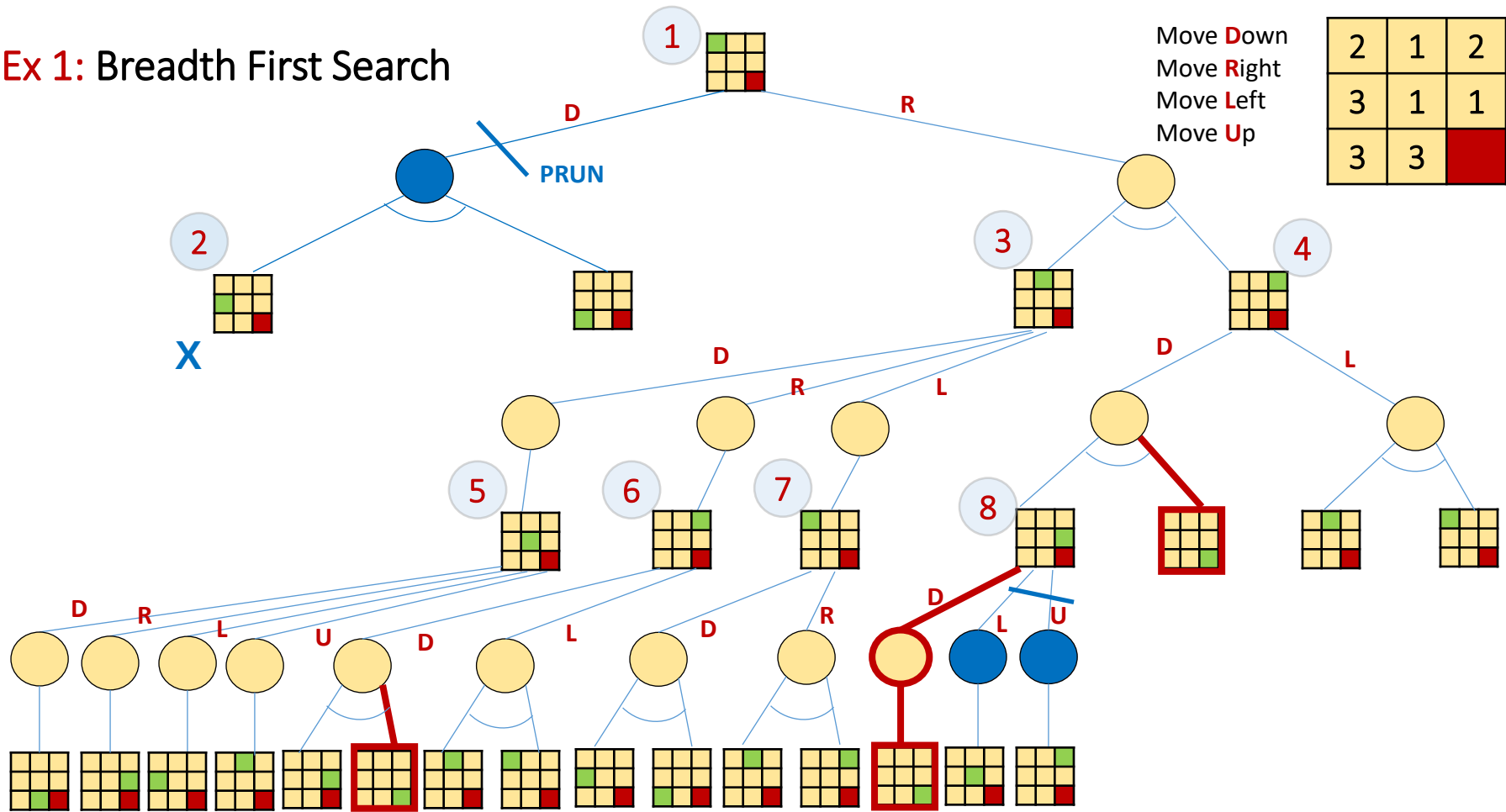
Ex 1: Breadth First Search



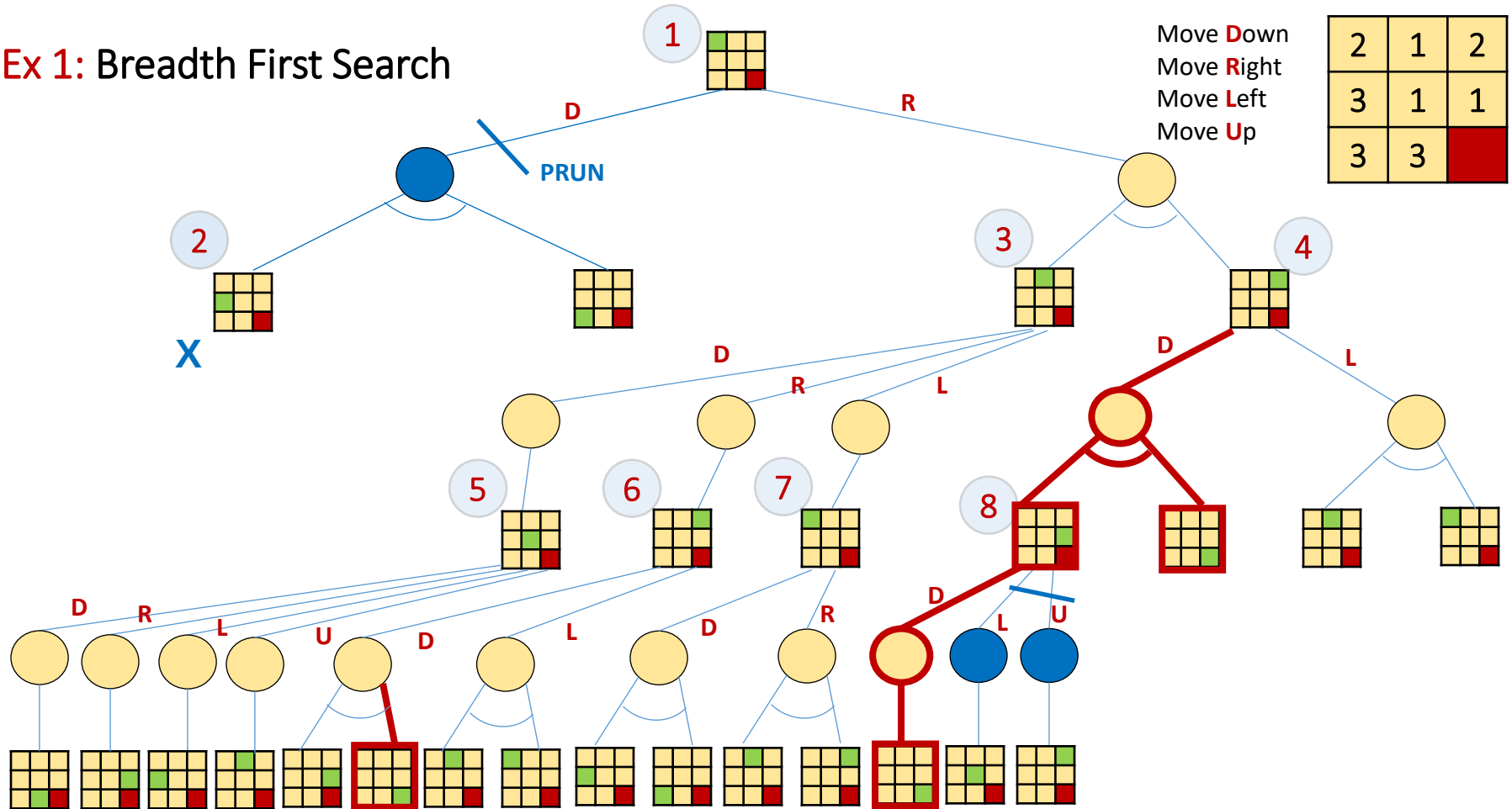
Ex 1: Breadth First Search



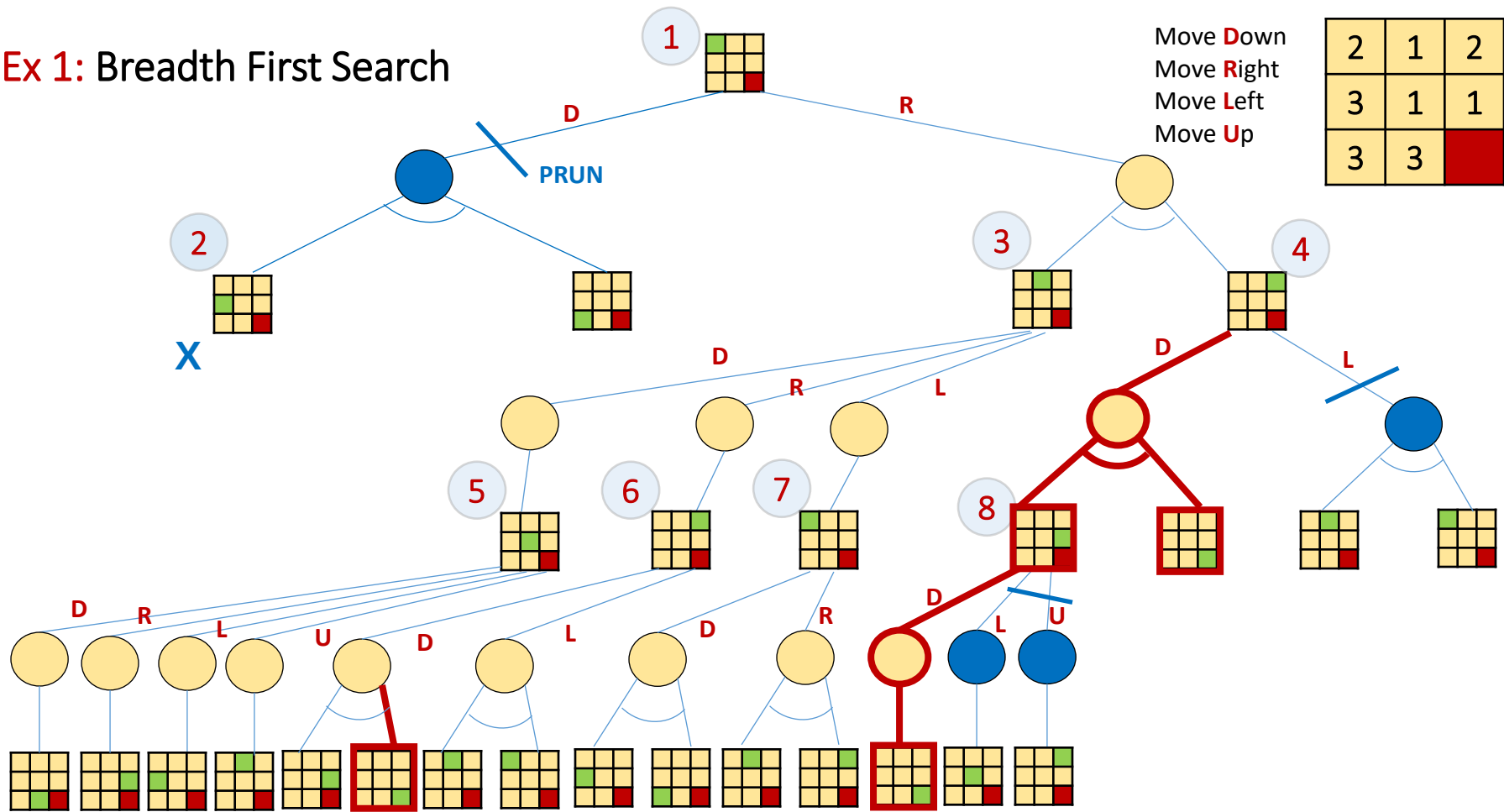
Ex 1: Breadth First Search



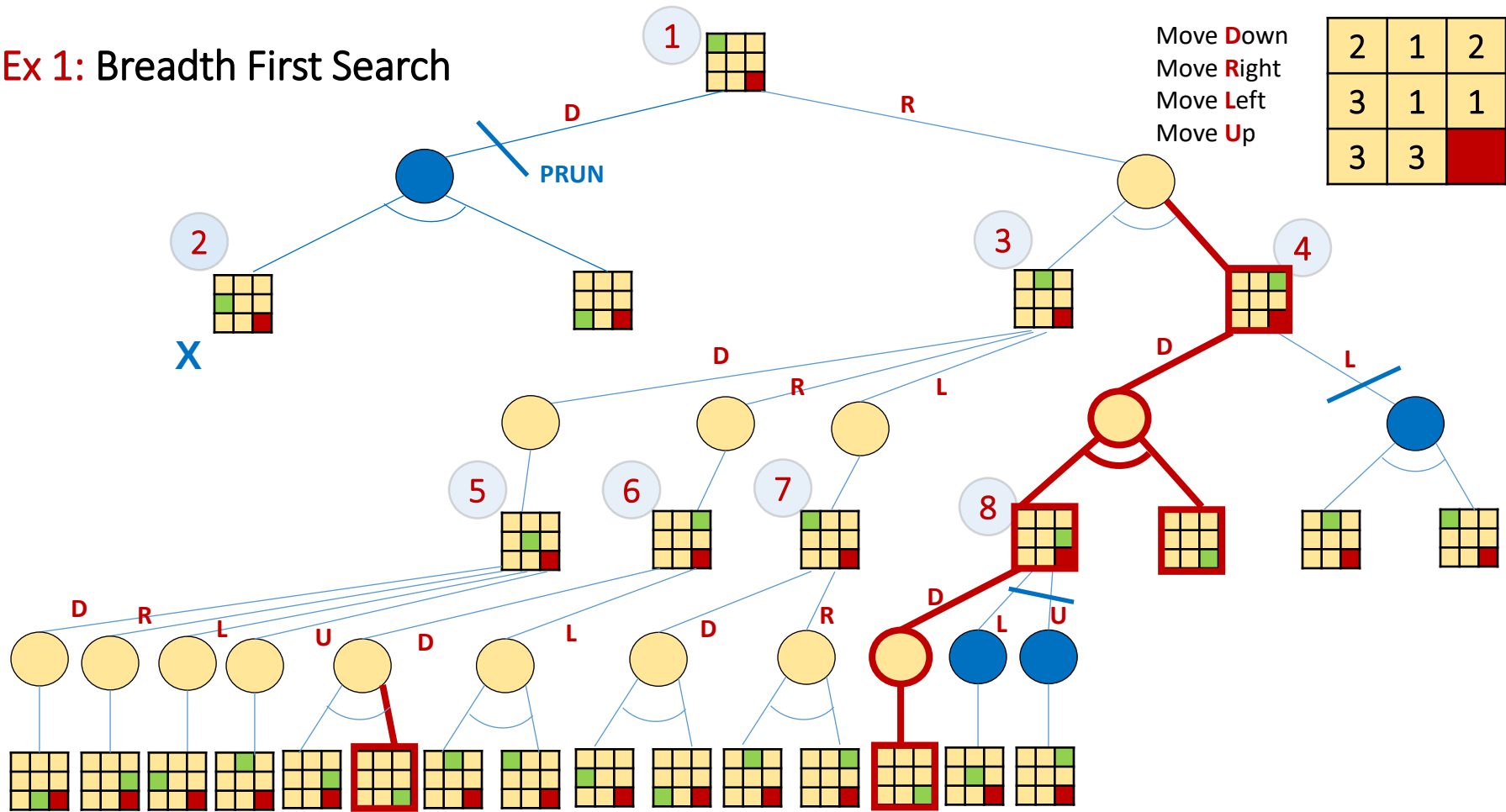
Ex 1: Breadth First Search



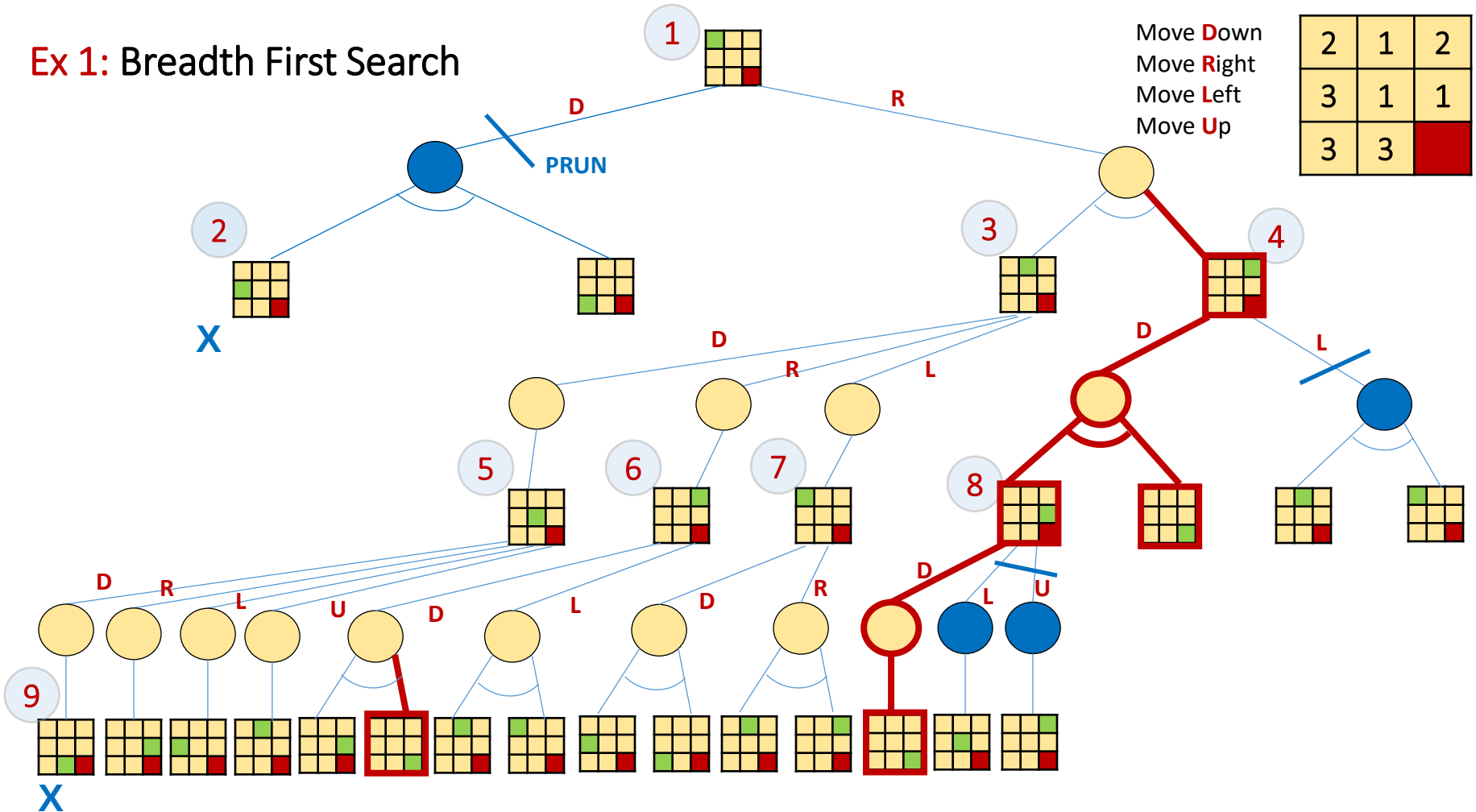
Ex 1: Breadth First Search



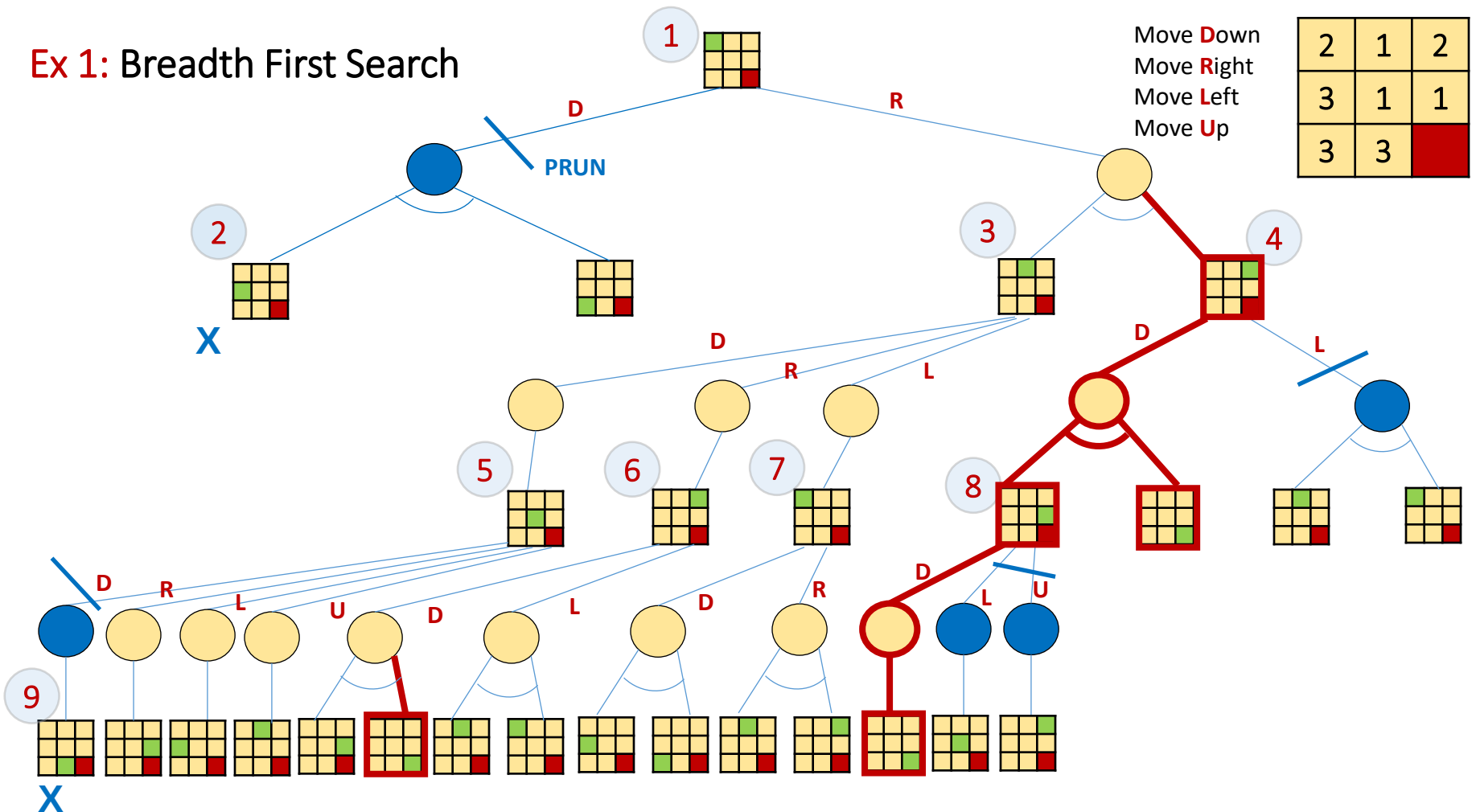
Ex 1: Breadth First Search



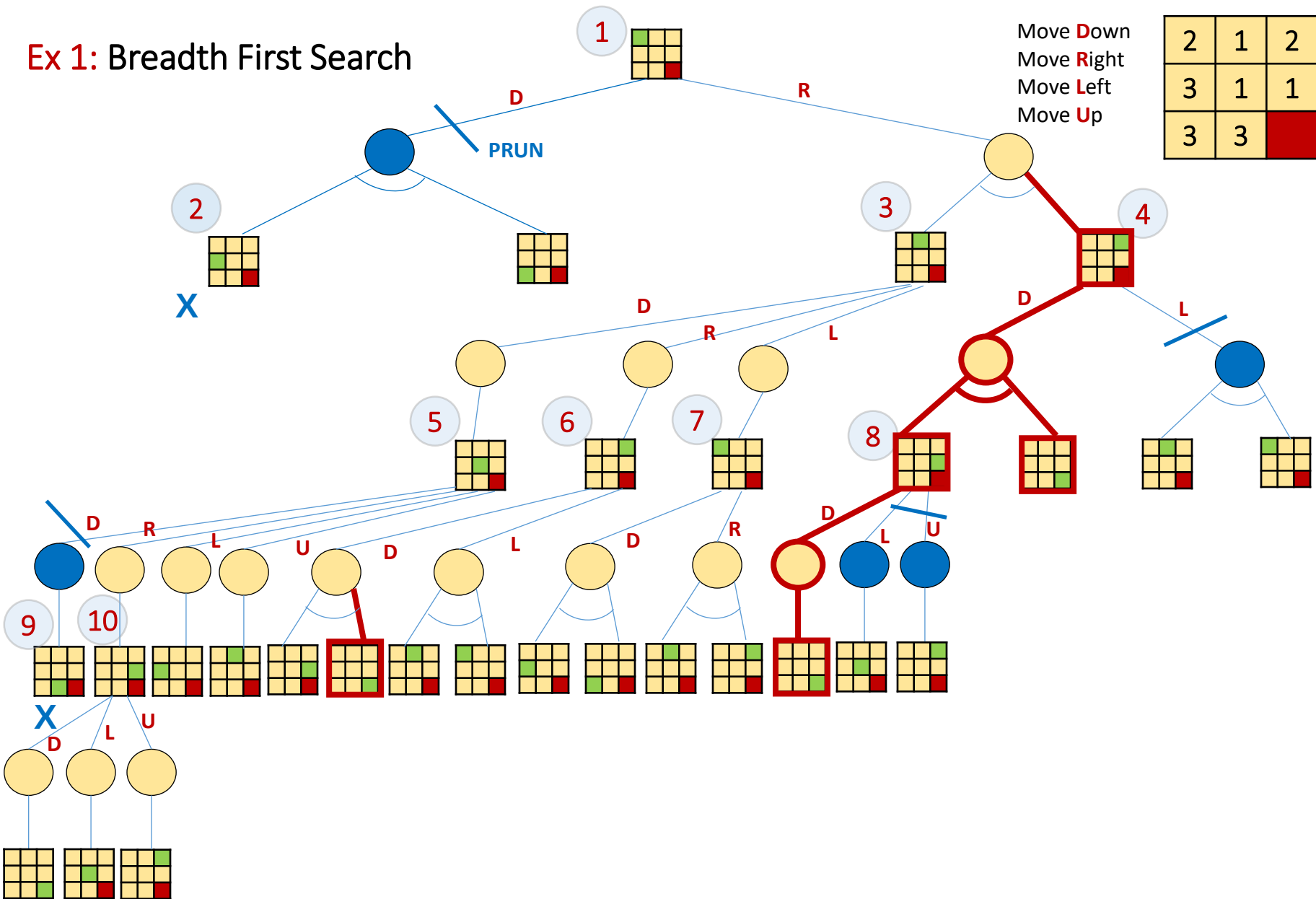
Ex 1: Breadth First Search



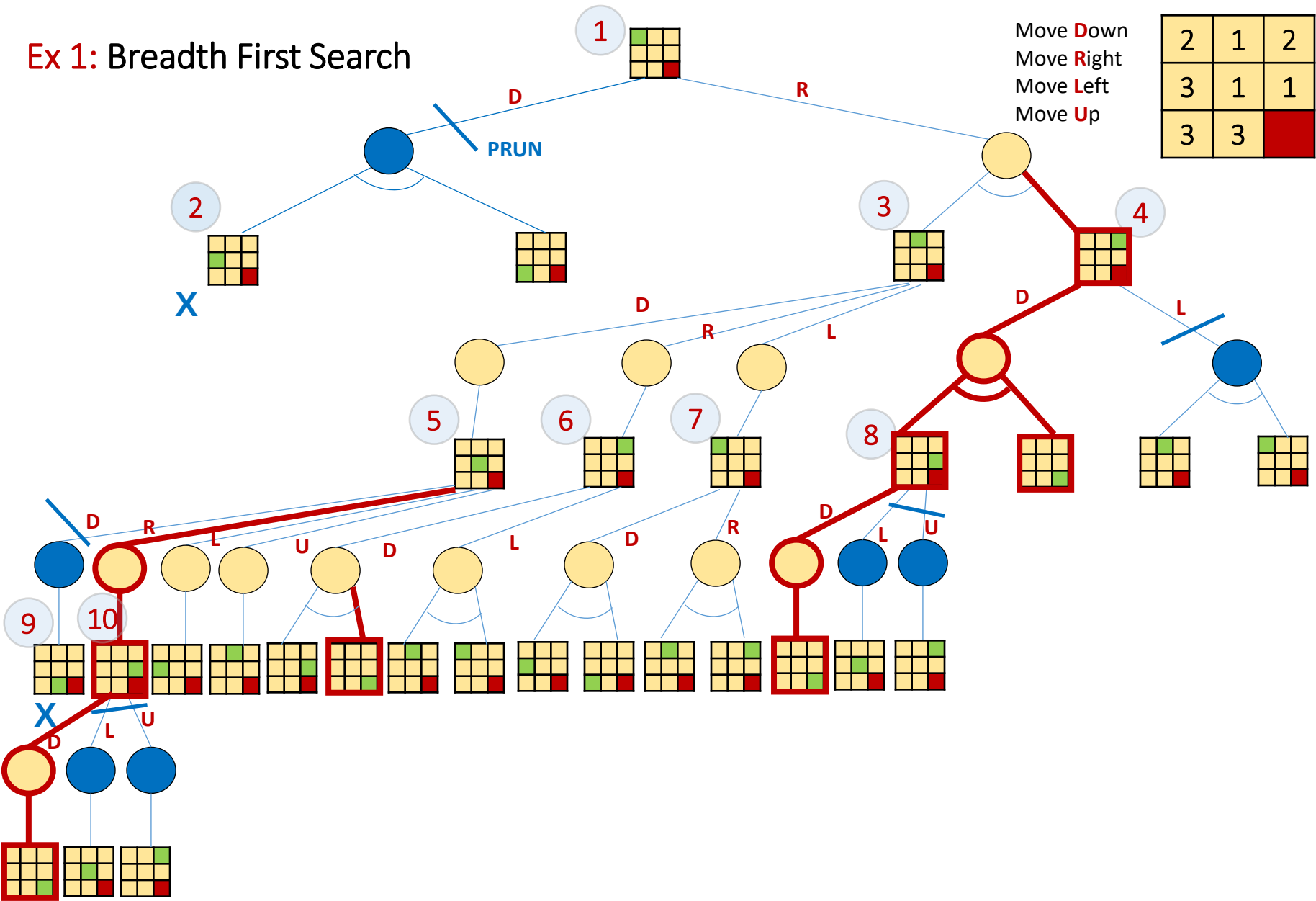
Ex 1: Breadth First Search



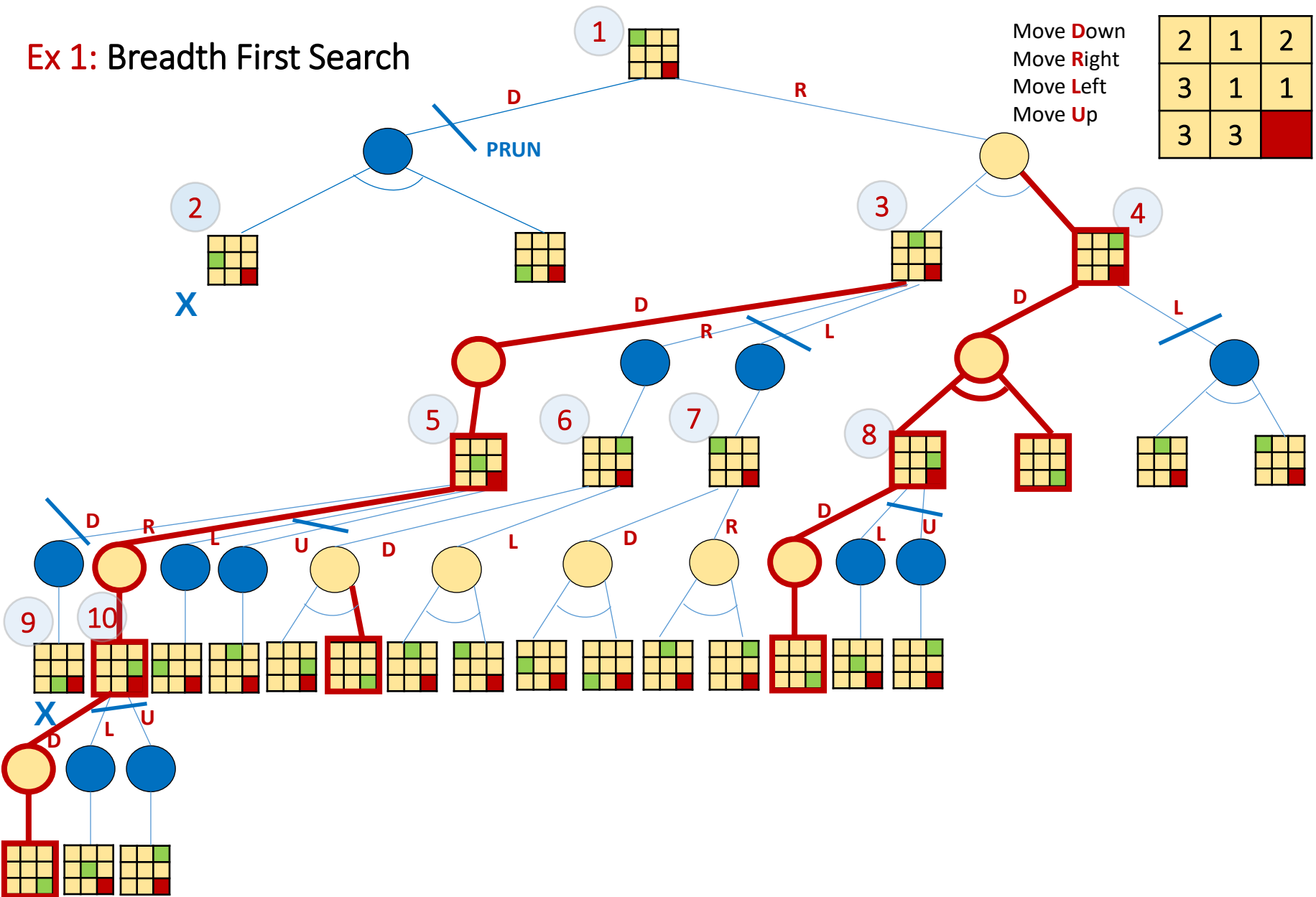
Ex 1: Breadth First Search



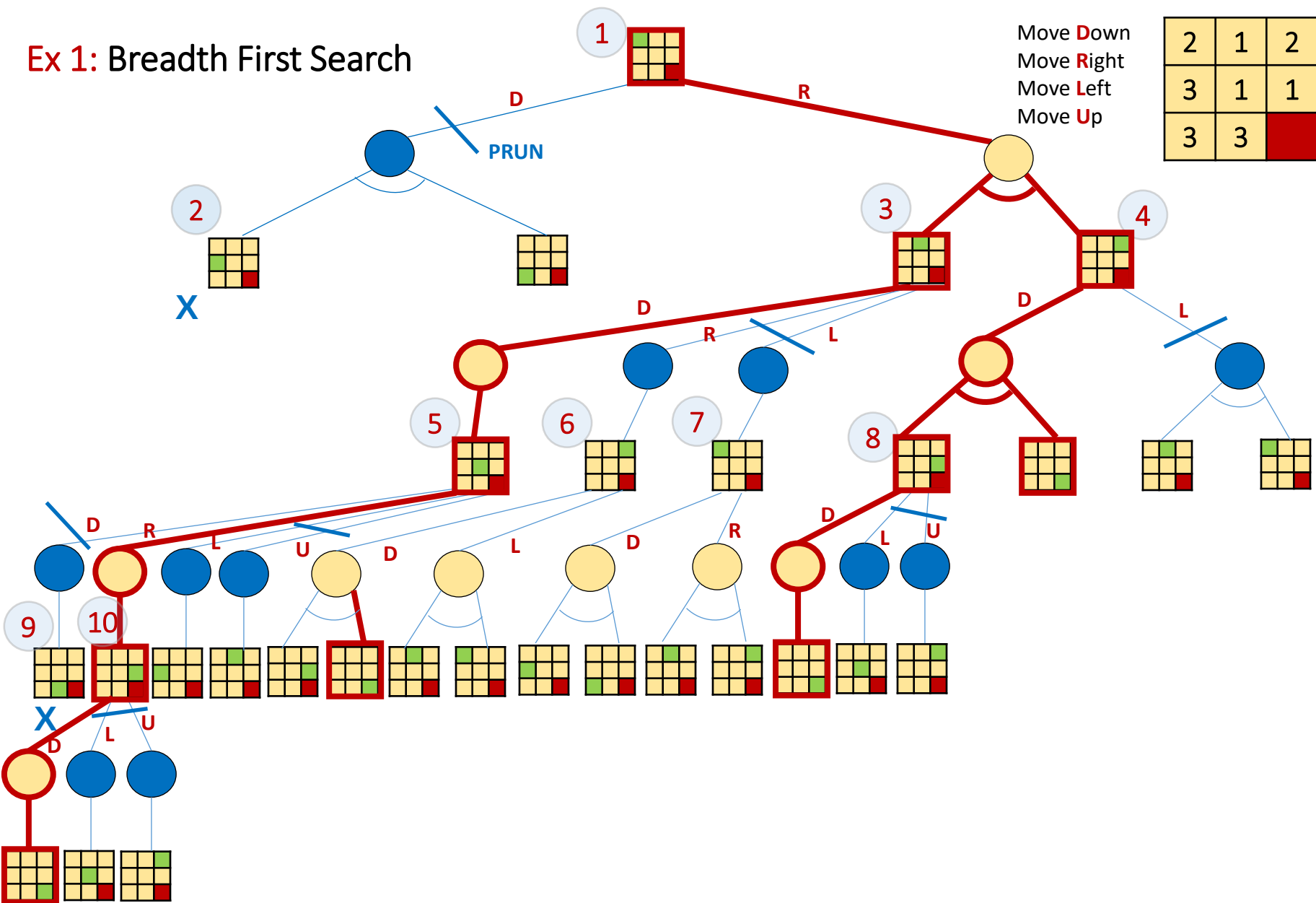
Ex 1: Breadth First Search



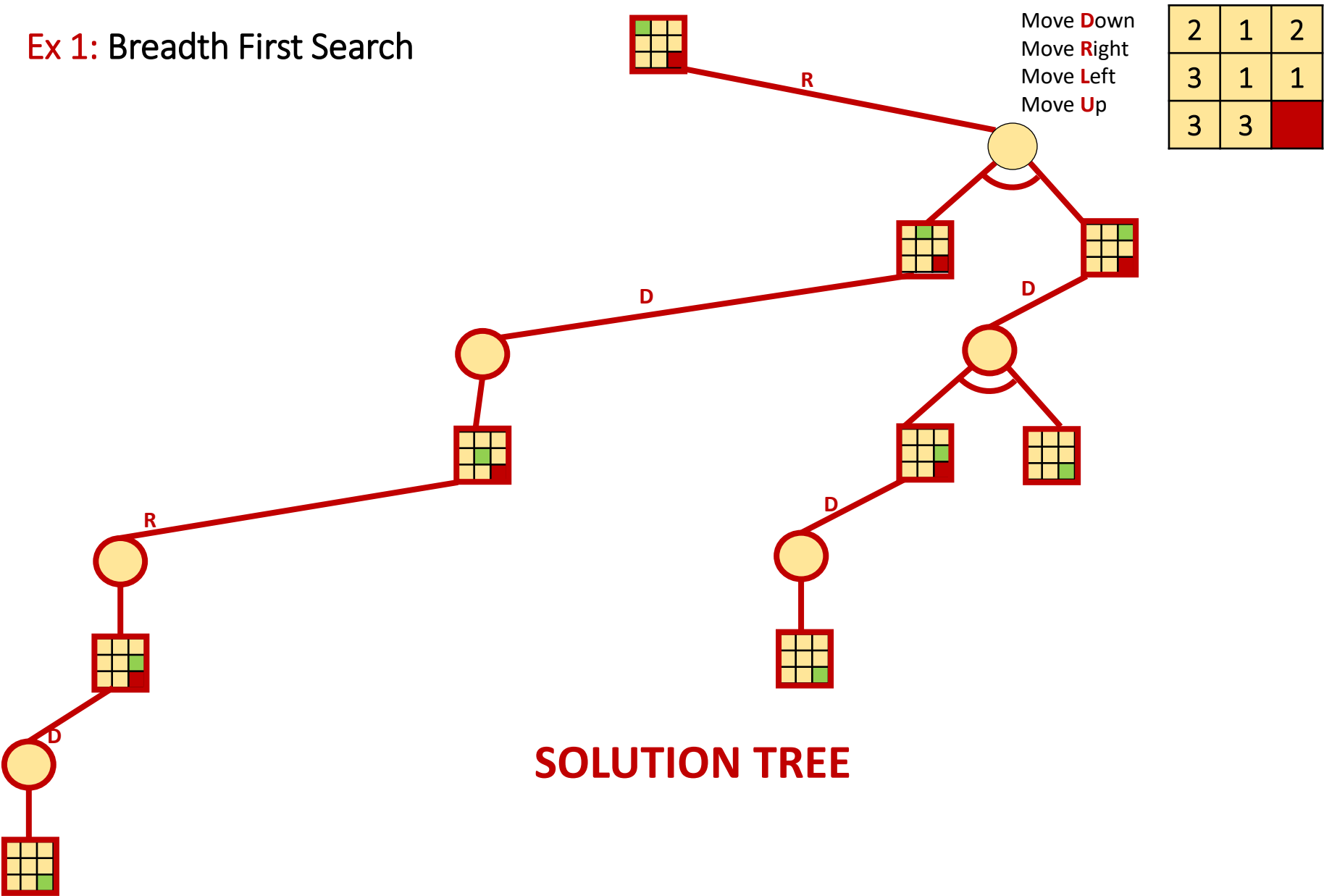
Ex 1: Breadth First Search



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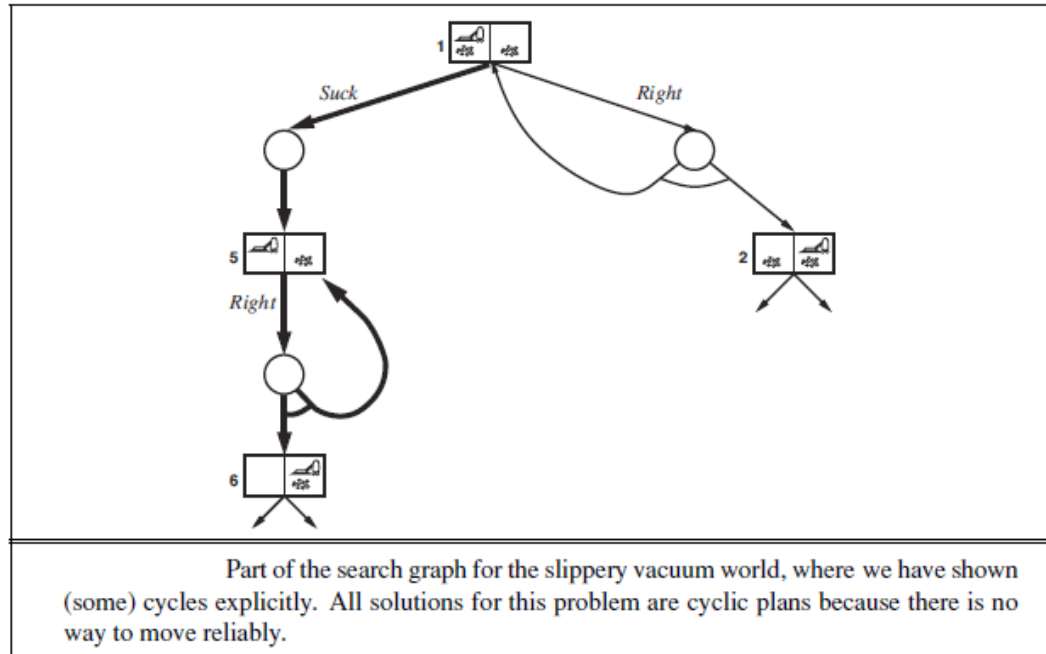
Ex 1: Breadth First Search



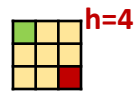
Consider the problem when the agent remembers its past!!



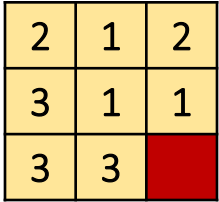
The problem becomes complicated. Assume that the following agent remembers its past. Does the agent find a solution?



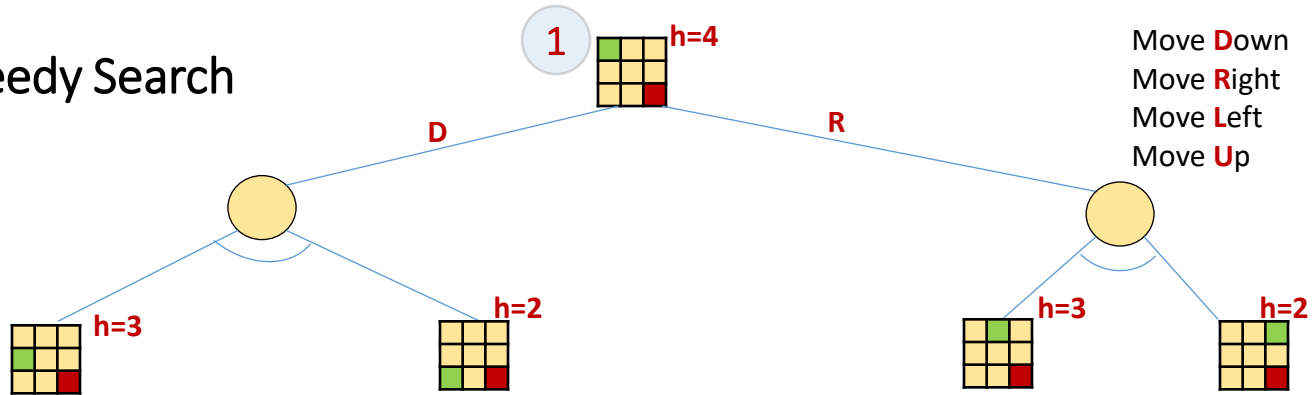
Ex 1: Greedy Search



- Move **D**own
- Move **R**ight
- Move **L**eft
- Move **U**p

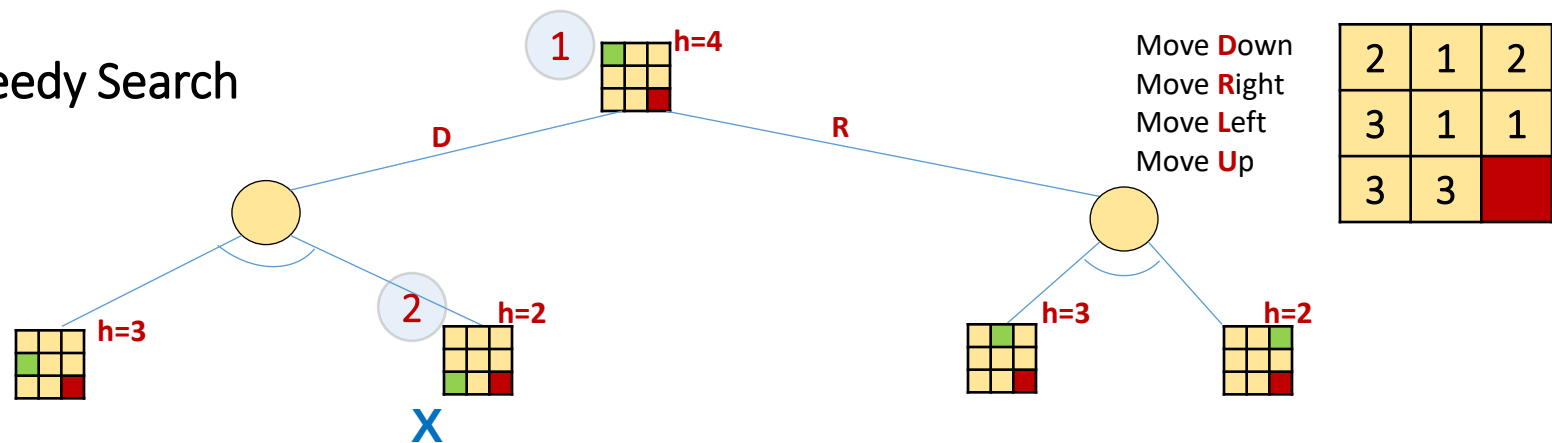


Ex 1: Greedy Search

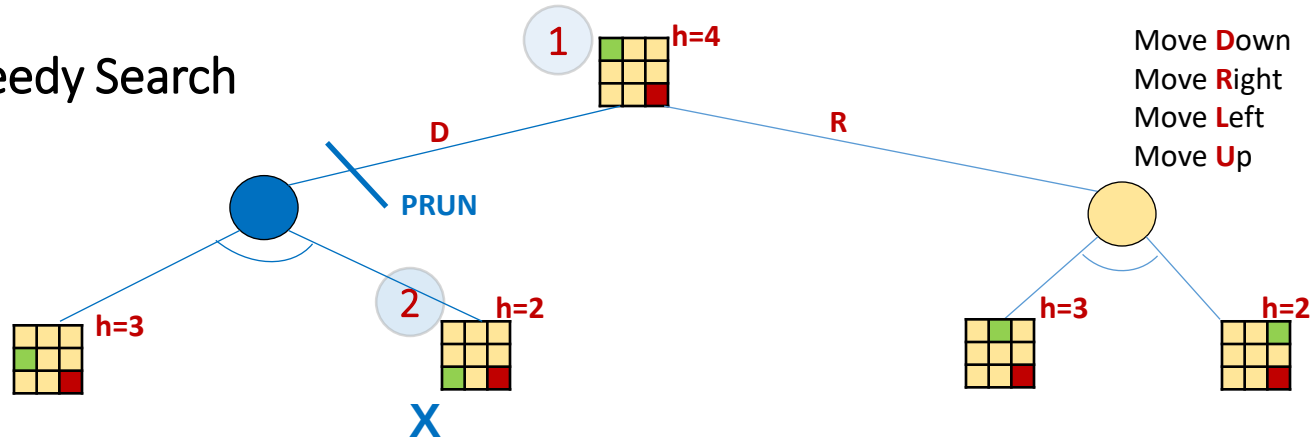


2	1	2
3	1	1
3	3	

Ex 1: Greedy Search

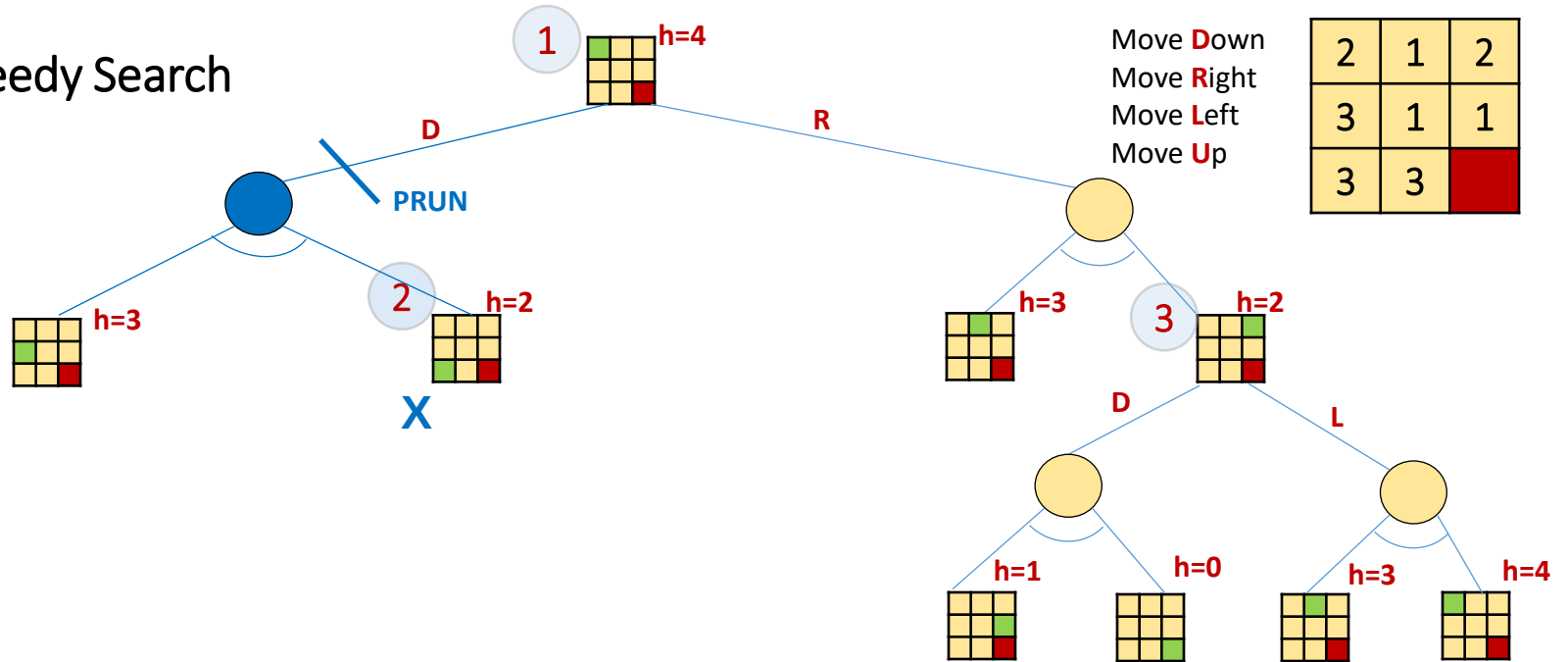


Ex 1: Greedy Search

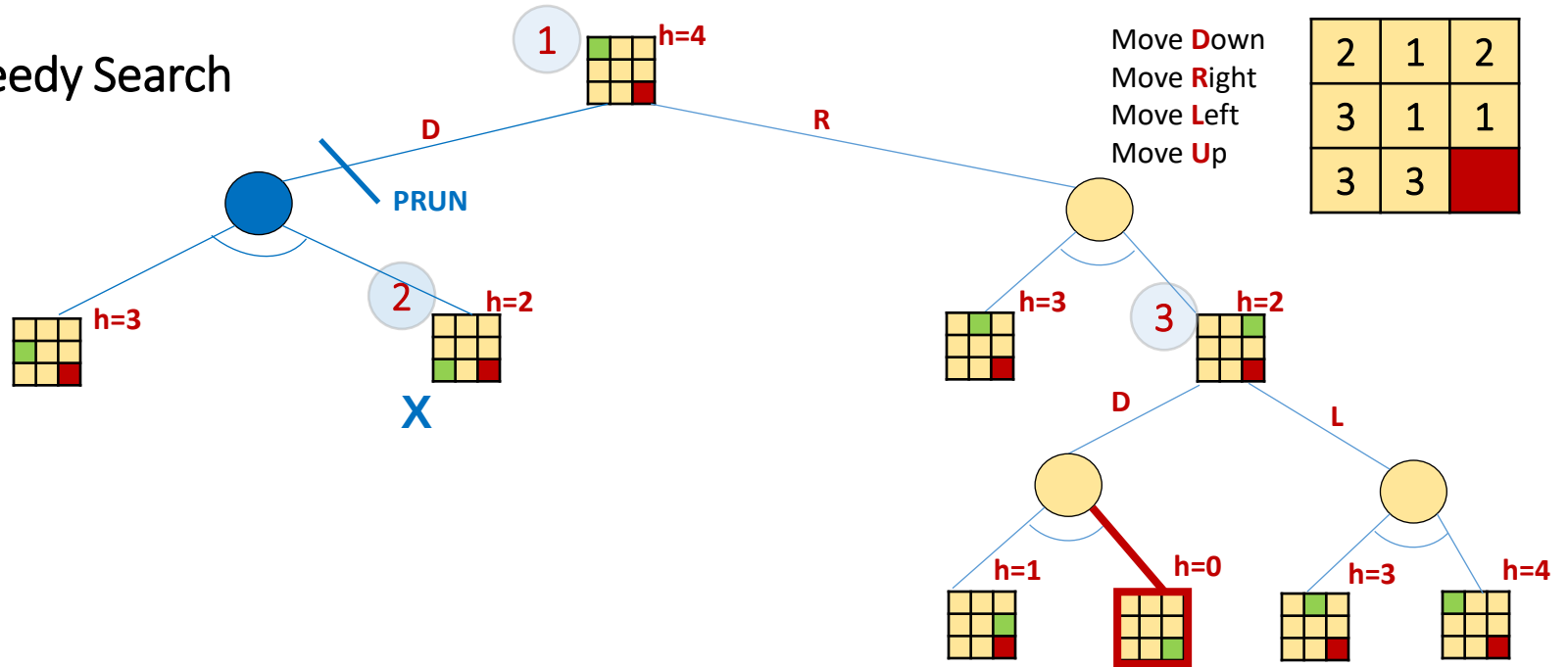


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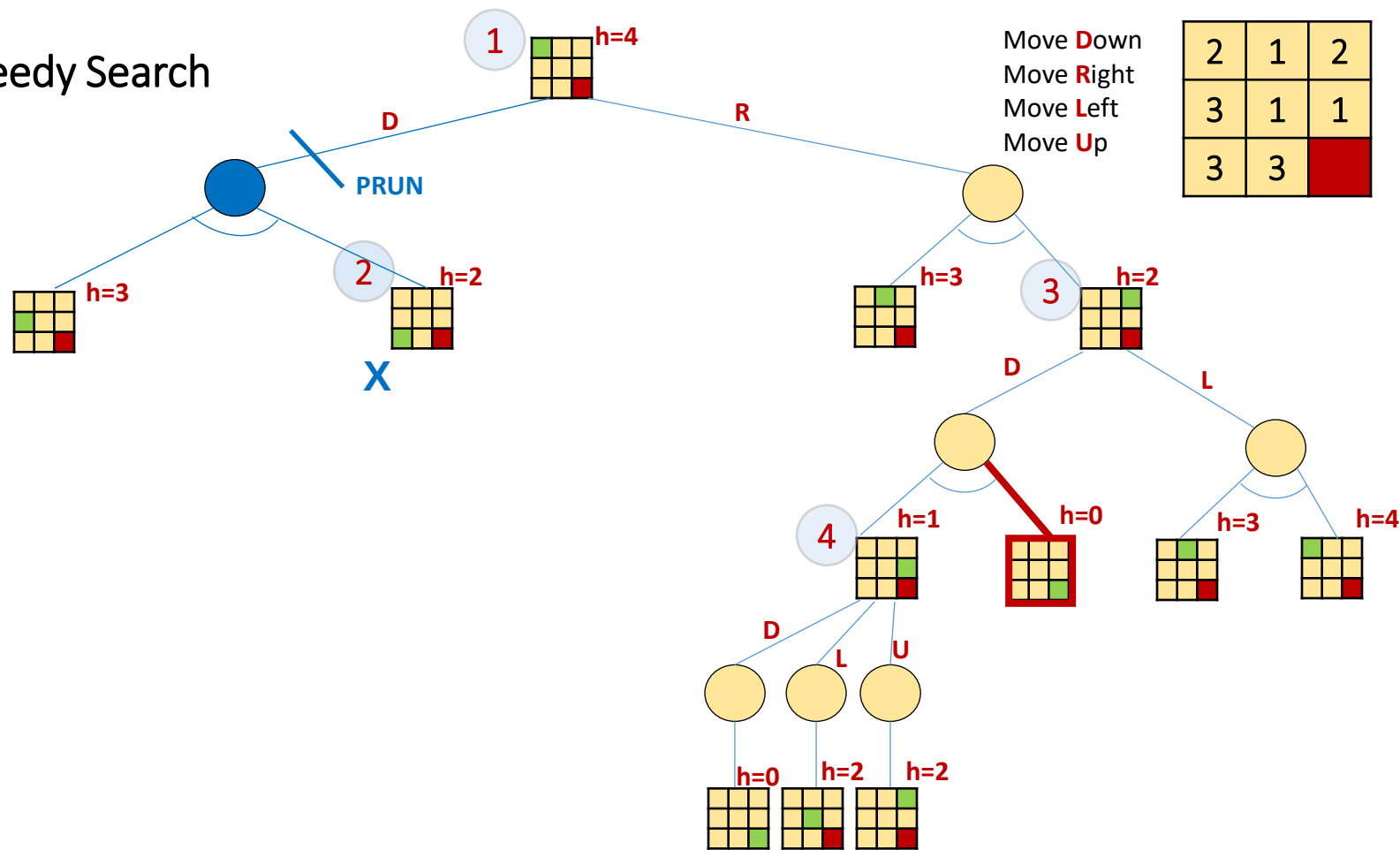
Ex 1: Greedy Search



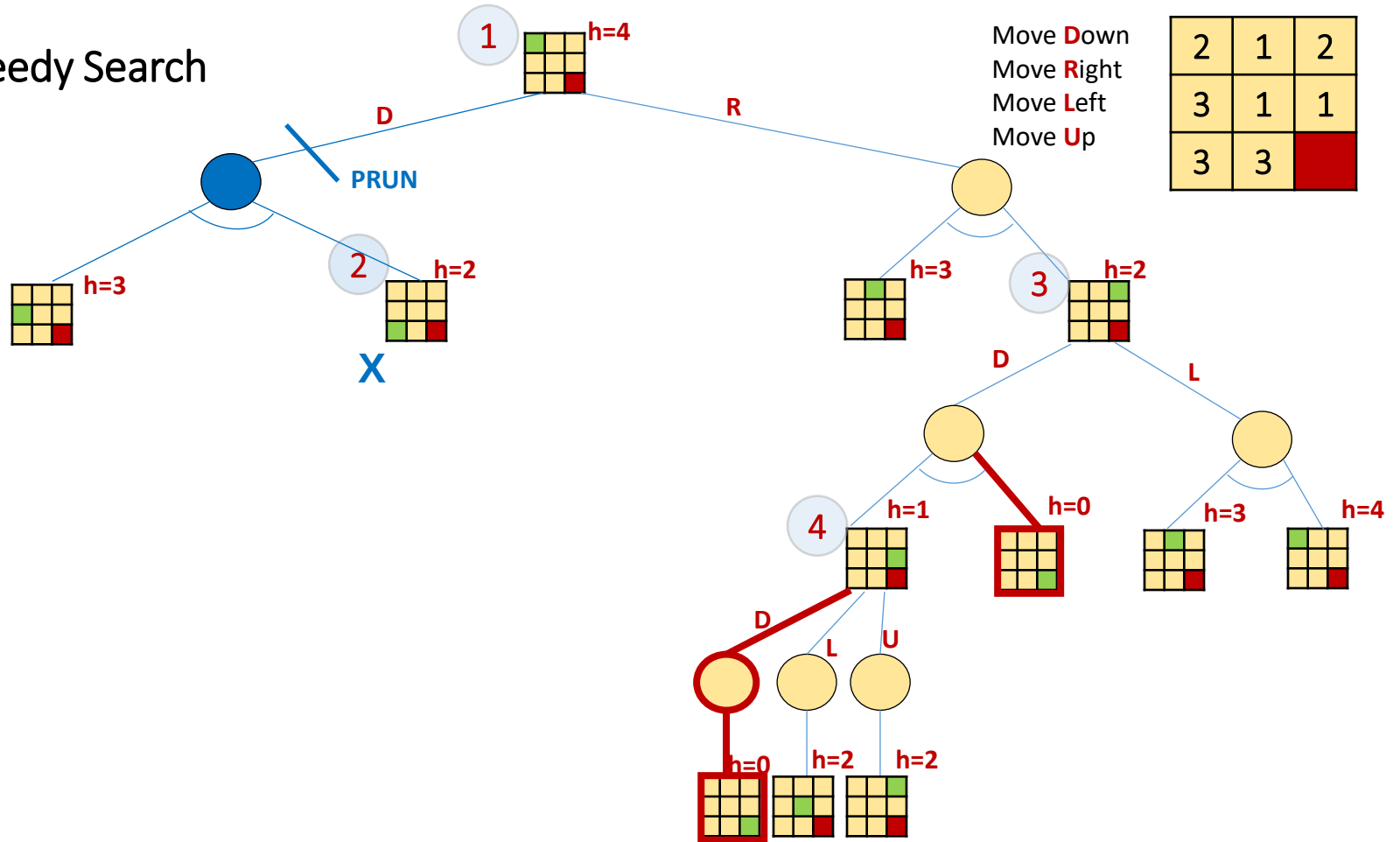
Ex 1: Greedy Search



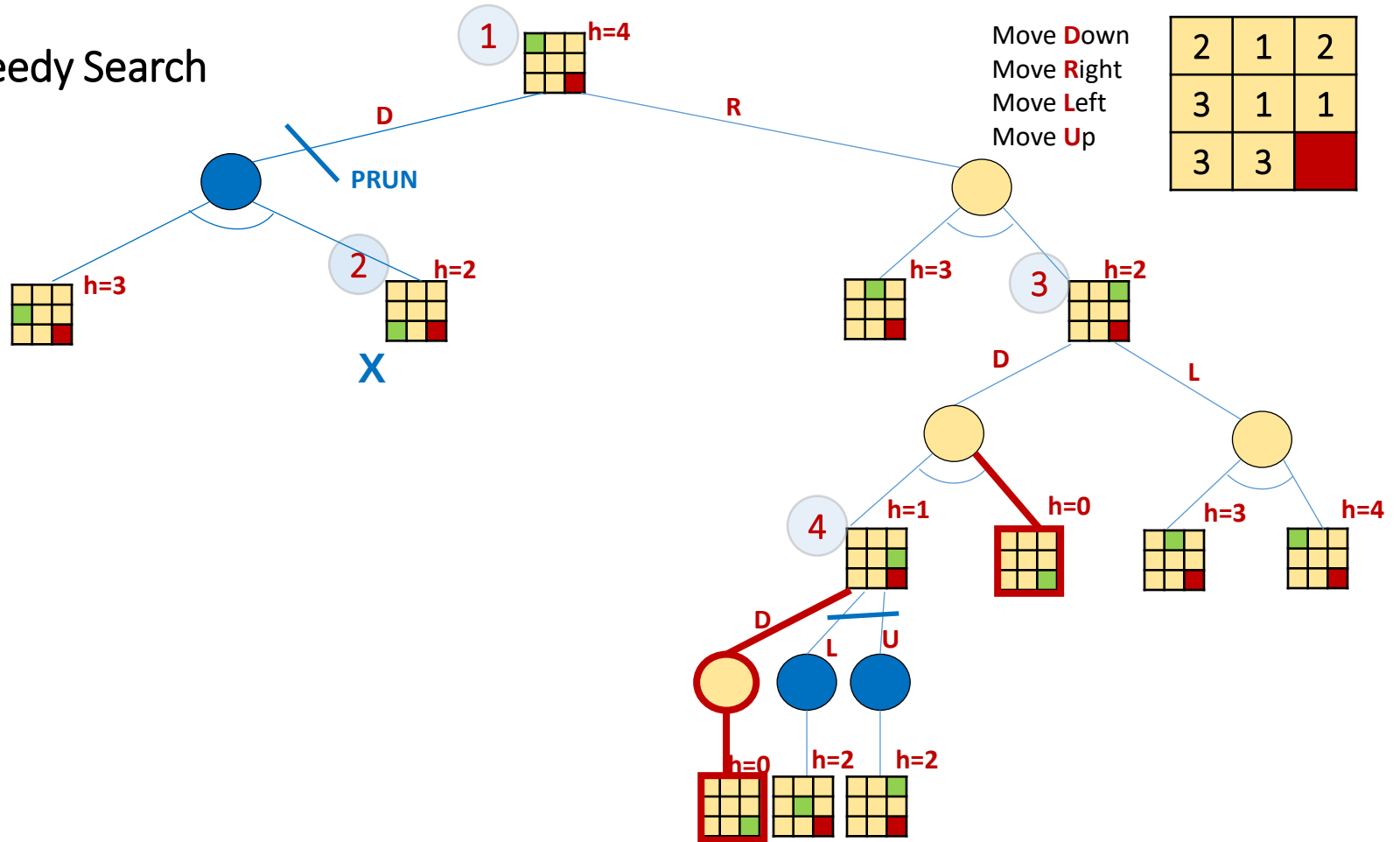
Ex 1: Greedy Search



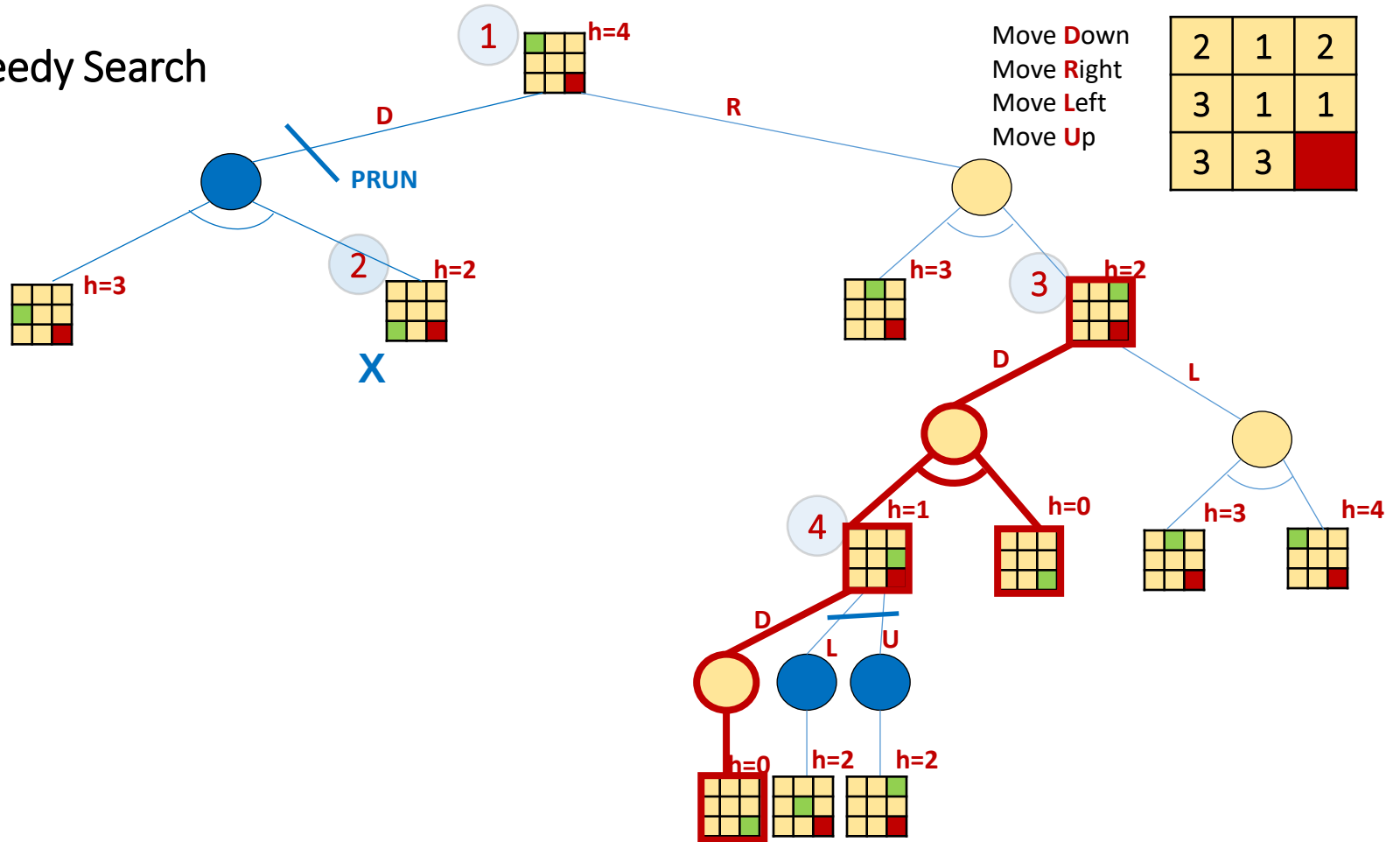
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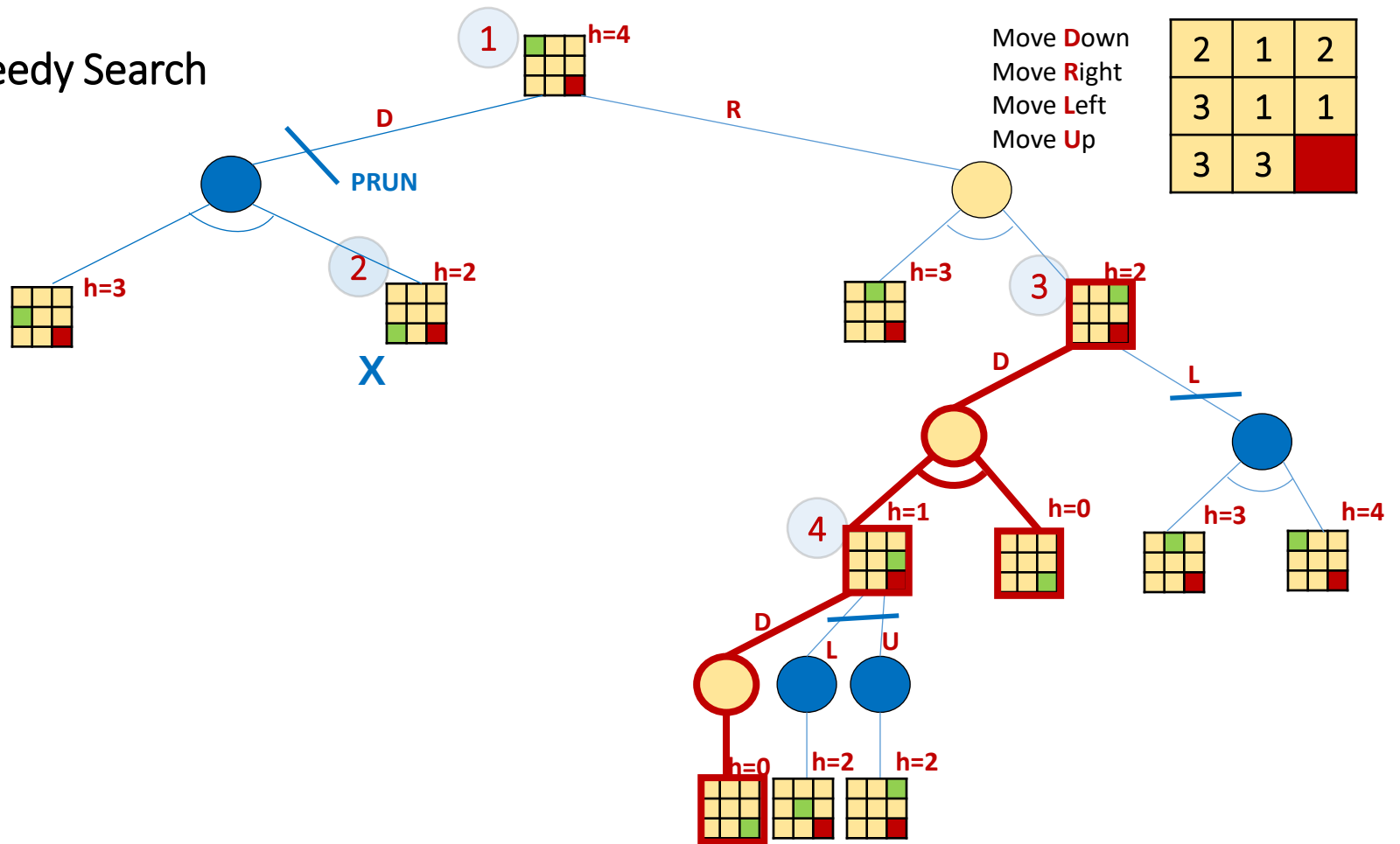
Ex 1: Greedy Search



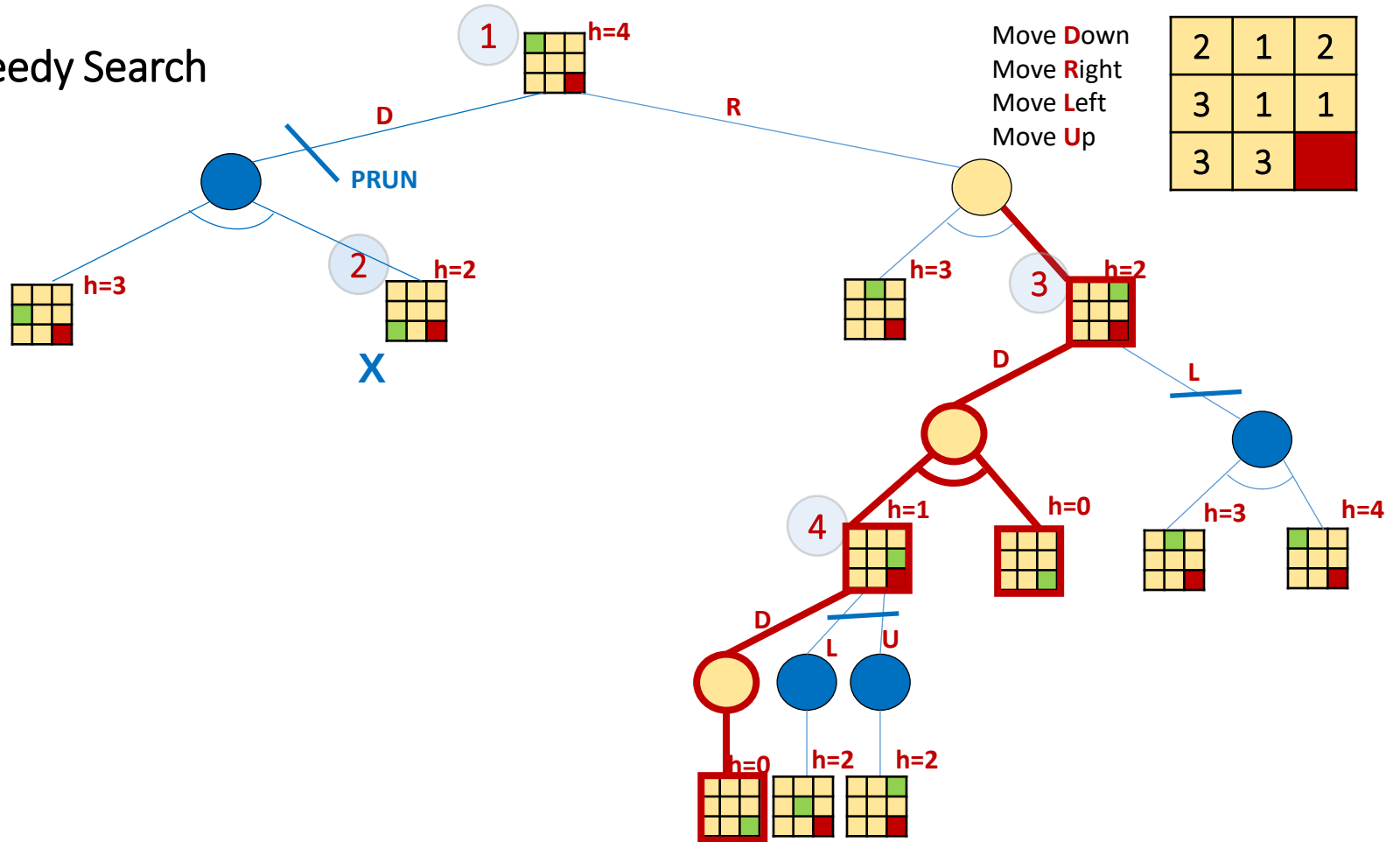
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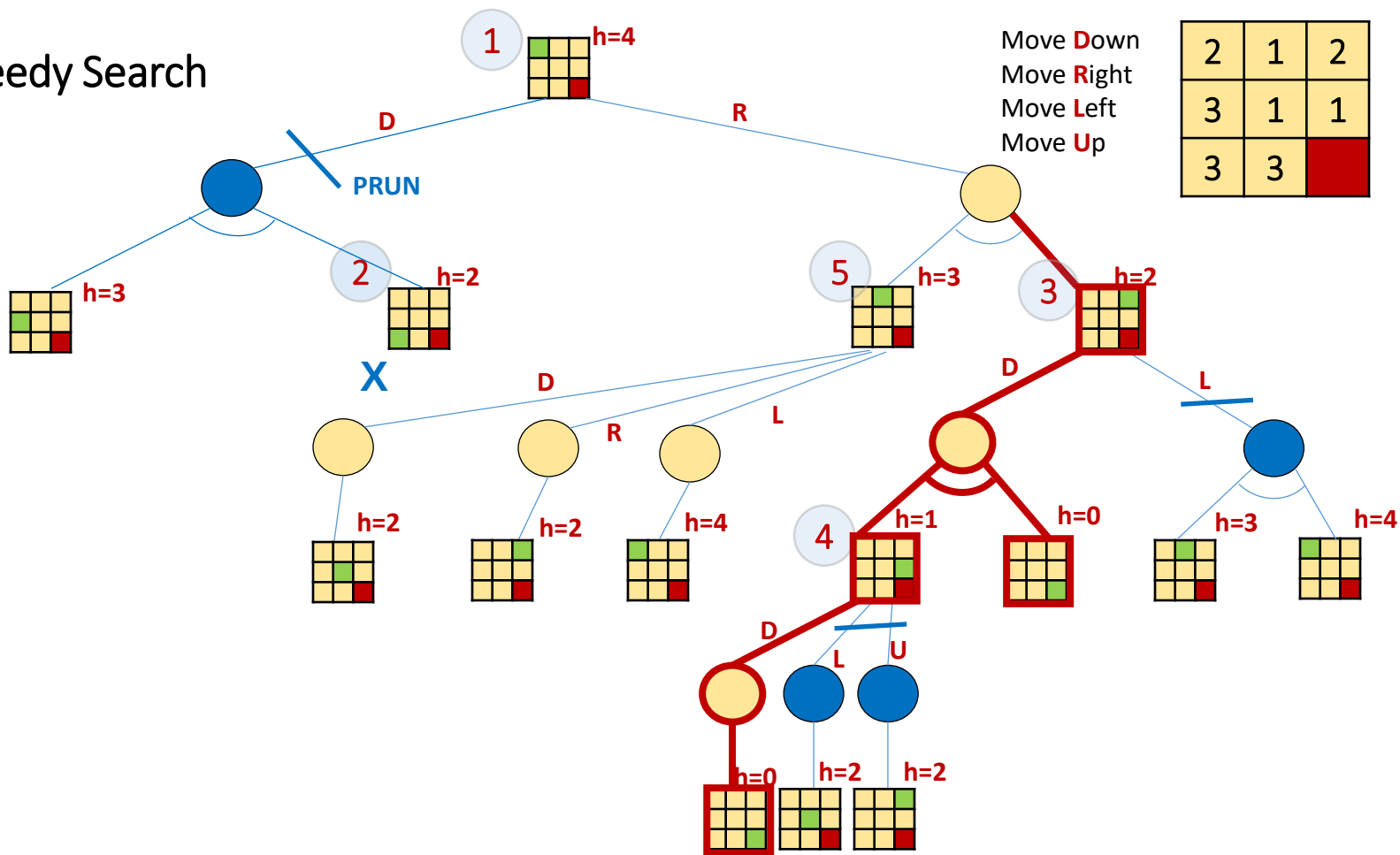
Ex 1: Greedy Search



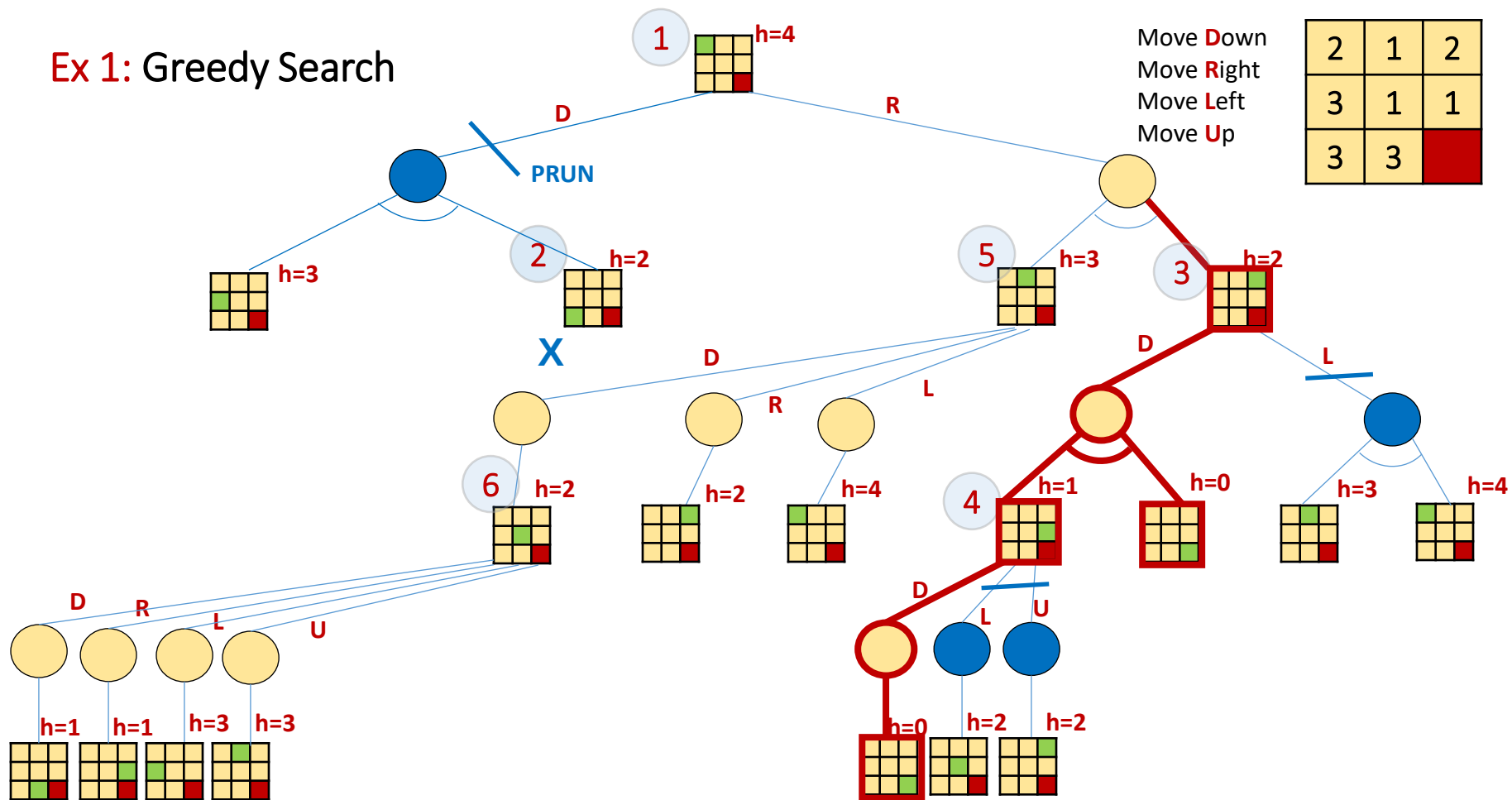
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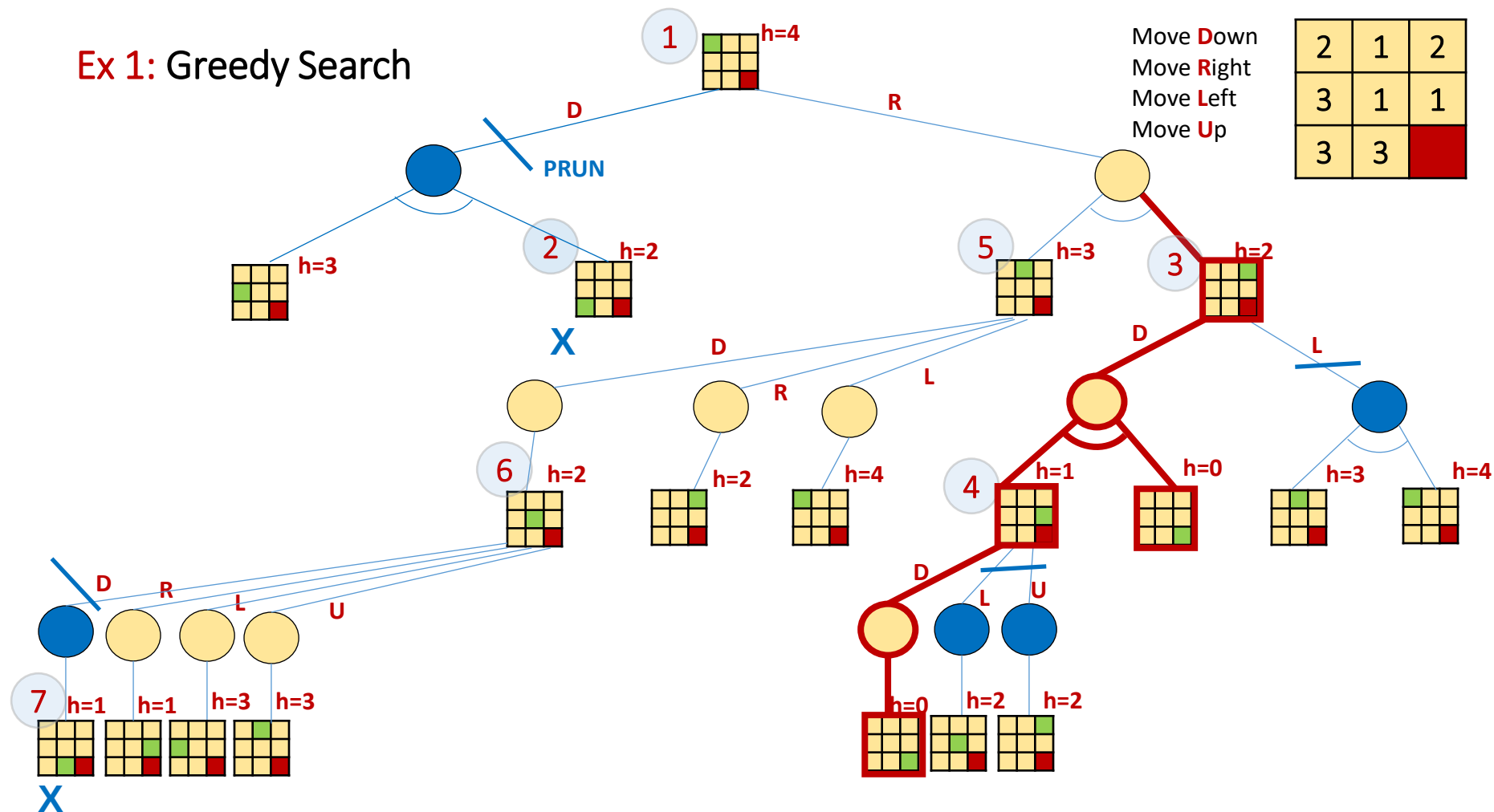
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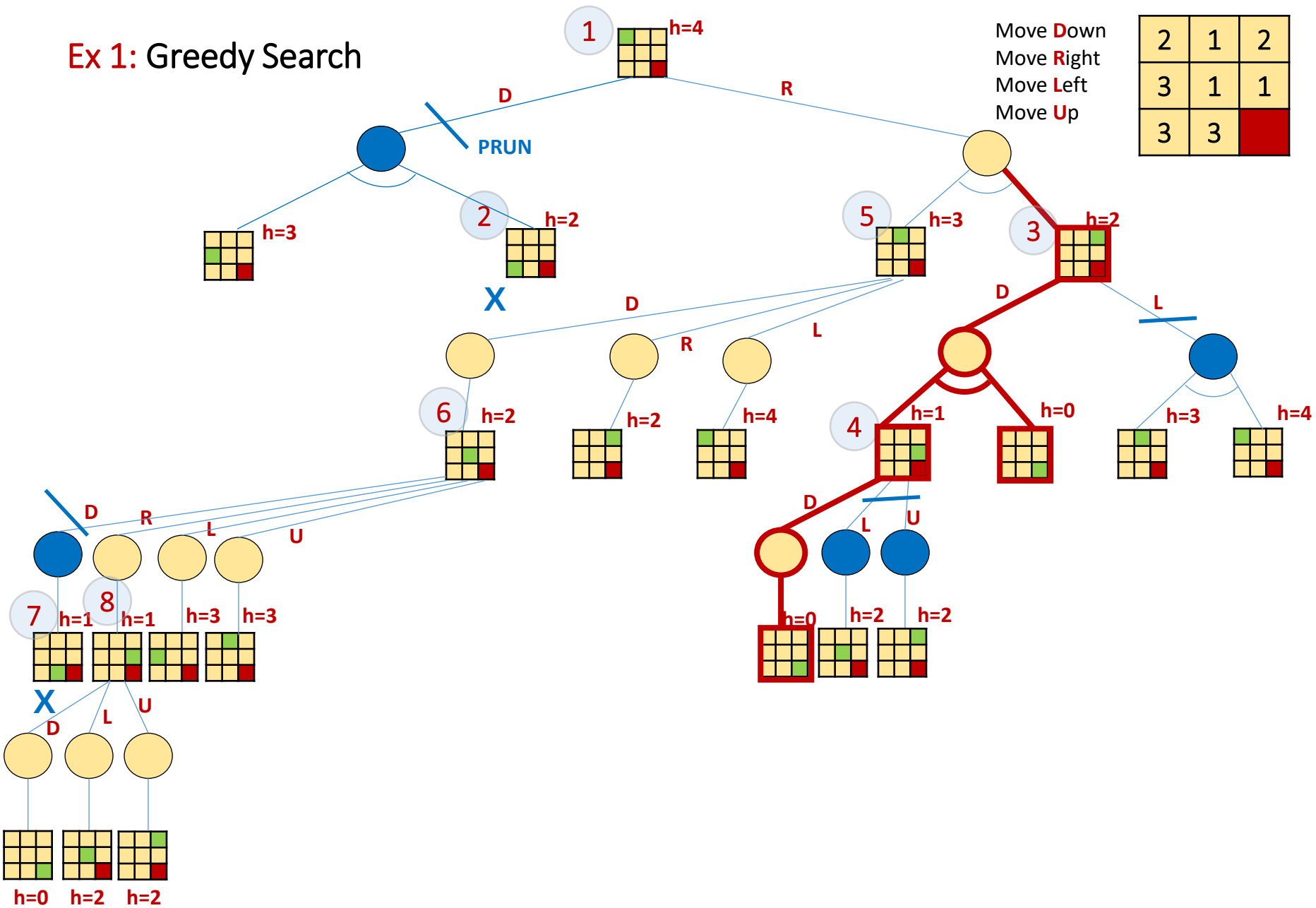
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Move **D**own
Move **R**ight
Move **L**eft
Move **U**p

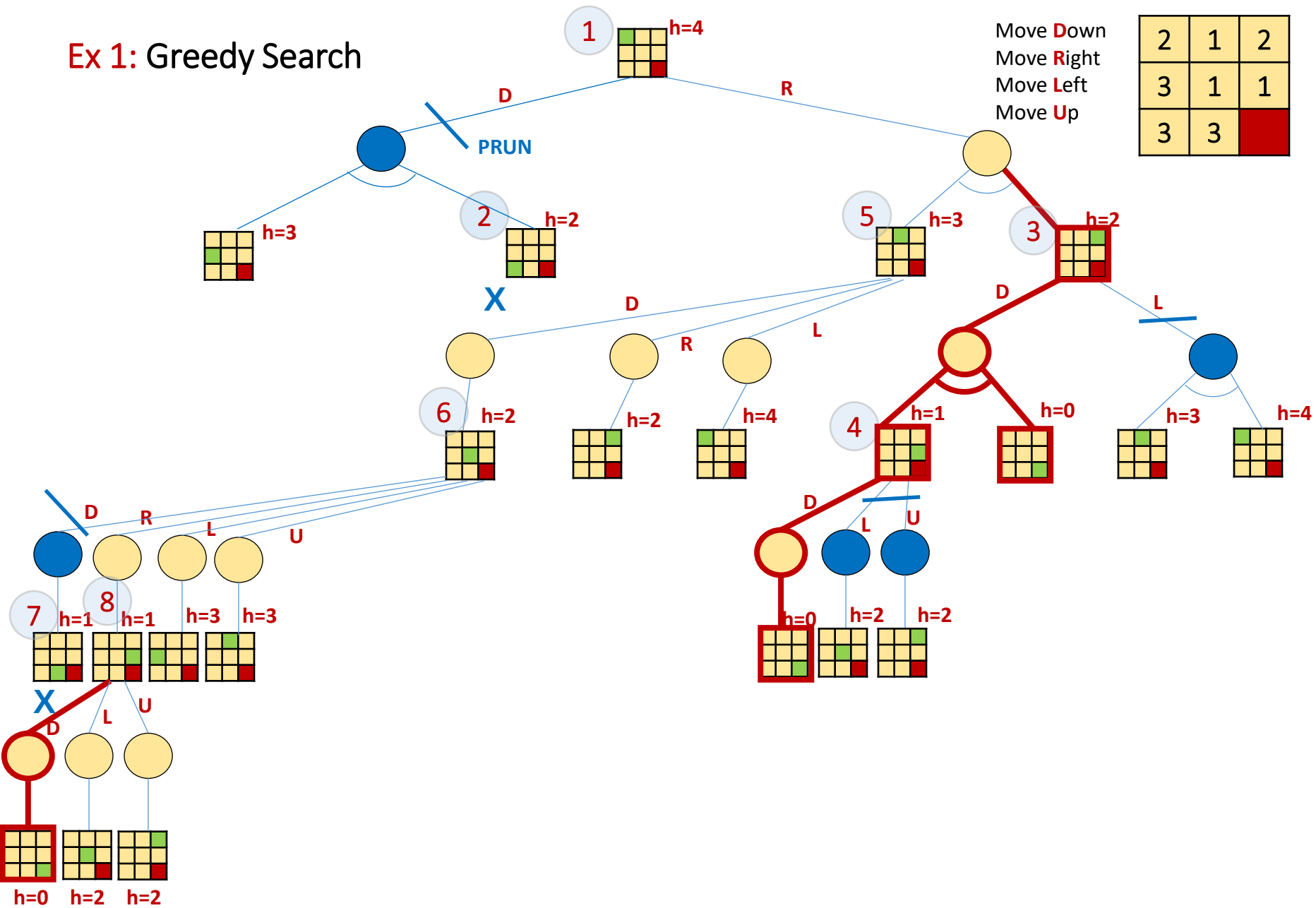
2	1	2
3	1	1
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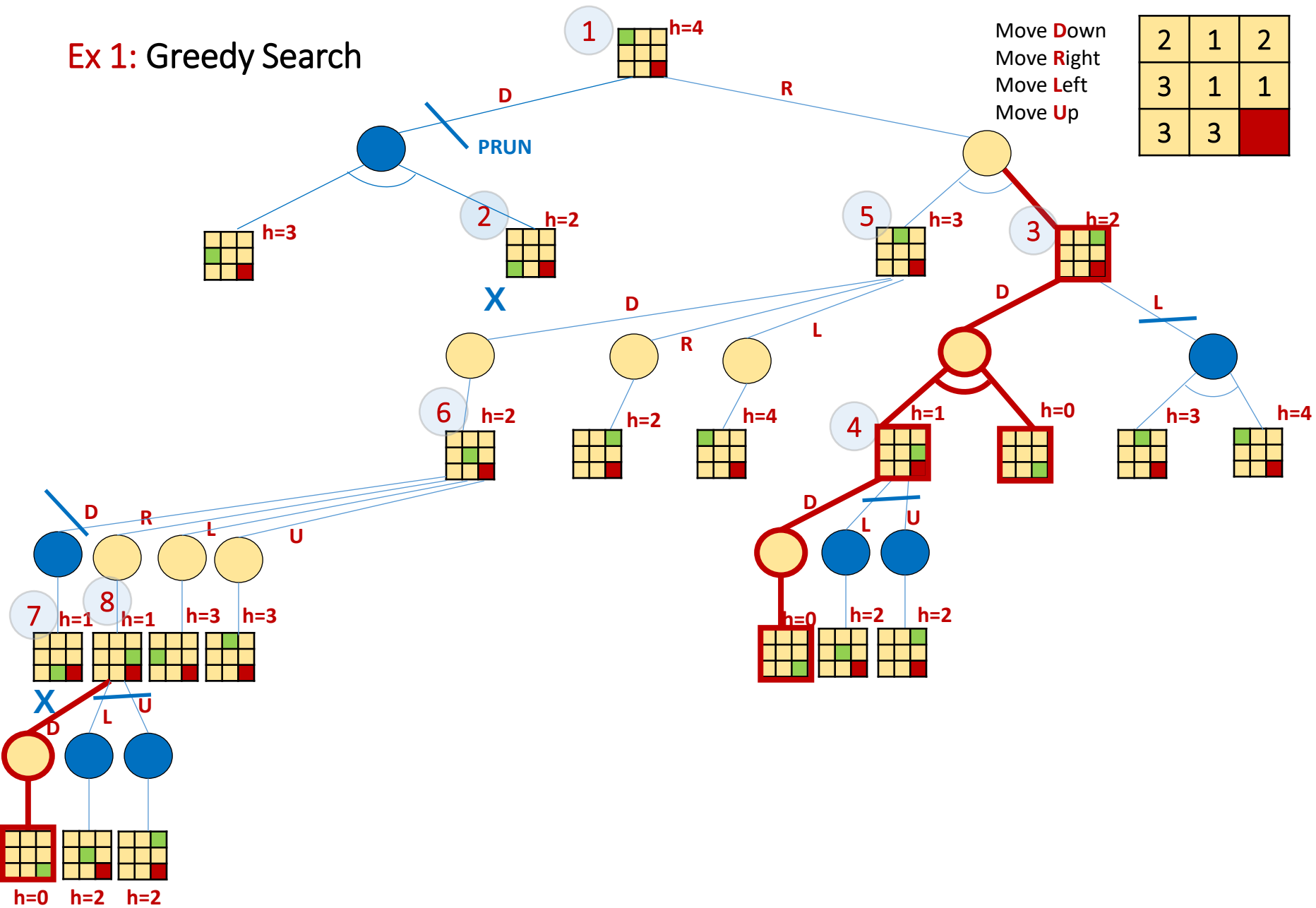
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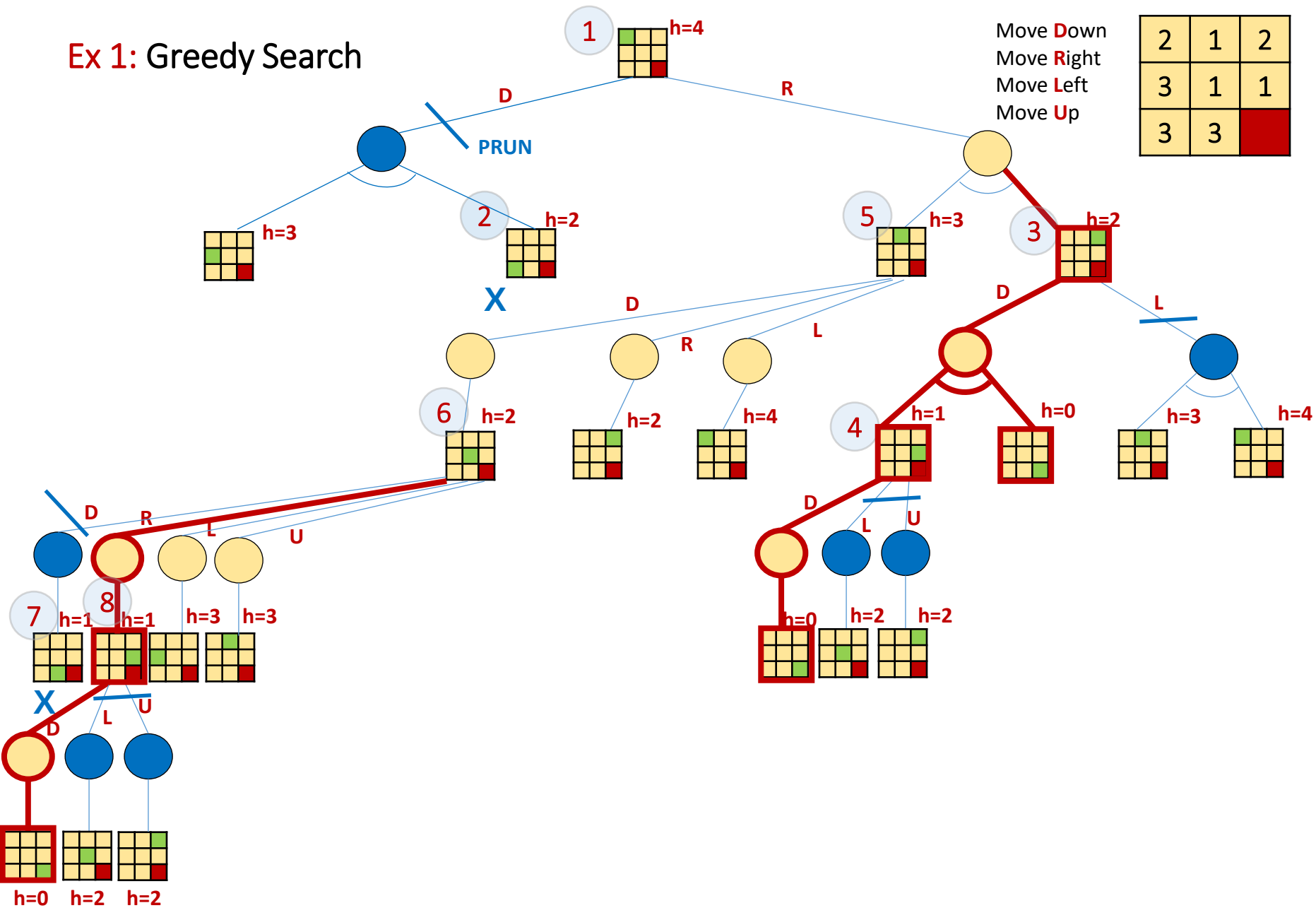
Ex 1: Greedy Search



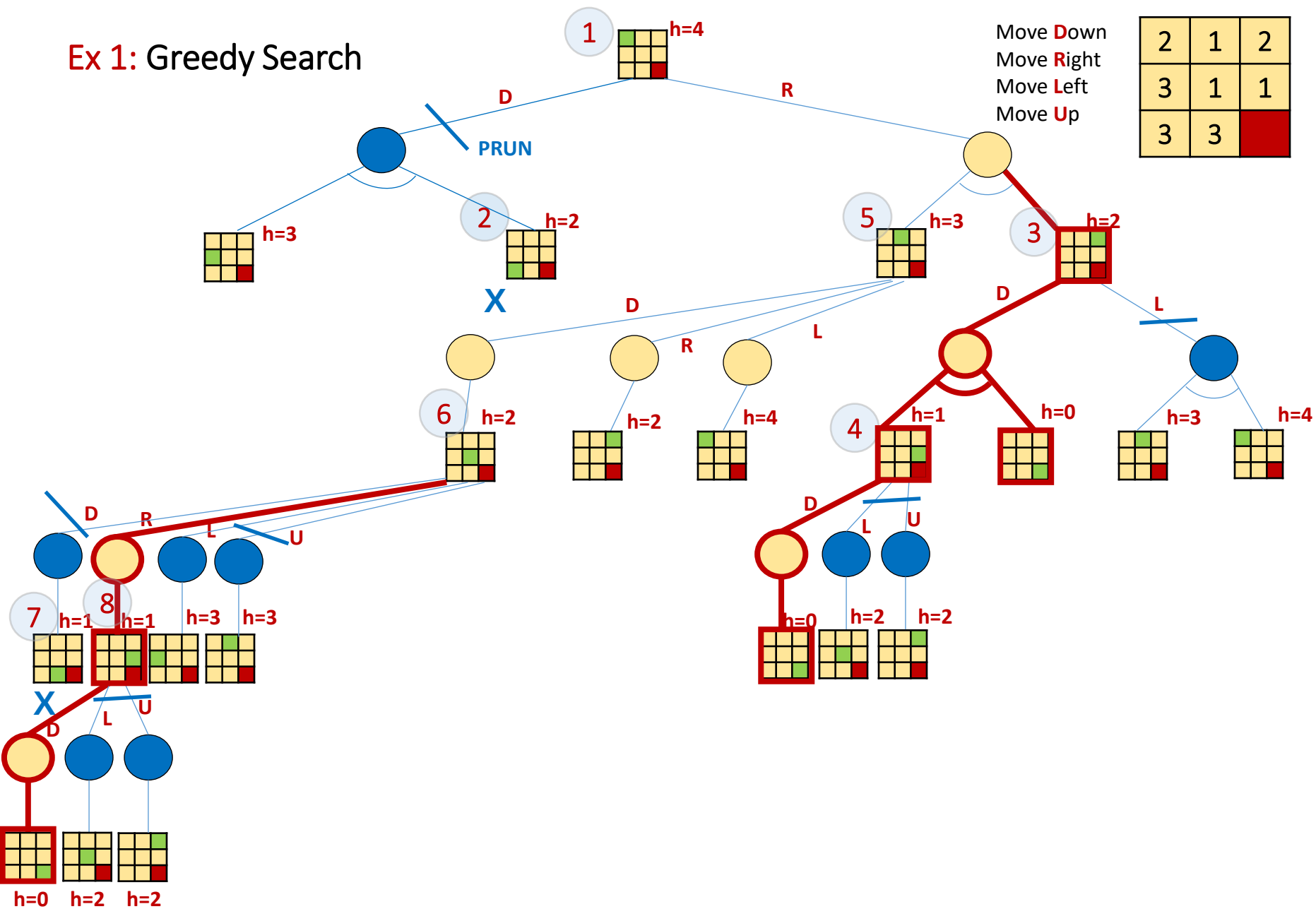
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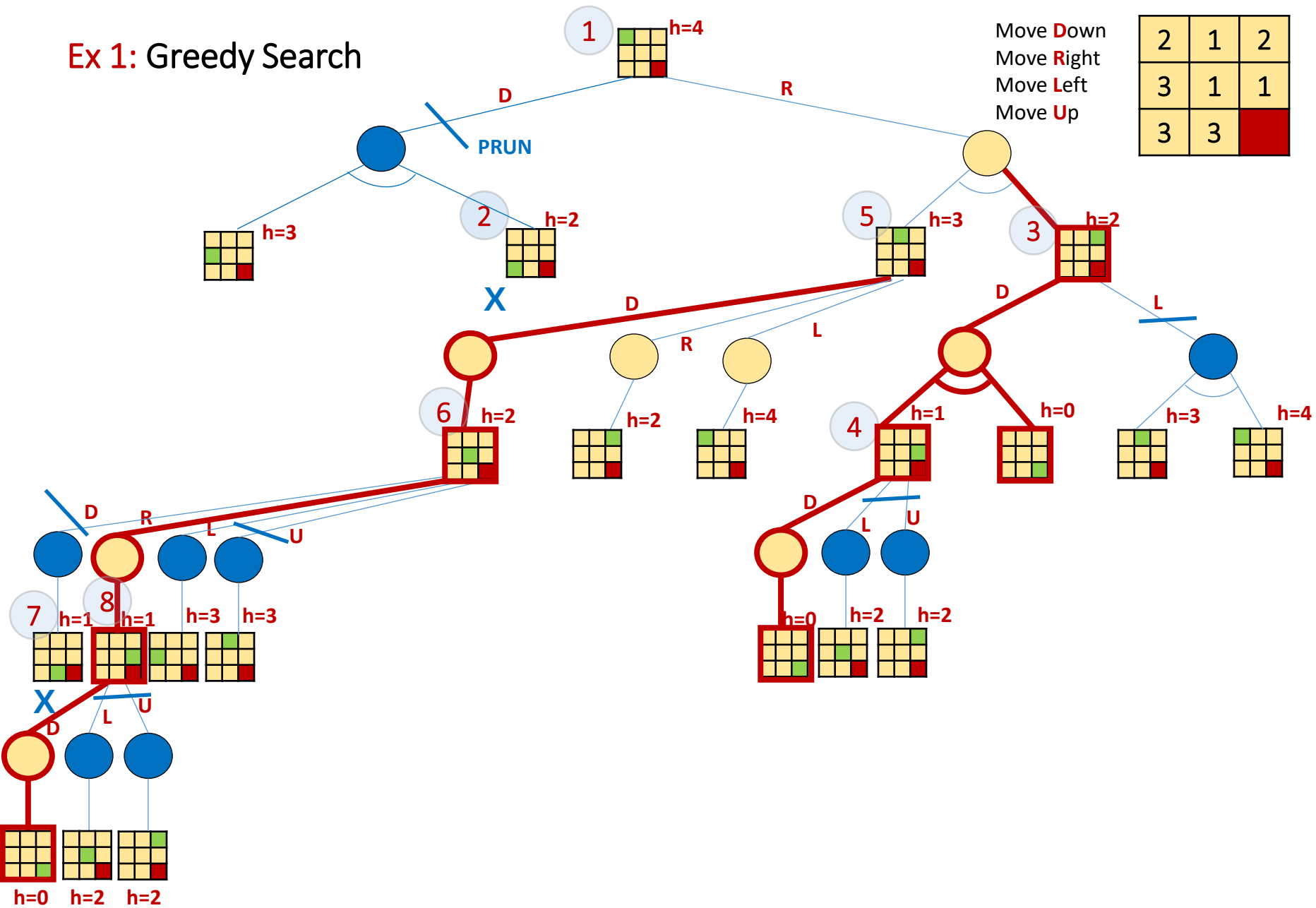
Ex 1: Greedy Search



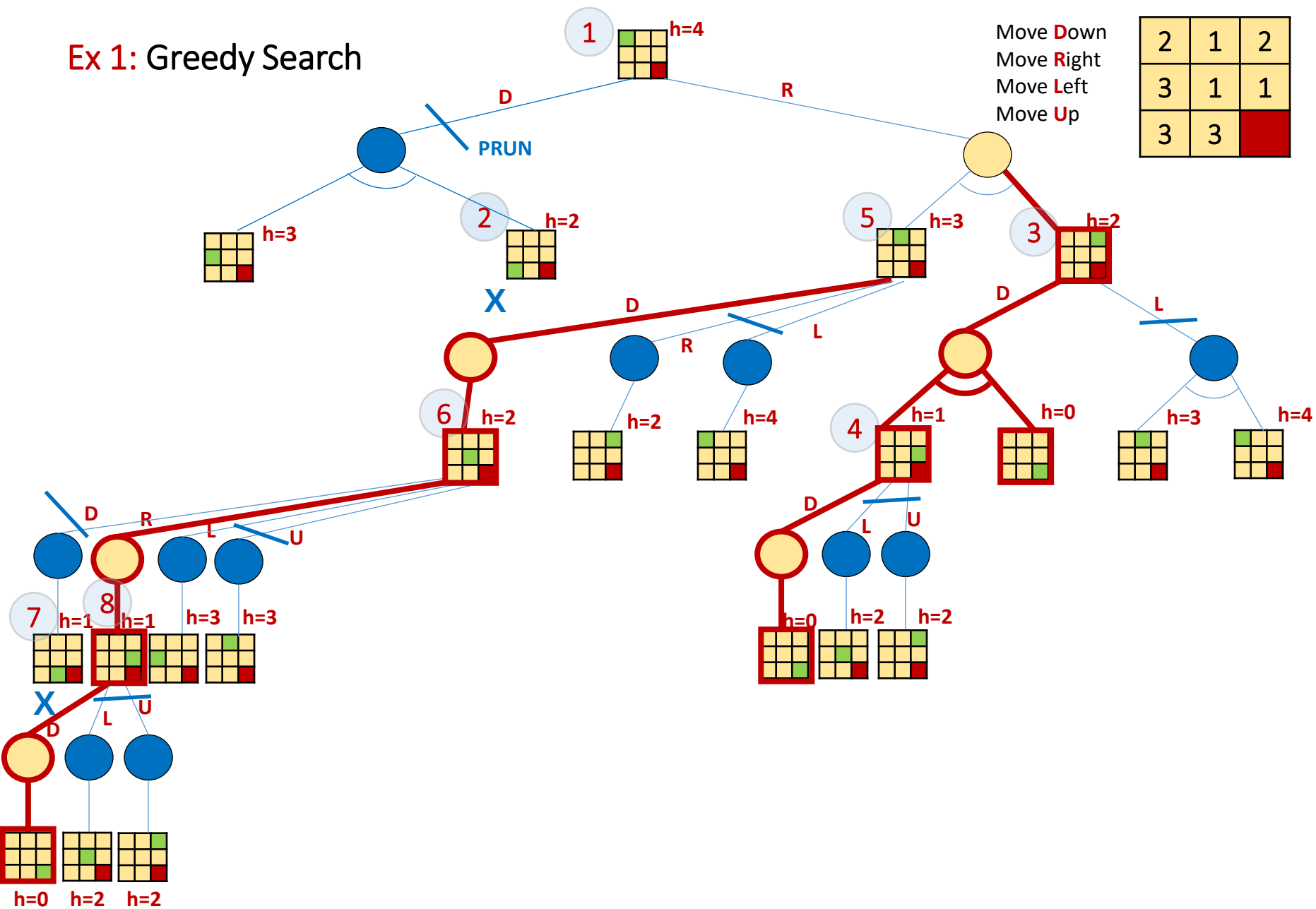
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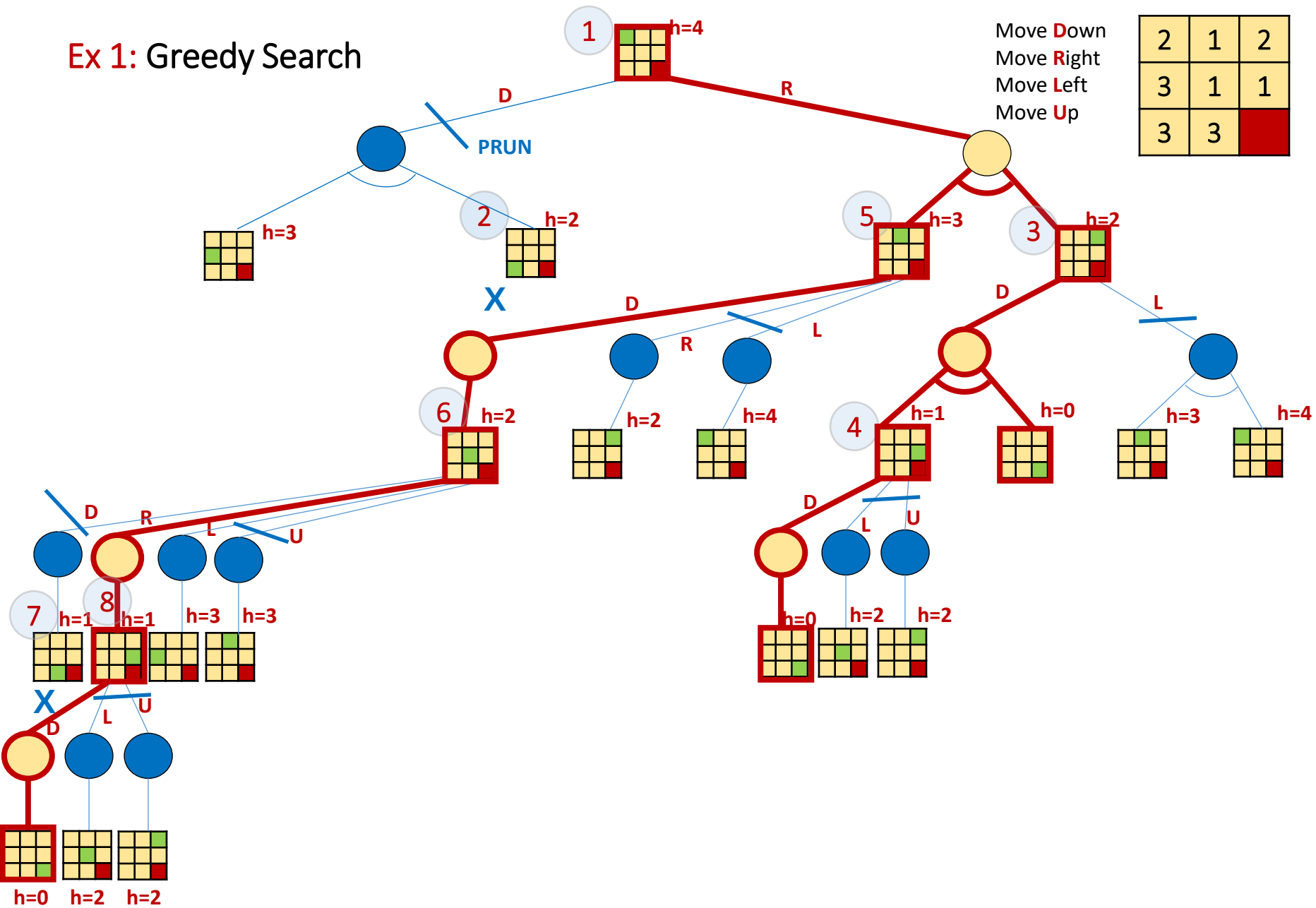
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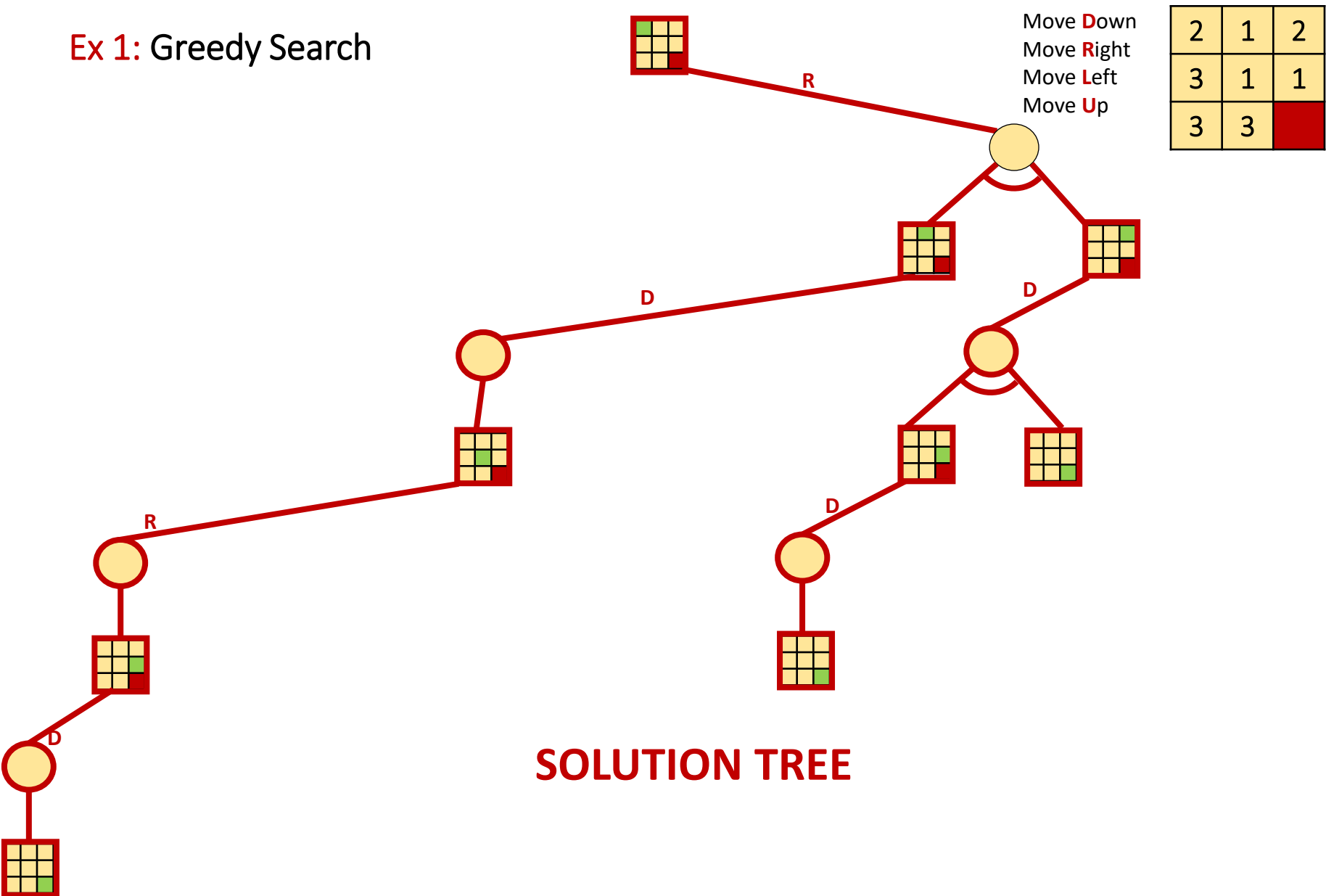
Ex 1: Greedy Search



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Ex 1: Greedy Search



What is the optimal solution for
undeterministic problems?



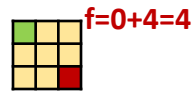
Consider the solution trees $T_1, T_2, T_3, \dots$.

Let C_n^* be the cost of a solution tree T_n and defined as the worst (the longest) path cost in tree T_n , $n = 1, 2, 3, \dots$. Then the optimal cost is

$$\min\{C_1^*, C_2^*, C_3^*, \dots\}$$

- For the above definition of optimality, solve the example through A* search with the heuristic function defined as Manhattan Distance.

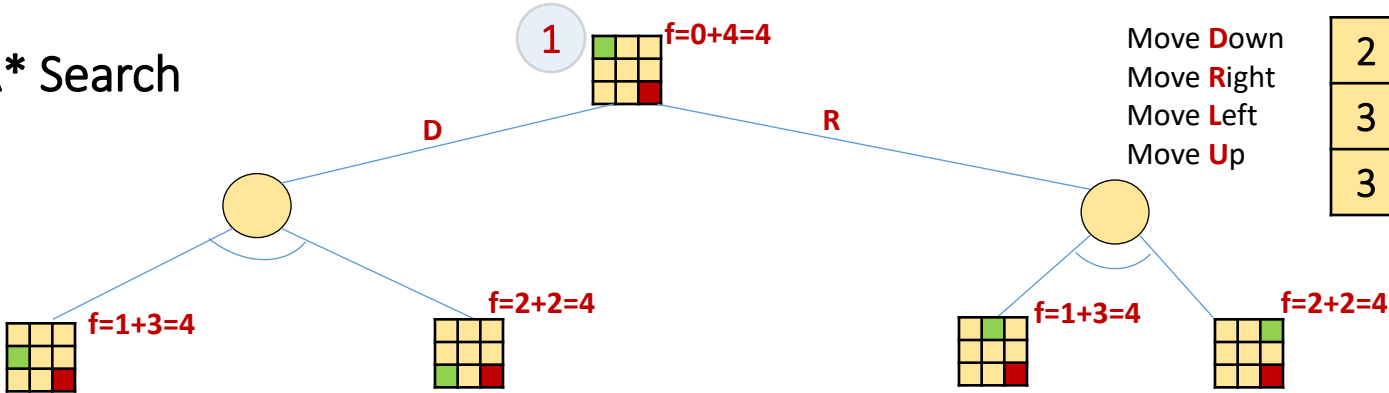
Ex 1: A* Search



Move **D**own
Move **R**ight
Move **L**eft
Move **U**p

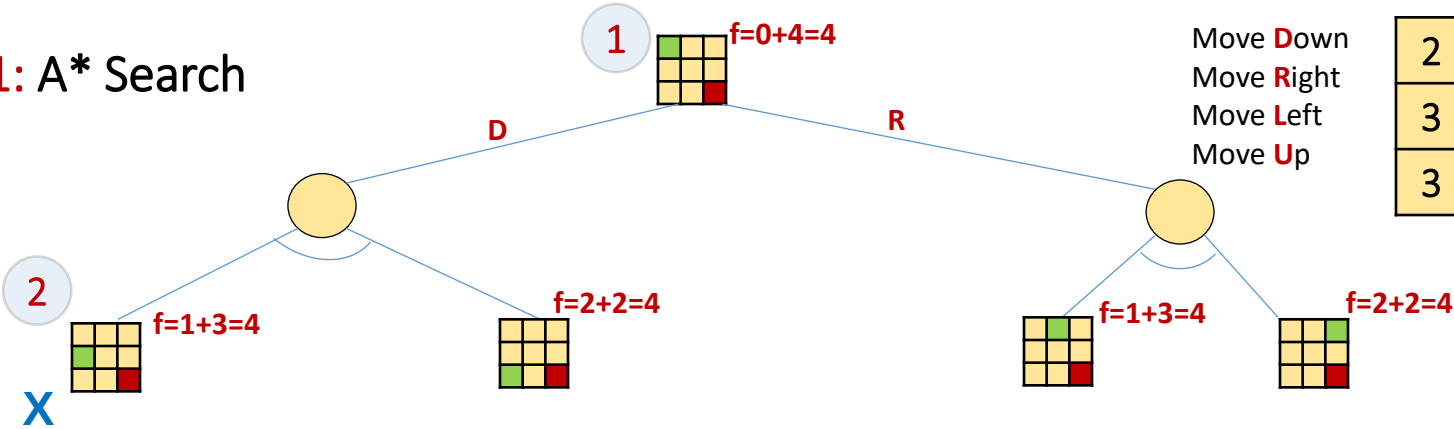
2	1	2
3	1	1
3	3	

Ex 1: A* Search



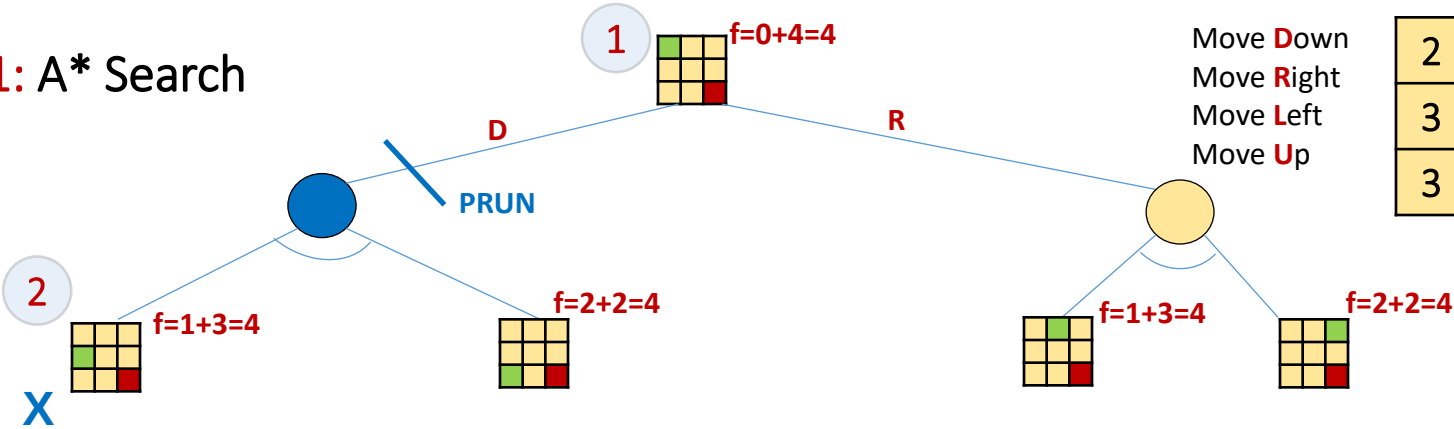
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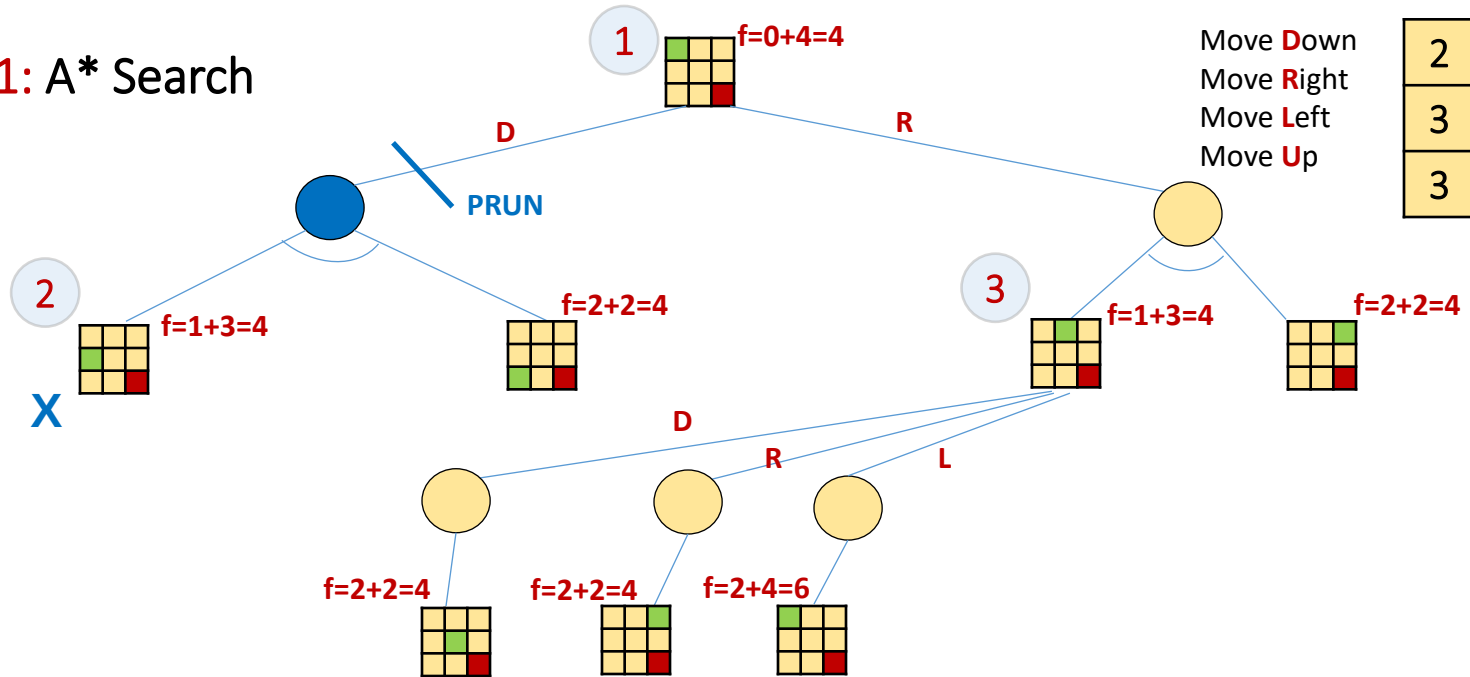
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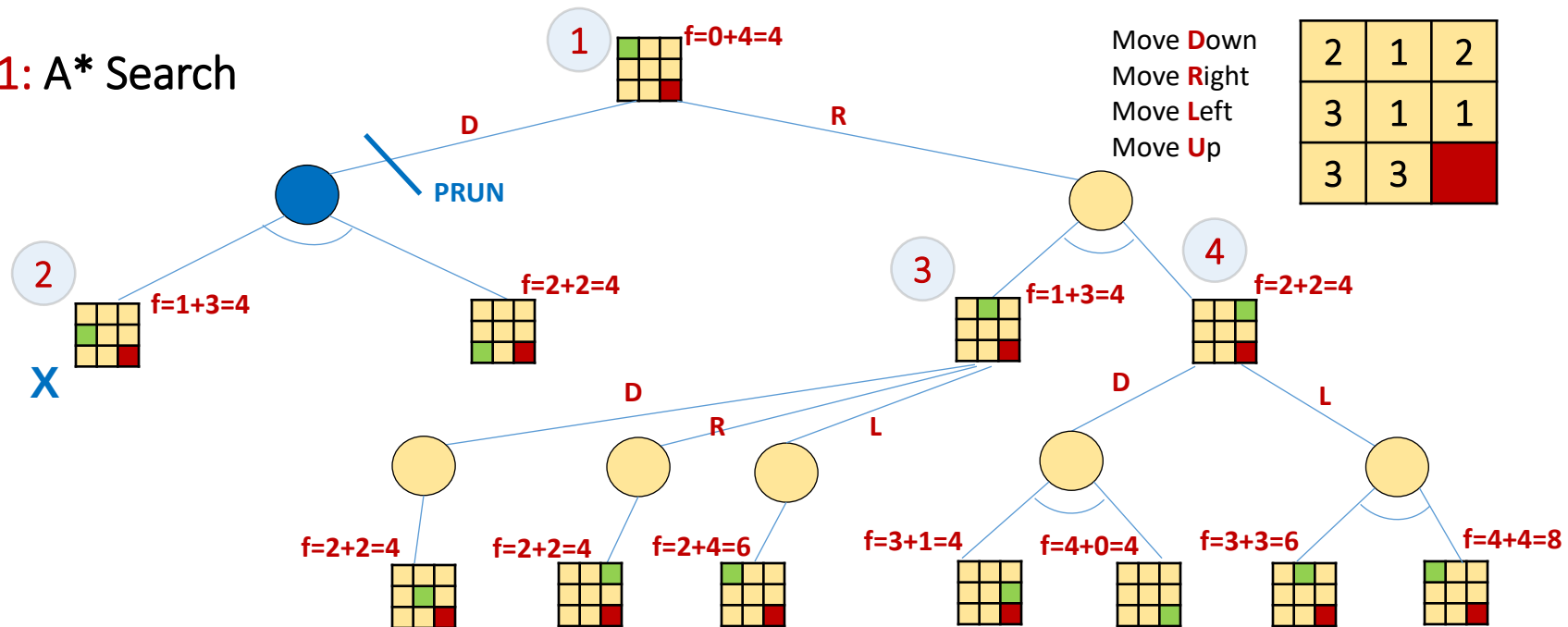


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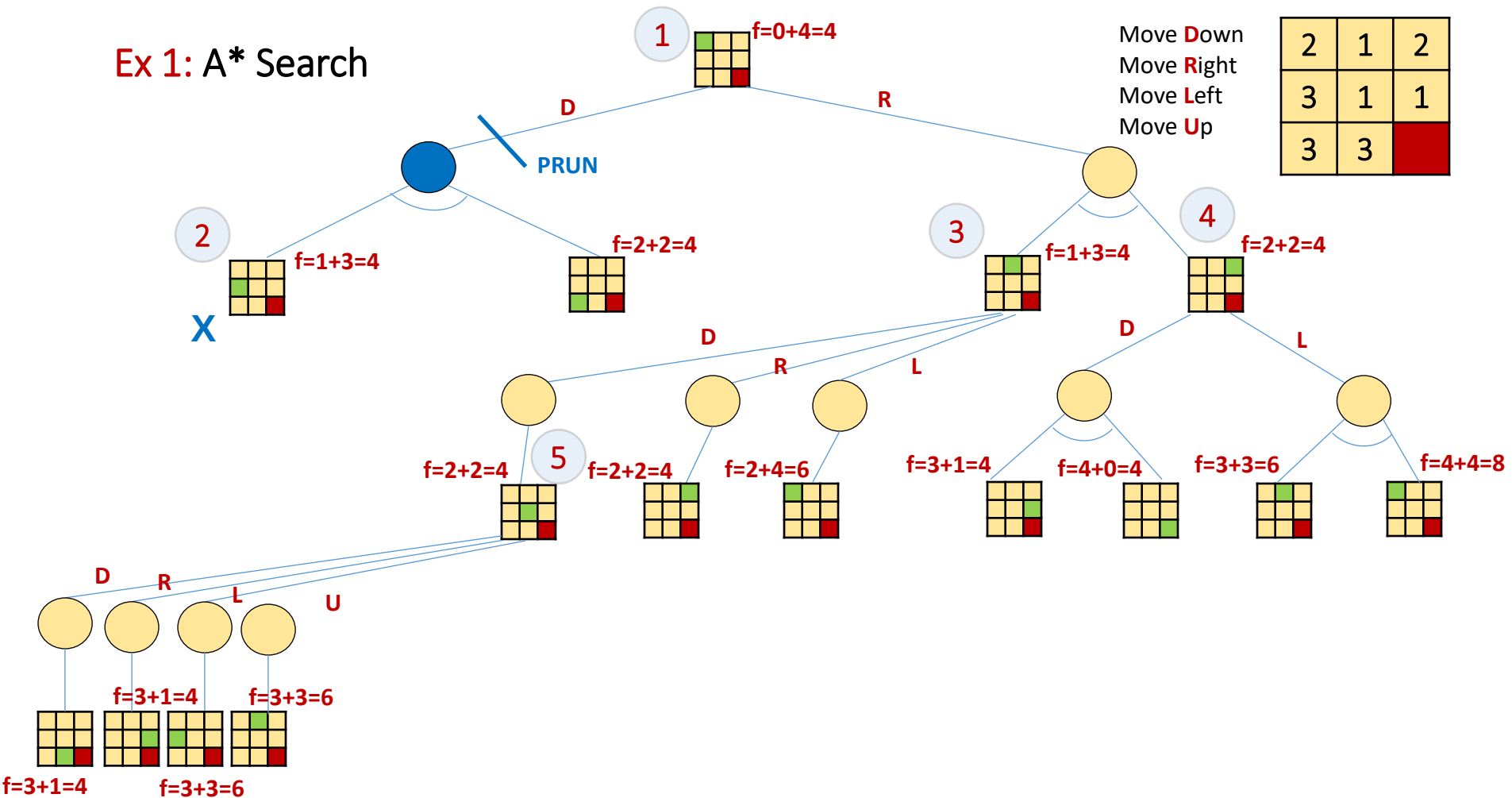
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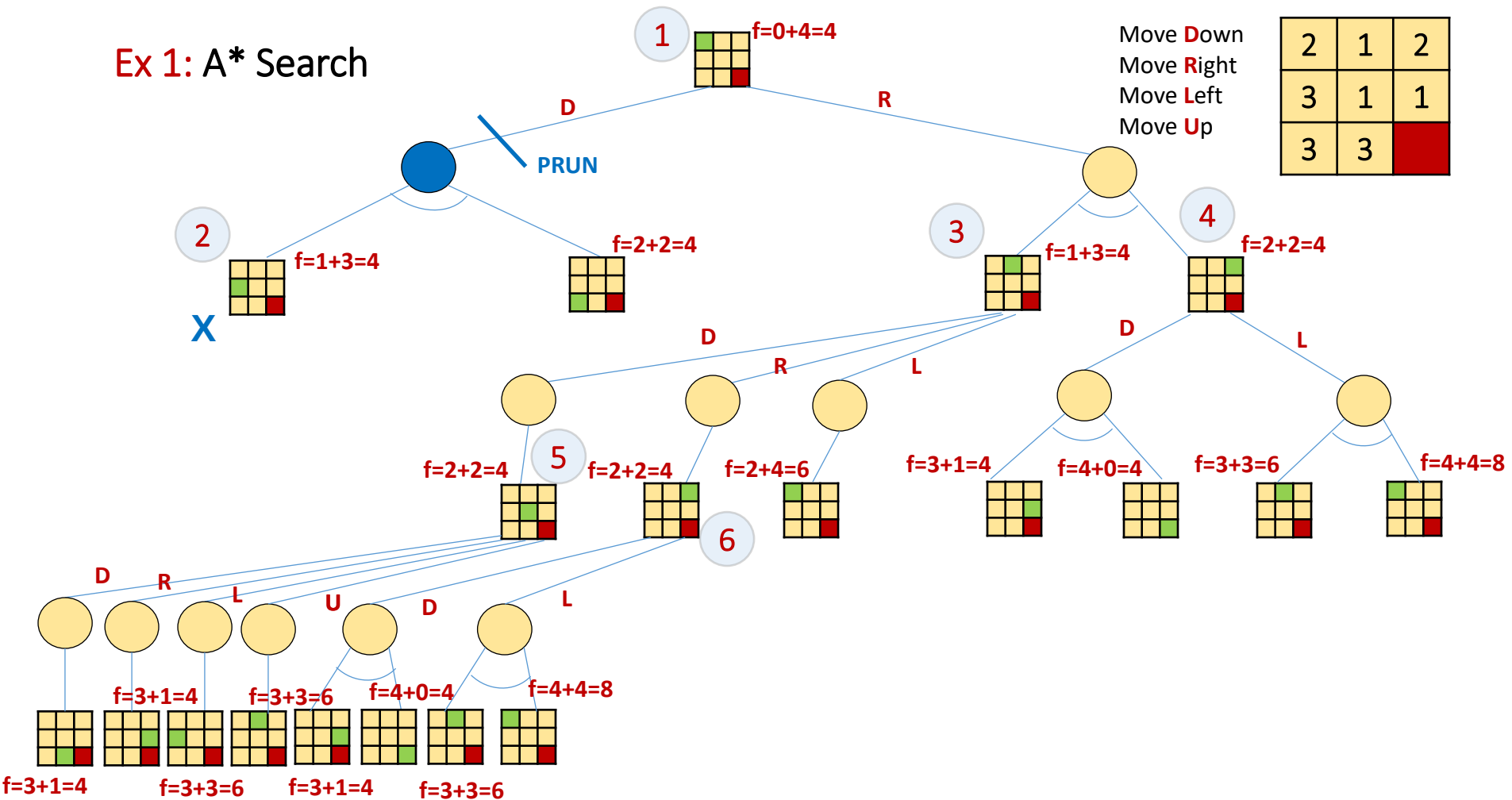
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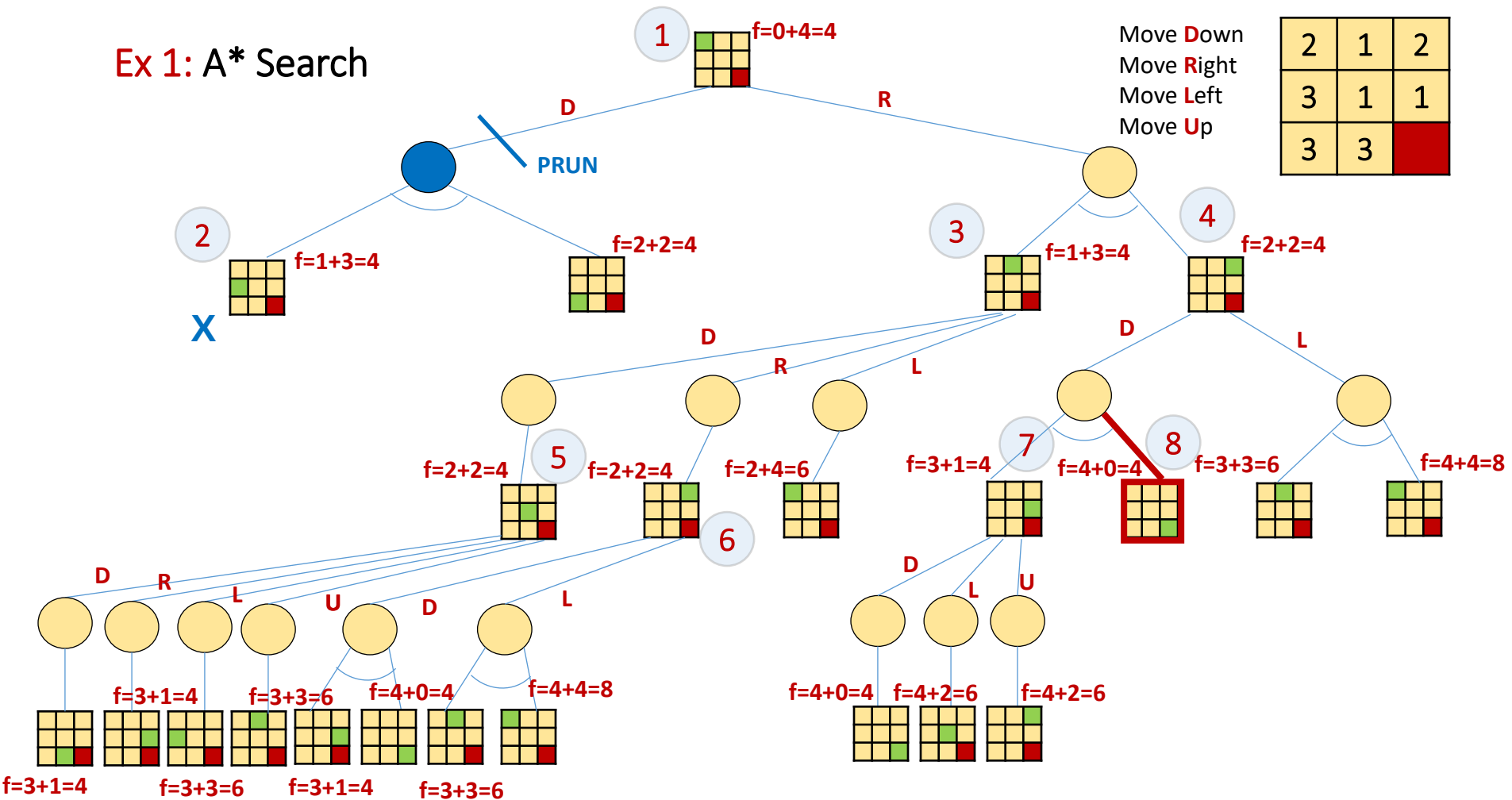
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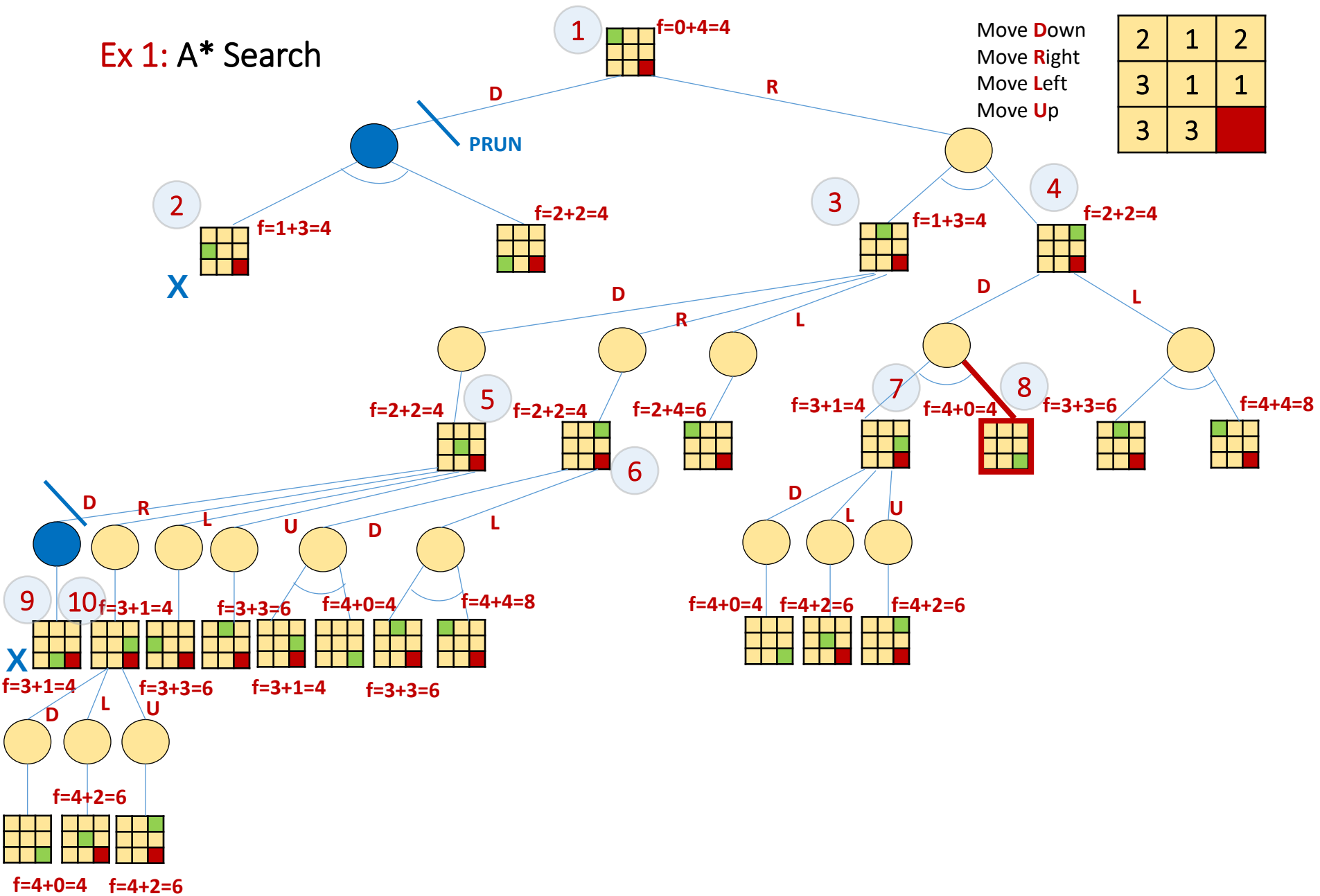
Ex 1: A* Search



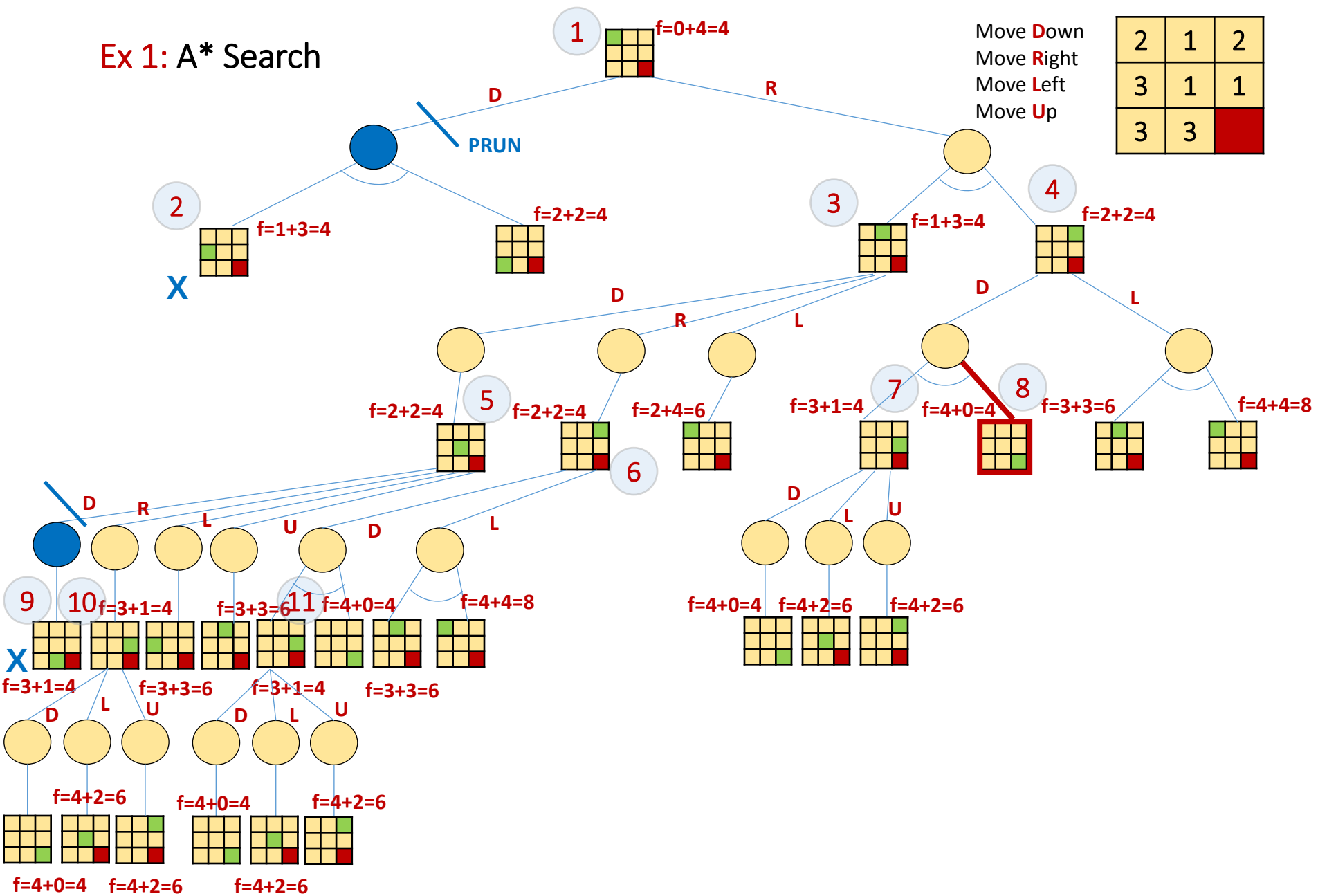
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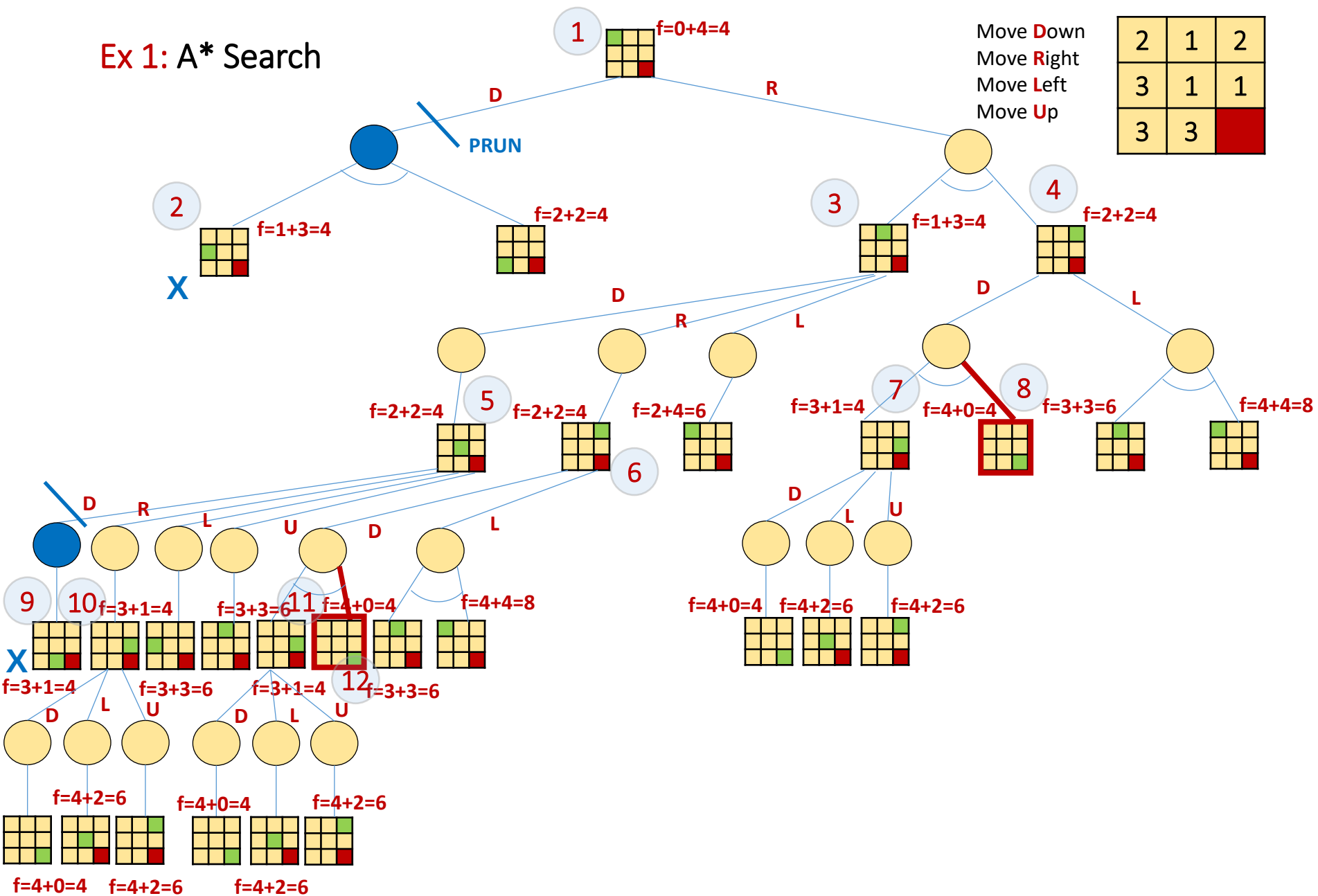
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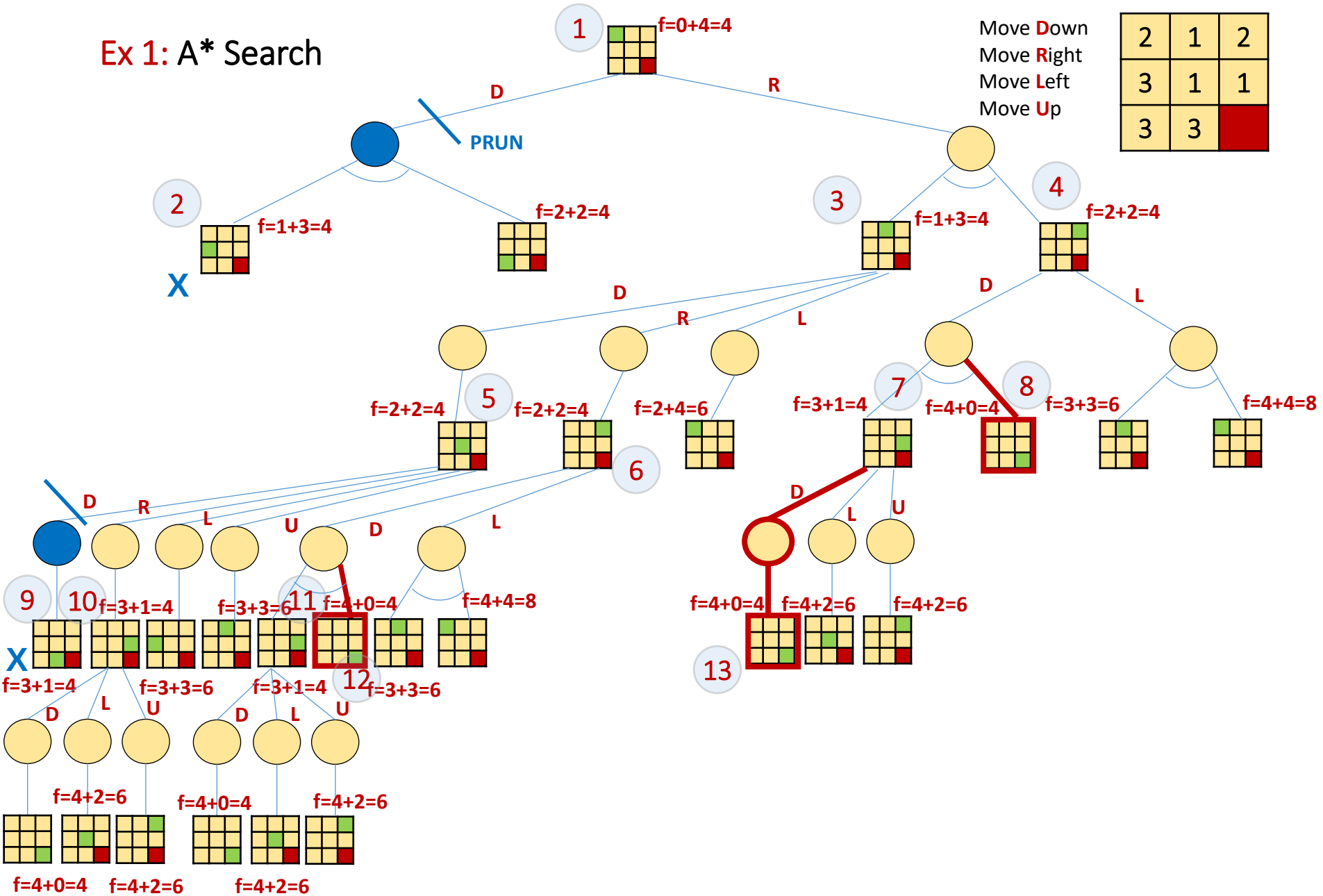
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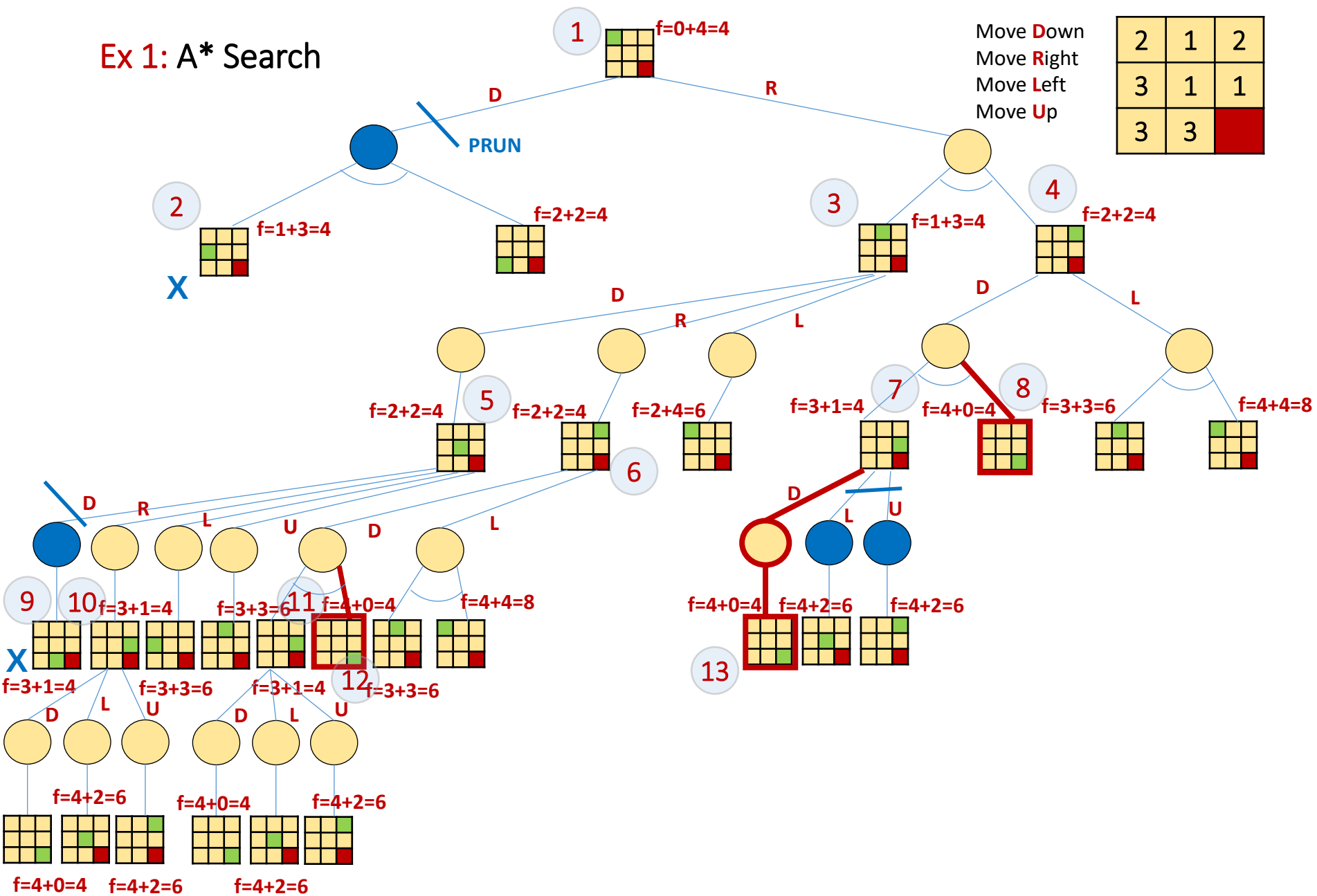
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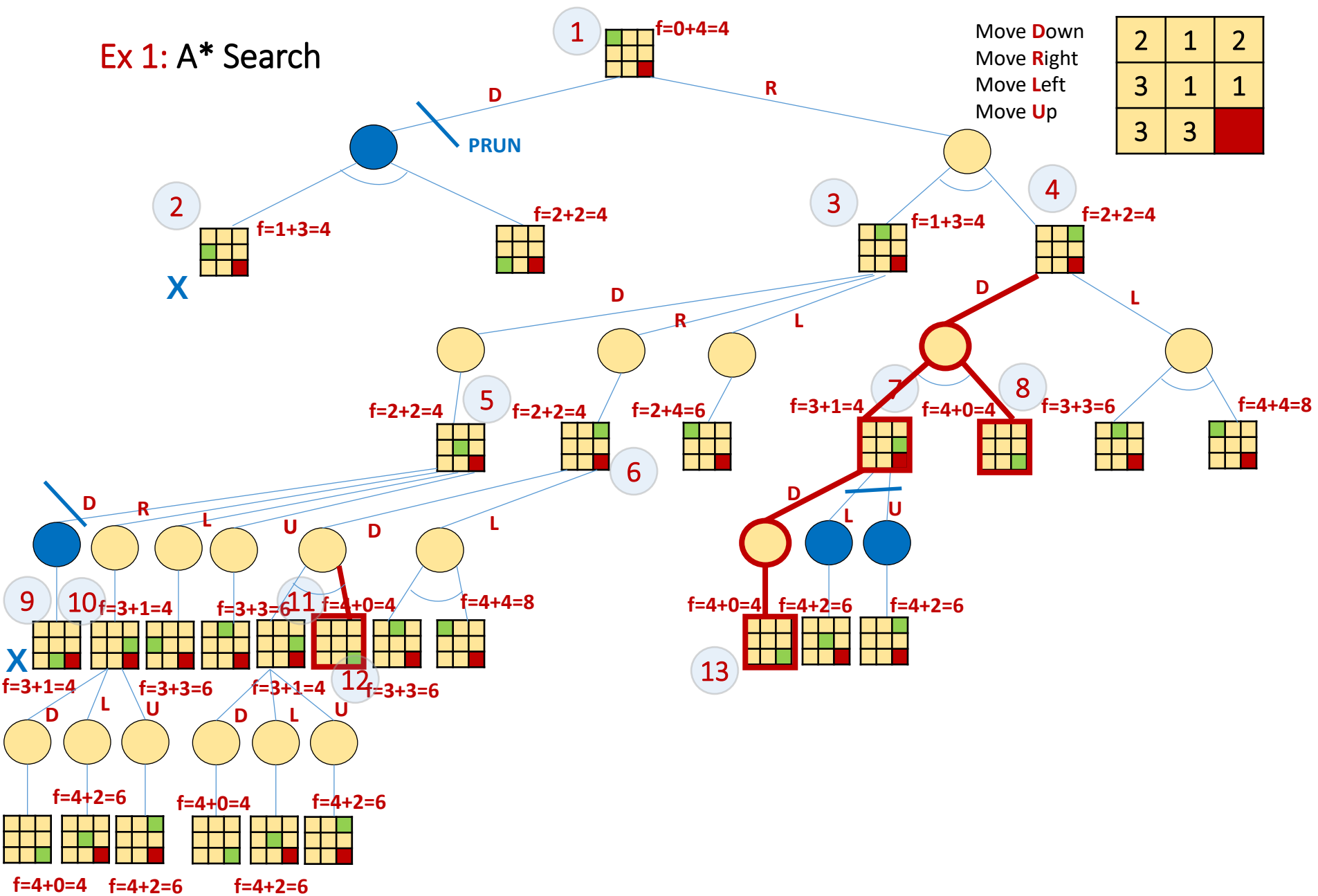
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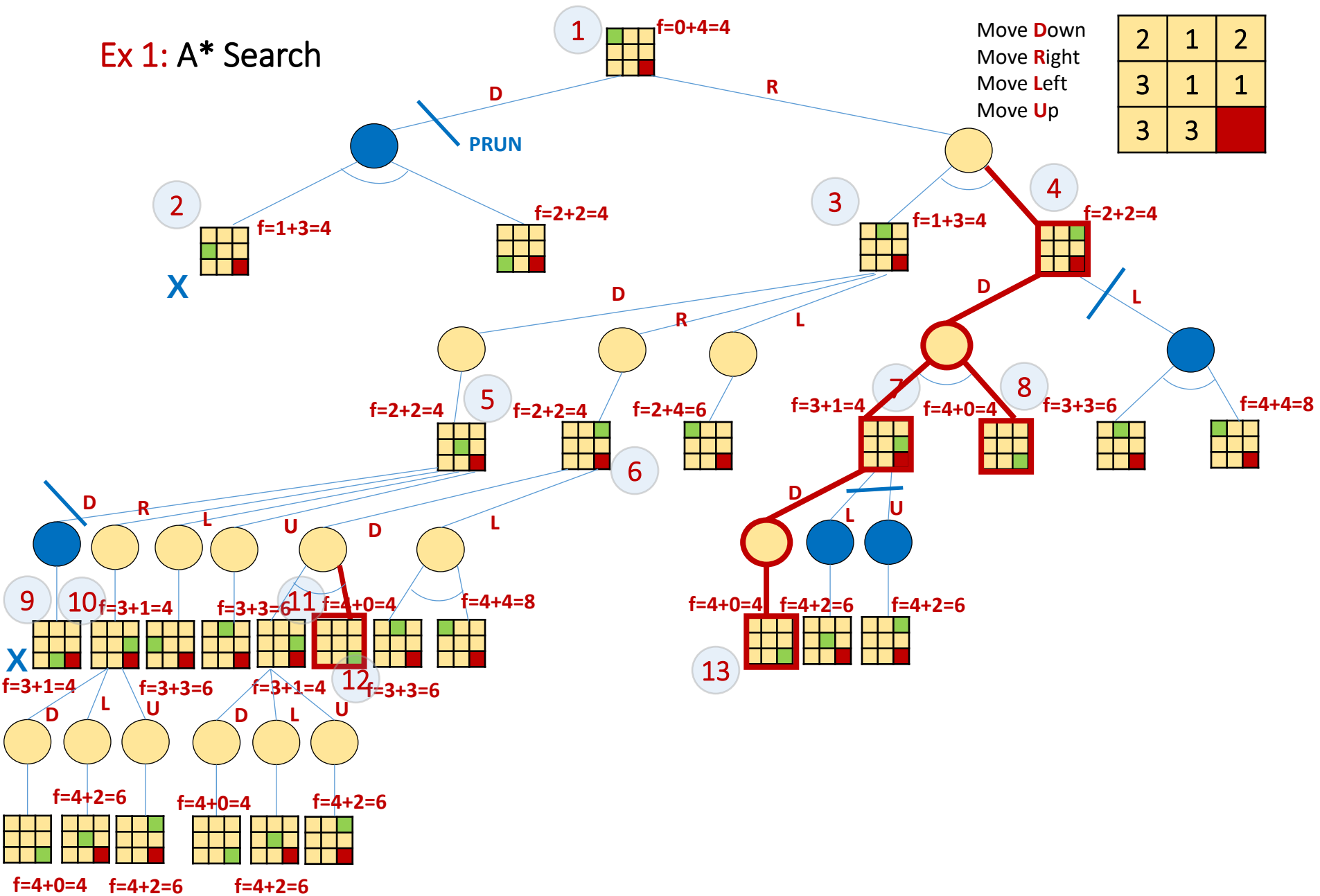
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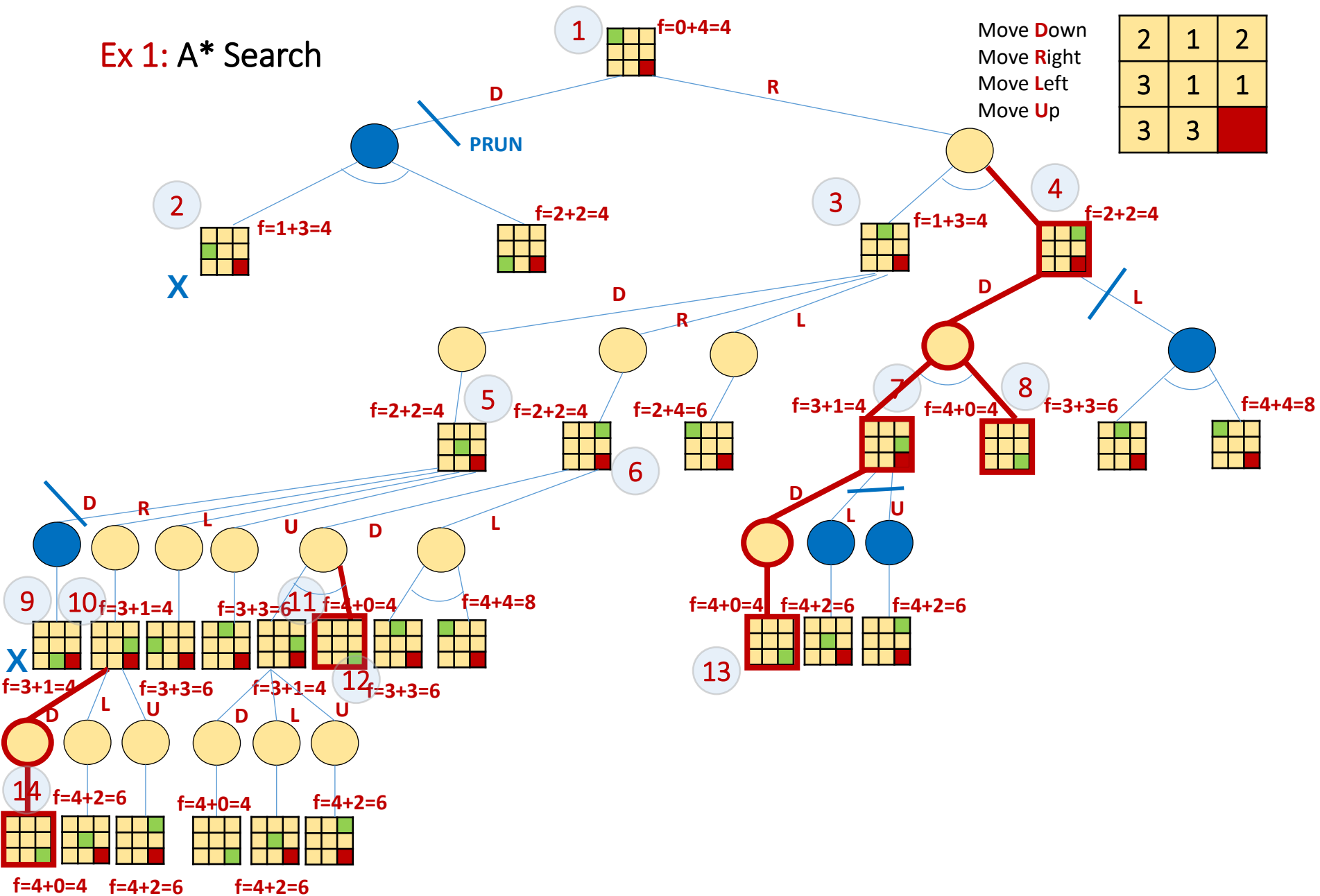
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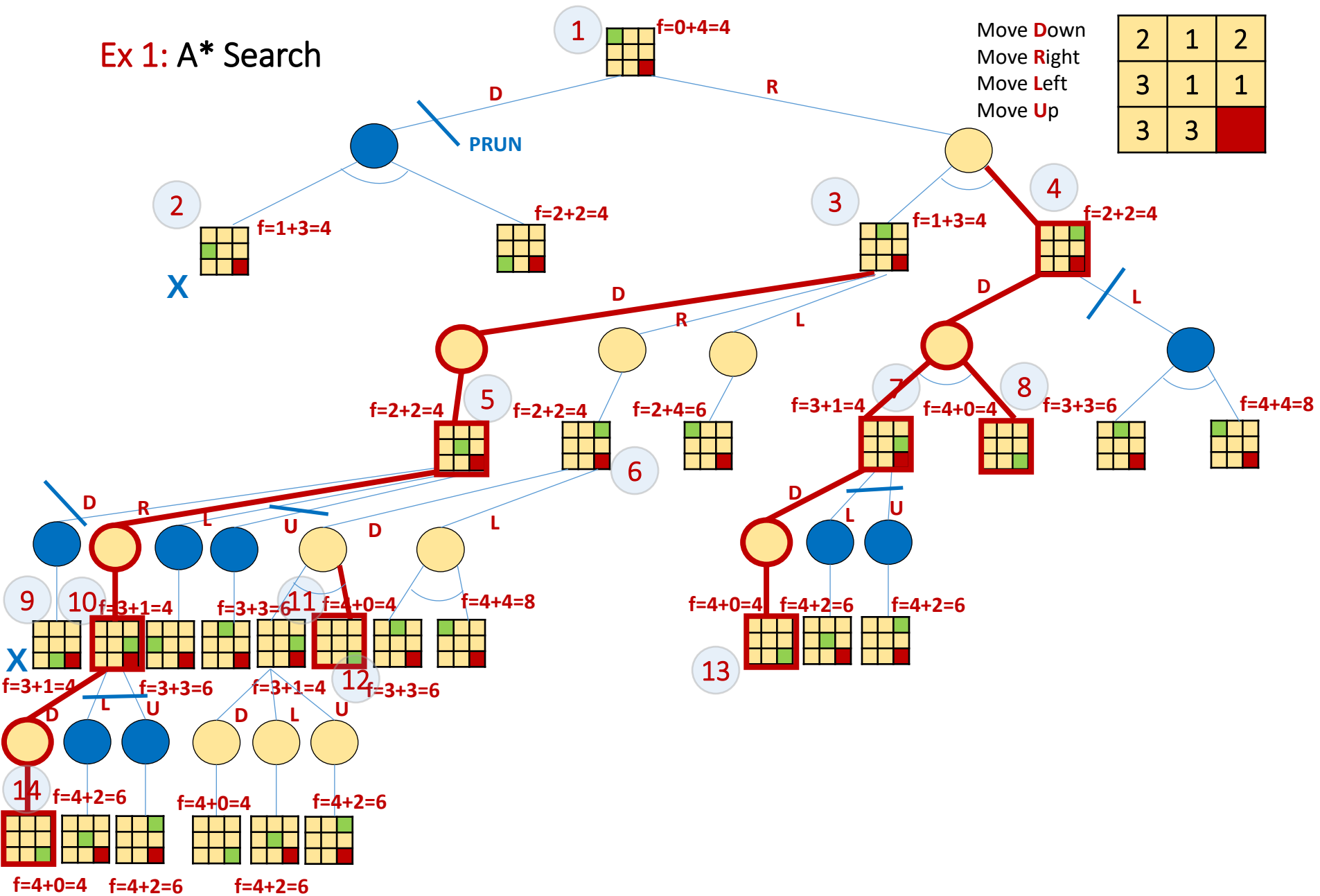
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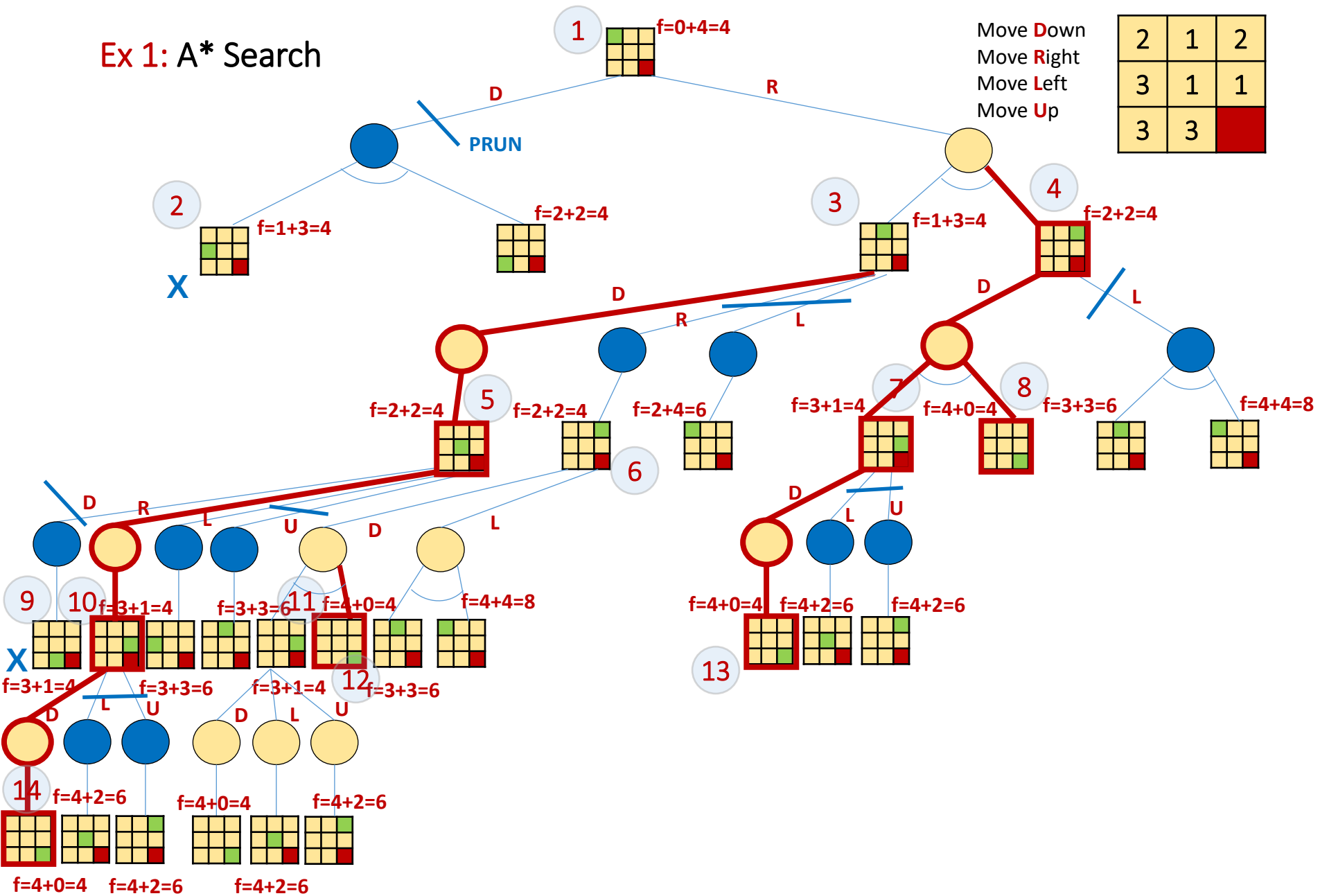
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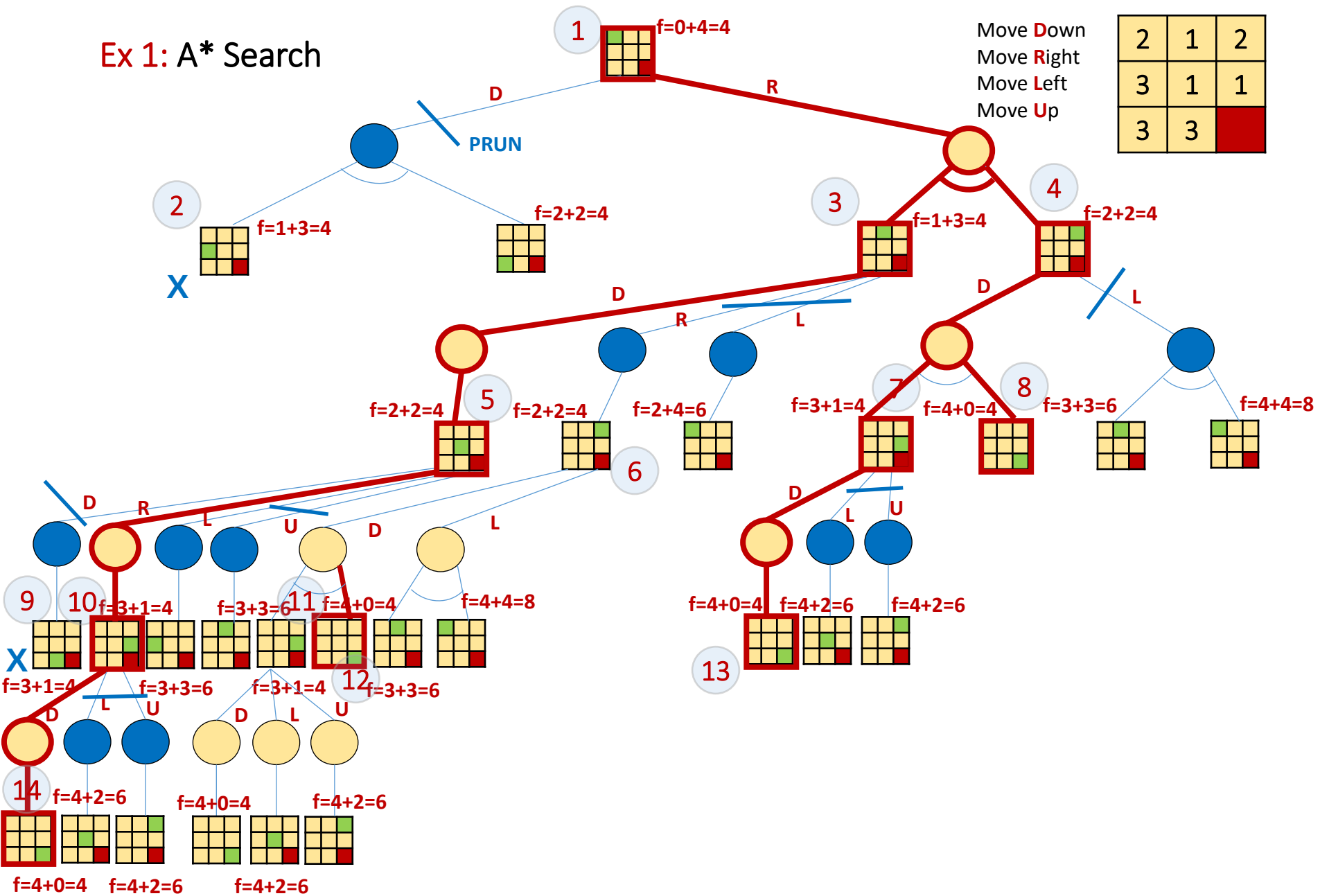
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